

**GRADE 5 COMMON CORE MATH**

# **Answer Key**

Main • Backward Branches • Forward 1 (Grade 6) • Forward 2 (Grade 7)

57 question sets • 1580 questions

Page numbers match the Question Collection.

Every page shows its question ID (for example S40-F2-Q3).

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# Evaluate expressions that use parentheses

|                   |                                            |                   |
|-------------------|--------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 5 main question                      | <b>5.OA.A.1</b>   |
| <b>BACKWARD 1</b> | Do the work in parentheses first           | <b>3.OA.D.8</b>   |
| <b>BACKWARD 2</b> | Multiplication and division facts          | <b>3.OA.C.7</b>   |
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| <b>FORWARD 1</b>  | Evaluate expressions with variables        | <b>6.EE.A.2.c</b> |
| <b>FORWARD 2</b>  | Evaluate expressions with negative numbers | <b>7.NS.A.3</b>   |

SET 1 • 5.OA.A.1

# MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$6 \times (8 + 5)$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 1 • MAIN • Q1/5

8

QUESTION ID

**S1-M-Q1**

TYPE

Short answer

CORRECT ANSWER

**78**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 1 • MAIN • Q1/5



Evaluate.

$$(35 - 11) \div 4$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 1 • MAIN • Q2/5

9

QUESTION ID

**S1-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**6**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 1 • MAIN • Q2/5

Evaluate.

$$50 - (6 \times 7)$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 1 • MAIN • Q3/5

10

QUESTION ID

**S1-M-Q3**

TYPE

Short answer

CORRECT ANSWER

**8**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 1 • MAIN • Q3/5

Evaluate.

$$3 + (24 \div 6) \times 5$$

- A. 35
- B. 135
- C. 27
- D. 23

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 1 • MAIN • Q4/5

11

QUESTION ID

**S1-M-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. 23**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 1 • MAIN • Q4/5

Evaluate.

$$(9 + 3) \times (10 - 4)$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 1 • MAIN • Q5/5

12

QUESTION ID

**S1-M-Q5**

TYPE

Short answer

CORRECT ANSWER

**72**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 1 • MAIN • Q5/5

SET 1 • 5.OA.A.1

# BACKWARD 1

Do the work in parentheses first • Grade 3 • Operations & Algebraic Thinking •  
3.OA.D.8

**True or false?**

In  $4 \times (2 + 5)$ , you add  $2 + 5$  first.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.D.8 • Set 1 • BACKWARD 1 • Q1/5

14

QUESTION ID

**S1-B1-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.D.8 • Set 1 • BACKWARD 1 • Q1/5

Evaluate.

$$(6 + 4) \times 2$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 1 • BACKWARD 1 • Q2/5

15

QUESTION ID

**S1-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**20**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 1 • BACKWARD 1 • Q2/5

Which step comes first in  $18 - (3 \times 4)$ ?

- A. Subtract  $18 - 4$
- B. Multiply  $3 \times 4$
- C. Subtract  $18 - 3$
- D. Add  $3 + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 1 • BACKWARD 1 • Q3/5

16

QUESTION ID

**S1-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. Multiply  $3 \times 4$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 1 • BACKWARD 1 • Q3/5



Evaluate.

$$20 - (5 + 5)$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 1 • BACKWARD 1 • Q4/5

17

QUESTION ID

**S1-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**10**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 1 • BACKWARD 1 • Q4/5

**True or false?**

$$(10 - 4) + 3 = 10 - 7$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 1 • BACKWARD 1 • Q5/5

18

QUESTION ID

**S1-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 1 • BACKWARD 1 • Q5/5

SET 1 • 5.OA.A.1

# BACKWARD 2

Multiplication and division facts • Grade 3 • Operations & Algebraic Thinking •  
3.OA.C.7

Multiply.

$$7 \times 8$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q1/5

20

QUESTION ID

**S1-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**56**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q1/5

Divide.

$$24 \div 6$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q2/5

21

QUESTION ID

**S1-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**4**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q2/5

**True or false?**

$$9 \times 7 = 56$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q3/5

22

QUESTION ID

**S1-B2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q3/5

$$48 \div 8 = ?$$

- A.** 8
- B.** 5
- C.** 7
- D.** 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q4/5

23

QUESTION ID

**S1-B2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. 6**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q4/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$3 \times 9$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q5/5

24

QUESTION ID

**S1-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**27**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q5/5



SET 1 • 5.OA.A.1

# BACKWARD 3

Adding and subtracting multi-digit numbers • Grade 4 • Number & Operations  
in Base Ten • 4.NBT.B.4

**ANSWER KEY**

Reduced copy of the question page, with its answer

Add.

$$348 + 275$$

Sum: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q1

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.4 • Set 1 • BACKWARD 3 • Q1/5

26

QUESTION ID

**S1-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**623**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q1

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.4 • Set 1 • BACKWARD 3 • Q1/5

26

**ANSWER KEY**

Reduced copy of the question page, with its answer

Subtract.

$$502 - 168$$

Difference: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q2

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.4 • Set 1 • BACKWARD 3 • Q2/5

27

QUESTION ID

**S1-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**334**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q2

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.4 • Set 1 • BACKWARD 3 • Q2/5

**True or false?**

$$1,250 - 375 = 985$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 1 • BACKWARD 3 • Q3/5

28

QUESTION ID

**S1-B3-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 1 • BACKWARD 3 • Q3/5

$$609 + 194 = ?$$

- A.** 793
- B.** 703
- C.** 813
- D.** 803

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q4

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.4 • Set 1 • BACKWARD 3 • Q4/5

29

QUESTION ID

**S1-B3-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. 803**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q4

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.4 • Set 1 • BACKWARD 3 • Q4/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Subtract.

$$1,000 - 432$$

Difference: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q5

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.4 • Set 1 • BACKWARD 3 • Q5/5

30

QUESTION ID

**S1-B3-Q5**

TYPE

Short answer

CORRECT ANSWER

**568**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q5

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.4 • Set 1 • BACKWARD 3 • Q5/5

SET 1 • 5.OA.A.1

# FORWARD 1

Evaluate expressions with variables • Grade 6 • Expressions & Equations •  
6.EE.A.2.c

Evaluate  $4(x + 3)$  when  $x = 5$ .

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 1 • FORWARD 1 • Q1/5

32

QUESTION ID

**S1-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**32**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 1 • FORWARD 1 • Q1/5



Evaluate  $20 - 2y$  when  $y = 6$ .

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 1 • FORWARD 1 • Q2/5

33

QUESTION ID

**S1-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**8**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 1 • FORWARD 1 • Q2/5

Evaluate  $(a + b) \div 3$  when  $a = 10$  and  $b = 8$ .

- A.** 4
- B.** 9
- C.** 54
- D.** 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 1 • FORWARD 1 • Q3/5

34

QUESTION ID

**S1-F1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. 6**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 1 • FORWARD 1 • Q3/5

Evaluate  $3m + 2$  when  $m = 7$ .

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 1 • FORWARD 1 • Q4/5

35

QUESTION ID

**S1-F1-Q4**

TYPE

Short answer

CORRECT ANSWER

**23**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 1 • FORWARD 1 • Q4/5

**True or false?**

When  $n = 4$ , the value of  $5(n - 1)$  is 19.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

Grade 6 • Expressions &amp; Equations • 6.EE.A.2.c • Set 1 • FORWARD 1 • Q5/5

36

QUESTION ID

**S1-F1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

Grade 6 • Expressions &amp; Equations • 6.EE.A.2.c • Set 1 • FORWARD 1 • Q5/5

SET 1 • 5.OA.A.1

# FORWARD 2

Evaluate expressions with negative numbers • Grade 7 • The Number System •  
7.NS.A.3

Find the value.

$$-3 \times (4 - 9)$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 1 • FORWARD 2 • Q1/5

38

QUESTION ID

**S1-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**15**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 1 • FORWARD 2 • Q1/5

Find the value.

$$(-12 + 4) \div 2$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 1 • FORWARD 2 • Q2/5

39

QUESTION ID

**S1-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**-4**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 1 • FORWARD 2 • Q2/5

Find the value.

$$6 - (2 - 10)$$

- A.** -2
- B.** 2
- C.** 14
- D.** 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 1 • FORWARD 2 • Q3/5

40

QUESTION ID

**S1-F2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. 14**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 1 • FORWARD 2 • Q3/5



Find the value.

$$(-5) \times (-2) + (-7)$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 1 • FORWARD 2 • Q4/5

41

QUESTION ID

**S1-F2-Q4**

TYPE

Short answer

CORRECT ANSWER

**3**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 1 • FORWARD 2 • Q4/5

**True or false?**

$$(-20) \div (3 + 2) = 4$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 1 • FORWARD 2 • Q5/5

42

QUESTION ID

**S1-F2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 1 • FORWARD 2 • Q5/5

# Evaluate expressions with brackets and braces

|                   |                                                  |                  |
|-------------------|--------------------------------------------------|------------------|
| <b>MAIN</b>       | Grade 5 main question                            | <b>5.OA.A.1</b>  |
| <b>BACKWARD 1</b> | Do the work in parentheses first                 | <b>3.OA.D.8</b>  |
| <b>BACKWARD 2</b> | Multiplication and division facts                | <b>3.OA.C.7</b>  |
| <b>BACKWARD 3</b> | Adding and subtracting within 100                | <b>2.NBT.B.5</b> |
| <b>FORWARD 1</b>  | Evaluate expressions with whole-number exponents | <b>6.EE.A.1</b>  |
| <b>FORWARD 2</b>  | Grouping symbols with negative numbers           | <b>7.NS.A.3</b>  |

SET 2 • 5.OA.A.1

# MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$2 \times [5 + (3 \times 4)]$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 2 • MAIN • Q1/5

45

QUESTION ID

**S2-M-Q1**

TYPE

Short answer

CORRECT ANSWER

**34**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 2 • MAIN • Q1/5

Evaluate.

$$[20 - (2 + 6)] \div 3$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 2 • MAIN • Q2/5

46

QUESTION ID

**S2-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**4**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 2 • MAIN • Q2/5

Evaluate.

$$4 \times \{3 + [10 - (2 \times 4)]\}$$

- A. 36
- B. 20
- C. 44
- D. 26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 2 • MAIN • Q3/5

47

QUESTION ID

**S2-M-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. 20**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 2 • MAIN • Q3/5

Evaluate.

$$60 \div [3 \times (7 - 5)]$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 2 • MAIN • Q4/5

48

QUESTION ID

**S2-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**10**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 2 • MAIN • Q4/5



Evaluate.

$$\{18 - [2 \times (4 + 1)]\} + 9$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 2 • MAIN • Q5/5

49

QUESTION ID

**S2-M-Q5**

TYPE

Short answer

CORRECT ANSWER

**17**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 2 • MAIN • Q5/5

SET 2 • 5.OA.A.1

# BACKWARD 1

Do the work in parentheses first • Grade 3 • Operations & Algebraic Thinking •  
3.OA.D.8

Evaluate.

$$(12 - 5) \times 3$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 2 • BACKWARD 1 • Q1/5

51

QUESTION ID

**S2-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**21**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 2 • BACKWARD 1 • Q1/5

**True or false?**

In  $(8 + 2) \div 5$ , you divide  $2 \div 5$  first.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.D.8 • Set 2 • BACKWARD 1 • Q2/5

52

QUESTION ID

**S2-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.D.8 • Set 2 • BACKWARD 1 • Q2/5

Evaluate.

$$4 \times (9 - 6)$$

- A.** 33
- B.** 10
- C.** 30
- D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 2 • BACKWARD 1 • Q3/5

53

QUESTION ID

**S2-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. 12**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 2 • BACKWARD 1 • Q3/5

Evaluate.

$$30 \div (2 + 4)$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 2 • BACKWARD 1 • Q4/5

54

QUESTION ID

**S2-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**5**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 2 • BACKWARD 1 • Q4/5

**True or false?**

$$(3 + 5) \times 2 = 16$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 2 • BACKWARD 1 • Q5/5

55

QUESTION ID

**S2-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • 5.OA.A.1

# BACKWARD 2

Multiplication and division facts • Grade 3 • Operations & Algebraic Thinking •  
3.OA.C.7



**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$8 \times 6$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.C.7 • Set 2 • BACKWARD 2 • Q1/5

57

QUESTION ID

**S2-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**48**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.C.7 • Set 2 • BACKWARD 2 • Q1/5

57

Divide.

$$56 \div 7$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 2 • BACKWARD 2 • Q2/5

58

QUESTION ID

**S2-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**8**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 2 • BACKWARD 2 • Q2/5

**True or false?**

$$36 \div 4 = 8$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 2 • BACKWARD 2 • Q3/5

59

QUESTION ID

**S2-B2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 2 • BACKWARD 2 • Q3/5

$$7 \times 9 = ?$$

- A.** 63
- B.** 56
- C.** 72
- D.** 49

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.C.7 • Set 2 • BACKWARD 2 • Q4/5

60

QUESTION ID

**S2-B2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. 63**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.C.7 • Set 2 • BACKWARD 2 • Q4/5

Divide.

$$81 \div 9$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 2 • BACKWARD 2 • Q5/5

61

QUESTION ID

**S2-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**9**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • 5.OA.A.1

# BACKWARD 3

Adding and subtracting within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Add.

$$46 + 37$$

Sum: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q1

Grade 2 • Number &amp; Operations in Base Ten • 2.NBT.B.5 • Set 2 • BACKWARD 3 • Q1/5

63

QUESTION ID

**S2-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**83**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q1

Grade 2 • Number &amp; Operations in Base Ten • 2.NBT.B.5 • Set 2 • BACKWARD 3 • Q1/5

63

**ANSWER KEY**

Reduced copy of the question page, with its answer

Subtract.

$$72 - 28$$

Difference: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q2

Grade 2 • Number &amp; Operations in Base Ten • 2.NBT.B.5 • Set 2 • BACKWARD 3 • Q2/5

64

QUESTION ID

**S2-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**44**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q2

Grade 2 • Number &amp; Operations in Base Ten • 2.NBT.B.5 • Set 2 • BACKWARD 3 • Q2/5



**True or false?**

$$55 + 29 = 74$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 2 • BACKWARD 3 • Q3/5

65

QUESTION ID

**S2-B3-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 2 • BACKWARD 3 • Q3/5

$$90 - 47 = ?$$

- A.** 47
- B.** 43
- C.** 53
- D.** 33

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q4

Grade 2 • Number &amp; Operations in Base Ten • 2.NBT.B.5 • Set 2 • BACKWARD 3 • Q4/5

66

QUESTION ID

**S2-B3-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. 43**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q4

Grade 2 • Number &amp; Operations in Base Ten • 2.NBT.B.5 • Set 2 • BACKWARD 3 • Q4/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Add.

$$38 + 54$$

Sum: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q5

Grade 2 • Number &amp; Operations in Base Ten • 2.NBT.B.5 • Set 2 • BACKWARD 3 • Q5/5

67

QUESTION ID

**S2-B3-Q5**

TYPE

Short answer

CORRECT ANSWER

**92**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q5

Grade 2 • Number &amp; Operations in Base Ten • 2.NBT.B.5 • Set 2 • BACKWARD 3 • Q5/5

67

SET 2 • 5.OA.A.1

# FORWARD 1

Evaluate expressions with whole-number exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Evaluate.

$$2 \times (3 + 1)^2$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 2 • FORWARD 1 • Q1/5

69

QUESTION ID

**S2-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**32**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 2 • FORWARD 1 • Q1/5

Evaluate.

$$[10 - 2^3] \times 5$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 2 • FORWARD 1 • Q2/5

70

QUESTION ID

**S2-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**10**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 2 • FORWARD 1 • Q2/5

Evaluate.

$$4^2 - (6 + 2)$$

- A. 8
- B. 2
- C. 0
- D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 2 • FORWARD 1 • Q3/5

71

QUESTION ID

**S2-F1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. 8**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 2 • FORWARD 1 • Q3/5

Evaluate.

$$(5^2 - 1) \div 6$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 2 • FORWARD 1 • Q4/5

72

QUESTION ID

**S2-F1-Q4**

TYPE

Short answer

CORRECT ANSWER

**4**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 2 • FORWARD 1 • Q4/5



**True or false?**

$$3 \times 2^2 = 36$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 2 • FORWARD 1 • Q5/5

73

QUESTION ID

**S2-F1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 2 • FORWARD 1 • Q5/5

SET 2 • 5.OA.A.1

# FORWARD 2

Grouping symbols with negative numbers • Grade 7 • The Number System •  
7.NS.A.3

Find the value.

$$2 \times [-3 + (8 \div (-4))]$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 2 • FORWARD 2 • Q1/5

75

QUESTION ID

**S2-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**-10**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 2 • FORWARD 2 • Q1/5

Find the value.

$$[(-6) + 15] \div (-3)$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 2 • FORWARD 2 • Q2/5

76

QUESTION ID

**S2-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**-3**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 2 • FORWARD 2 • Q2/5

Find the value.

$$-4 \times [5 - (-1)]$$

- A.** 16
- B.** -16
- C.** -24
- D.** 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 2 • FORWARD 2 • Q3/5

77

QUESTION ID

**S2-F2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. -24**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 2 • FORWARD 2 • Q3/5

Find the value.

$$\{10 - [2 \times (-3)]\} \div 4$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 2 • FORWARD 2 • Q4/5

78

QUESTION ID

**S2-F2-Q4**

TYPE

Short answer

CORRECT ANSWER

**4**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 2 • FORWARD 2 • Q4/5

**True or false?**

$$[-8 - (-2)] \times 3 = -30$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 2 • FORWARD 2 • Q5/5

79

QUESTION ID

**S2-F2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 2 • FORWARD 2 • Q5/5

# Place grouping symbols to make an equation true

|                   |                                                          |                 |
|-------------------|----------------------------------------------------------|-----------------|
| <b>MAIN</b>       | Grade 5 main question                                    | <b>5.OA.A.1</b> |
| <b>BACKWARD 1</b> | Do the work in parentheses first                         | <b>3.OA.D.8</b> |
| <b>BACKWARD 2</b> | When regrouping keeps the same value                     | <b>3.OA.B.5</b> |
| <b>BACKWARD 3</b> | Multiplication and division facts                        | <b>3.OA.C.7</b> |
| <b>FORWARD 1</b>  | Use parentheses to write equivalent expressions          | <b>6.EE.A.3</b> |
| <b>FORWARD 2</b>  | Expand and factor expressions with rational coefficients | <b>7.EE.A.1</b> |



SET 3 • 5.OA.A.1

# MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Insert parentheses to make the equation true.

$$4 + 6 \times 3 = 30$$

Equation: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 3 • MAIN • Q1/5

82

QUESTION ID

**S3-M-Q1**

TYPE

Short answer

CORRECT ANSWER

$$(4 + 6) \times 3 = 30$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 3 • MAIN • Q1/5

Insert parentheses to make the equation true.

$$20 - 8 - 2 = 14$$

Equation: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 3 • MAIN • Q2/5

83

QUESTION ID

**S3-M-Q2**

TYPE

Short answer

CORRECT ANSWER

$$20 - (8 - 2) = 14$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 3 • MAIN • Q2/5

Which placement of parentheses makes the equation true?

$$2 \times 5 + 3 \times 4 = 52$$

**A.**  $2 \times (5 + 3) \times 4 = 52$       **B.**  $(2 \times 5 + 3) \times 4 = 52$

**C.**  $2 \times (5 + 3 \times 4) = 52$       **D.**  $2 \times 5 + (3 \times 4) = 52$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 3 • MAIN • Q3/5

84

QUESTION ID

**S3-M-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B.**

**$(2 \times 5 + 3) \times 4 = 52$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 3 • MAIN • Q3/5

Insert parentheses to make the equation true.

$$36 \div 6 - 2 = 9$$

Equation: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 3 • MAIN • Q4/5

85

QUESTION ID

**S3-M-Q4**

TYPE

Short answer

CORRECT ANSWER

$$36 \div (6 - 2) = 9$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 3 • MAIN • Q4/5

Insert brackets and parentheses to make the equation true.

$$2 \times 9 - 3 + 1 = 10$$

Equation: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 3 • MAIN • Q5/5

86

QUESTION ID

**S3-M-Q5**

TYPE

Short answer

CORRECT ANSWER

$$2 \times [9 - (3 + 1)] = 10$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 3 • MAIN • Q5/5

SET 3 • 5.OA.A.1

# BACKWARD 1

Do the work in parentheses first • Grade 3 • Operations & Algebraic Thinking •  
3.OA.D.8

Evaluate.

$$(4 + 6) \times 3$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 3 • BACKWARD 1 • Q1/5

88

QUESTION ID

**S3-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**30**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 3 • BACKWARD 1 • Q1/5



Evaluate.

$$4 + (6 \times 3)$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 3 • BACKWARD 1 • Q2/5

89

QUESTION ID

**S3-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**22**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 3 • BACKWARD 1 • Q2/5

**True or false?**

$$(20 - 8) - 2 = 20 - (8 - 2)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 3 • BACKWARD 1 • Q3/5

90

QUESTION ID

**S3-B1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 3 • BACKWARD 1 • Q3/5

Evaluate.

$$36 \div (6 - 2)$$

- A.** 4
- B.** 34
- C.** 9
- D.** 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 3 • BACKWARD 1 • Q4/5

91

QUESTION ID

**S3-B1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. 9**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 3 • BACKWARD 1 • Q4/5

Evaluate.

$$(36 \div 6) - 2$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 3 • BACKWARD 1 • Q5/5

92

QUESTION ID

**S3-B1-Q5**

TYPE

Short answer

CORRECT ANSWER

**4**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • 5.OA.A.1

# BACKWARD 2

When regrouping keeps the same value • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

**True or false?**

$$(2 \times 3) \times 4 = 2 \times (3 \times 4)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q1/5

94

QUESTION ID

**S3-B2-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q1/5

**True or false?**

$$(12 \div 6) \div 2 = 12 \div (6 \div 2)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q2/5

95

QUESTION ID

**S3-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q2/5

Which equation is true?

- A.  $(5 + 2) + 3 = 5 + (2 + 3)$
- B.  $(9 - 4) - 1 = 9 - (4 - 1)$
- C.  $(16 \div 4) \div 2 = 16 \div (4 \div 2)$
- D.  $(7 - 3) - 2 = 7 - (3 - 2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q3/5

96

QUESTION ID

**S3-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A.**

**$(5 + 2) + 3 = 5 + (2 + 3)$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q3/5



**True or false?**

$$4 \times (5 \times 2) = (4 \times 5) \times 2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q4/5

97

QUESTION ID

**S3-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q4/5

Fill in the blank.

$$(6 \times 2) \times 5 = 6 \times (2 \times \underline{\quad})$$

Blank: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q5/5

98

QUESTION ID

**S3-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**5**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q5/5

SET 3 • 5.OA.A.1

# BACKWARD 3

Multiplication and division facts • Grade 3 • Operations & Algebraic Thinking •  
3.OA.C.7

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$9 \times 4$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q1

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.C.7 • Set 3 • BACKWARD 3 • Q1/5

100

QUESTION ID

**S3-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**36**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q1

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.C.7 • Set 3 • BACKWARD 3 • Q1/5

Divide.

$$42 \div 6$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 3 • BACKWARD 3 • Q2/5

101

QUESTION ID

**S3-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**7**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 3 • BACKWARD 3 • Q2/5

**True or false?**

$$8 \times 7 = 54$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 3 • BACKWARD 3 • Q3/5

102

QUESTION ID

**S3-B3-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 3 • BACKWARD 3 • Q3/5

$$64 \div 8 = ?$$

- A.** 6
- B.** 7
- C.** 8
- D.** 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q4

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.C.7 • Set 3 • BACKWARD 3 • Q4/5

103

QUESTION ID

**S3-B3-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. 8**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q4

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.C.7 • Set 3 • BACKWARD 3 • Q4/5

Multiply.

$$6 \times 7$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 3 • BACKWARD 3 • Q5/5

104

QUESTION ID

**S3-B3-Q5**

TYPE

Short answer

CORRECT ANSWER

**42**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 3 • BACKWARD 3 • Q5/5



SET 3 • 5.OA.A.1

# FORWARD 1

Use parentheses to write equivalent expressions • Grade 6 • Expressions & Equations • 6.EE.A.3

Expand.

$$3(x + 5)$$

Expression: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 3 • FORWARD 1 • Q1/5

106

QUESTION ID

**S3-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**$3x + 15$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 3 • FORWARD 1 • Q1/5

Expand.

$$4(2y - 1)$$

Expression: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 3 • FORWARD 1 • Q2/5

107

QUESTION ID

**S3-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**$8y - 4$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 3 • FORWARD 1 • Q2/5

Which expression is equivalent to  $6(a + 2)$ ?

- A.  $6a + 2$
- B.  $a + 12$
- C.  $6a + 12$
- D.  $8a$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 3 • FORWARD 1 • Q3/5

108

QUESTION ID

**S3-F1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C.  $6a + 12$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 3 • FORWARD 1 • Q3/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Factor.

$$5m + 20$$

Expression: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q4

Grade 6 • Expressions &amp; Equations • 6.EE.A.3 • Set 3 • FORWARD 1 • Q4/5

109

QUESTION ID

**S3-F1-Q4**

TYPE

Short answer

CORRECT ANSWER

**5(m + 4)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q4

Grade 6 • Expressions &amp; Equations • 6.EE.A.3 • Set 3 • FORWARD 1 • Q4/5

109

**True or false?**

$$2(x + 7) = 2x + 7$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 3 • FORWARD 1 • Q5/5

110

QUESTION ID

**S3-F1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 3 • FORWARD 1 • Q5/5

SET 3 • 5.OA.A.1

# FORWARD 2

Expand and factor expressions with rational coefficients • Grade 7 •  
Expressions & Equations • 7.EE.A.1

**ANSWER KEY**

Reduced copy of the question page, with its answer

Expand.

$$-2(3x - 4)$$

Expression: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q1

Grade 7 • Expressions &amp; Equations • 7.EE.A.1 • Set 3 • FORWARD 2 • Q1/5

112

QUESTION ID

**S3-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**$$-6x + 8$$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q1

Grade 7 • Expressions &amp; Equations • 7.EE.A.1 • Set 3 • FORWARD 2 • Q1/5

112



Expand.

$$\frac{1}{2}(6y + 10)$$

Expression: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 3 • FORWARD 2 • Q2/5

113

QUESTION ID

**S3-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**$3y + 5$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 3 • FORWARD 2 • Q2/5

Which expression is equivalent to  $-5(n + 2)$ ?

- A.  $-5n + 2$
- B.  $-5n - 10$
- C.  $-5n + 10$
- D.  $5n - 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 3 • FORWARD 2 • Q3/5

114

QUESTION ID

**S3-F2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B.  $-5n - 10$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 3 • FORWARD 2 • Q3/5

Factor out 3.

$$3x - 12$$

Expression: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 3 • FORWARD 2 • Q4/5

115

QUESTION ID

**S3-F2-Q4**

TYPE

Short answer

CORRECT ANSWER

**$3(x - 4)$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 3 • FORWARD 2 • Q4/5

**True or false?**

$$0.5(4a - 8) = 2a - 8$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 3 • FORWARD 2 • Q5/5

116

QUESTION ID

**S3-F2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 3 • FORWARD 2 • Q5/5

# Write numerical expressions from words

|                   |                                      |                   |
|-------------------|--------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 5 main question                | <b>5.OA.A.2</b>   |
| <b>BACKWARD 1</b> | Products as equal groups             | <b>3.OA.A.1</b>   |
| <b>BACKWARD 2</b> | Quotients as equal shares            | <b>3.OA.A.2</b>   |
| <b>BACKWARD 3</b> | Sums and differences                 | <b>2.OA.A.1</b>   |
| <b>FORWARD 1</b>  | Write expressions with variables     | <b>6.EE.A.2.a</b> |
| <b>FORWARD 2</b>  | Write and solve equations from words | <b>7.EE.B.4.a</b> |

SET 4 • 5.OA.A.2

# MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

Write an expression for "add 9 and 6, then multiply by 4."

Expression: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 4 • MAIN • Q1/5

119

QUESTION ID

**S4-M-Q1**

TYPE

Short answer

CORRECT ANSWER

**$(9 + 6) \times 4$**

**GRADING NOTE**  $4 \times (9 + 6)$  is also correct. Parentheses are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 4 • MAIN • Q1/5

Write an expression for "subtract 5 from 30, then divide by 5."

Expression: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 4 • MAIN • Q2/5

120

QUESTION ID

**S4-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**$(30 - 5) \div 5$**

**GRADING NOTE** Parentheses are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 4 • MAIN • Q2/5



Which expression shows "triple the sum of 12 and 8"?

- A.  $12 + 8 \times 3$
- B.  $3 \times 12 + 8$
- C.  $3 \times (12 + 8)$
- D.  $(3 + 12) \times 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 4 • MAIN • Q3/5

121

QUESTION ID

**S4-M-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C.  $3 \times (12 + 8)$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 4 • MAIN • Q3/5

Write an expression for "divide 40 by 8, then add 7."

Expression: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 4 • MAIN • Q4/5

122

QUESTION ID

**S4-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**$(40 \div 8) + 7$**

**GRADING NOTE**  $40 \div 8 + 7$  is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 4 • MAIN • Q4/5

Write an expression for "the product of 6 and the difference of 15 and 9."

Expression: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 4 • MAIN • Q5/5

123

QUESTION ID

**S4-M-Q5**

TYPE

Short answer

CORRECT ANSWER

**$6 \times (15 - 9)$**

**GRADING NOTE**  $(15 - 9) \times 6$  is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 4 • MAIN • Q5/5

SET 4 • 5.OA.A.2

# BACKWARD 1

Products as equal groups • Grade 3 • Operations & Algebraic Thinking •  
3.OA.A.1

**True or false?**

$5 \times 7$  can mean 5 groups of 7.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q1

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.A.1 • Set 4 • BACKWARD 1 • Q1/5

125

QUESTION ID

**S4-B1-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q1

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.A.1 • Set 4 • BACKWARD 1 • Q1/5

Which expression shows 4 groups of 6?

- A. 46
- B.  $6 - 4$
- C.  $4 + 6$
- D.  $4 \times 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 4 • BACKWARD 1 • Q2/5

126

QUESTION ID

**S4-B1-Q2**

TYPE

Multiple choice

CORRECT ANSWER

**D.  $4 \times 6$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 4 • BACKWARD 1 • Q2/5

Write a multiplication expression for 8 groups of 3.

Expression: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 4 • BACKWARD 1 • Q3/5

127

QUESTION ID

**S4-B1-Q3**

TYPE

Short answer

CORRECT ANSWER

**8 × 3**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 4 • BACKWARD 1 • Q3/5

**True or false?**

$$3 + 3 + 3 + 3 = 3 \times 3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 4 • BACKWARD 1 • Q4/5

128

QUESTION ID

**S4-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 4 • BACKWARD 1 • Q4/5



What is the product of 7 and 5?

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 4 • BACKWARD 1 • Q5/5

129

QUESTION ID

**S4-B1-Q5**

TYPE

Short answer

CORRECT ANSWER

**35**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 4 • BACKWARD 1 • Q5/5

SET 4 • 5.OA.A.2

# BACKWARD 2

Quotients as equal shares • Grade 3 • Operations & Algebraic Thinking •  
3.OA.A.2

**True or false?**

15 ÷ 3 can mean 15 shared equally into 3 groups.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.A.2 • Set 4 • BACKWARD 2 • Q1/5

131

QUESTION ID

**S4-B2-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.A.2 • Set 4 • BACKWARD 2 • Q1/5

Which expression shows 20 apples shared equally among 4 bags?

- A.  $20 \times 4$
- B.  $4 \div 20$
- C.  $20 \div 4$
- D.  $20 - 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.A.2 • Set 4 • BACKWARD 2 • Q2/5

132

QUESTION ID

**S4-B2-Q2**

TYPE

Multiple choice

CORRECT ANSWER

**C.  $20 \div 4$** 

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.A.2 • Set 4 • BACKWARD 2 • Q2/5

What is the quotient of 18 and 6?

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 4 • BACKWARD 2 • Q3/5

133

QUESTION ID

**S4-B2-Q3**

TYPE

Short answer

CORRECT ANSWER

**3**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 4 • BACKWARD 2 • Q3/5

Write a division expression for 32 split into groups of 8.

Expression: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 4 • BACKWARD 2 • Q4/5

134

QUESTION ID

**S4-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**32 ÷ 8**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 4 • BACKWARD 2 • Q4/5

**True or false?**

The quotient of 12 and 2 is 24.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.A.2 • Set 4 • BACKWARD 2 • Q5/5

135

QUESTION ID

**S4-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.A.2 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • 5.OA.A.2

# BACKWARD 3

Sums and differences • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1



What is the sum of 14 and 9?

Sum: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 4 • BACKWARD 3 • Q1/5

137

QUESTION ID

**S4-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**23**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 4 • BACKWARD 3 • Q1/5

What is the difference of 20 and 7?

Difference: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 4 • BACKWARD 3 • Q2/5

138

QUESTION ID

**S4-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**13**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 4 • BACKWARD 3 • Q2/5

**True or false?**

The difference of 15 and 6 is 21.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q3

Grade 2 • Operations &amp; Algebraic Thinking • 2.OA.A.1 • Set 4 • BACKWARD 3 • Q3/5

139

QUESTION ID

**S4-B3-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q3

Grade 2 • Operations &amp; Algebraic Thinking • 2.OA.A.1 • Set 4 • BACKWARD 3 • Q3/5

Which word names the answer to an addition problem?

- A.** Difference
- B.** Quotient
- C.** Product
- D.** Sum

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 4 • BACKWARD 3 • Q4/5

140

QUESTION ID

**S4-B3-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. Sum**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 4 • BACKWARD 3 • Q4/5

Write an expression for "7 more than 25."

Expression: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 4 • BACKWARD 3 • Q5/5

141

QUESTION ID

**S4-B3-Q5**

TYPE

Short answer

CORRECT ANSWER

**25 + 7**

**GRADING NOTE** 7 + 25 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 4 • BACKWARD 3 • Q5/5

SET 4 • 5.OA.A.2

# FORWARD 1

Write expressions with variables • Grade 6 • Expressions & Equations •  
6.EE.A.2.a

Write an expression for "the sum of  $n$  and 8, times 3."

Expression: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 4 • FORWARD 1 • Q1/5

143

QUESTION ID

**S4-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**$3(n + 8)$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 4 • FORWARD 1 • Q1/5

Write an expression for "y divided by 5, then minus 2."

Expression: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 4 • FORWARD 1 • Q2/5

144

QUESTION ID

**S4-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**$y \div 5 - 2$**

**GRADING NOTE**  $y/5 - 2$  is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 4 • FORWARD 1 • Q2/5



Which expression means "4 less than the product of 2 and  $x$ "?

- A.  $4 - 2x$
- B.  $2x - 4$
- C.  $2(x - 4)$
- D.  $2 + x - 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 4 • FORWARD 1 • Q3/5

145

QUESTION ID

**S4-F1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B.  $2x - 4$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 4 • FORWARD 1 • Q3/5

Write an expression for "twice the difference of  $m$  and 6."

Expression: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 4 • FORWARD 1 • Q4/5

146

QUESTION ID

**S4-F1-Q4**

TYPE

Short answer

CORRECT ANSWER

**$2(m - 6)$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 4 • FORWARD 1 • Q4/5

**True or false?**

"9 more than  $k$ " can be written as  $k + 9$ .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q5

Grade 6 • Expressions &amp; Equations • 6.EE.A.2.a • Set 4 • FORWARD 1 • Q5/5

147

QUESTION ID

**S4-F1-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q5

Grade 6 • Expressions &amp; Equations • 6.EE.A.2.a • Set 4 • FORWARD 1 • Q5/5

SET 4 • 5.OA.A.2

# FORWARD 2

Write and solve equations from words • Grade 7 • Expressions & Equations •  
7.EE.B.4.a

Three times the sum of a number and 4 is 27.  
Write and solve an equation to find the number.

Equation and solution: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 4 • FORWARD 2 • Q1/5

149

QUESTION ID

**S4-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**$3(n + 4) = 27; n = 5$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 4 • FORWARD 2 • Q1/5

Half of a number, minus 6, is 10.

Write and solve an equation to find the number.

Equation and solution: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 4 • FORWARD 2 • Q2/5

150

QUESTION ID

**S4-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

$$(1/2)n - 6 = 10;$$
$$n = 32$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 4 • FORWARD 2 • Q2/5

Which equation represents "2 less than 5 times a number is 33"?

- A.  $5n + 2 = 33$
- B.  $5n - 2 = 33$
- C.  $2 - 5n = 33$
- D.  $5(n - 2) = 33$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 4 • FORWARD 2 • Q3/5

151

QUESTION ID

**S4-F2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B.  $5n - 2 = 33$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 4 • FORWARD 2 • Q3/5

A number divided by 4, plus 3, is 9.

Write and solve an equation to find the number.

Equation and solution: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 4 • FORWARD 2 • Q4/5

152

QUESTION ID

**S4-F2-Q4**

TYPE

Short answer

CORRECT ANSWER

$$n \div 4 + 3 = 9; n = 24$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 4 • FORWARD 2 • Q4/5



**True or false?**

The solution of  $2(x + 3) = 18$  is  $x = 7$ .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q5

Grade 7 • Expressions &amp; Equations • 7.EE.B.4.a • Set 4 • FORWARD 2 • Q5/5

153

QUESTION ID

**S4-F2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q5

Grade 7 • Expressions &amp; Equations • 7.EE.B.4.a • Set 4 • FORWARD 2 • Q5/5

# Interpret numerical expressions without evaluating them

|                   |                                                        |                   |
|-------------------|--------------------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 5 main question                                  | <b>5.OA.A.2</b>   |
| <b>BACKWARD 1</b> | Multiplicative comparison                              | <b>4.OA.A.1</b>   |
| <b>BACKWARD 2</b> | Groups of a sum                                        | <b>3.OA.A.1</b>   |
| <b>BACKWARD 3</b> | Equal shares                                           | <b>3.OA.A.2</b>   |
| <b>FORWARD 1</b>  | Identify parts of an expression                        | <b>6.EE.A.2.b</b> |
| <b>FORWARD 2</b>  | Rewrite expressions to show how quantities are related | <b>7.EE.A.2</b>   |

SET 5 • 5.OA.A.2

# MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

How does  $4 \times (563 + 218)$  compare to  $563 + 218$ ?

- A. It is 4 more.
- B. It is 4 times as large.
- C. It is 4 less.
- D. It is one-fourth as large.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 5 • MAIN • Q1/5

156

QUESTION ID

**S5-M-Q1**

TYPE

Multiple choice

CORRECT ANSWER

**B. It is 4 times as large.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 5 • MAIN • Q1/5

**True or false?**

$(7,215 - 1,005) \div 5$  is one-fifth as large as  $7,215 - 1,005$ .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

Grade 5 • Operations &amp; Algebraic Thinking • 5.OA.A.2 • Set 5 • MAIN • Q2/5

157

QUESTION ID

**S5-M-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

Grade 5 • Operations &amp; Algebraic Thinking • 5.OA.A.2 • Set 5 • MAIN • Q2/5

Without calculating, how many times as large is  
 $9 \times (48 + 27)$  as  $48 + 27$ ?

Times as large: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 5 • MAIN • Q3/5

158

QUESTION ID

**S5-M-Q3**

TYPE

Short answer

CORRECT ANSWER

**9**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 5 • MAIN • Q3/5

Which expression is twice as large as  $356 \times 12$ ?

- A.  $(356 + 2) \times 12$
- B.  $356 \times 12 + 2$
- C.  $356 \times 14$
- D.  $2 \times (356 \times 12)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 5 • MAIN • Q4/5

159

QUESTION ID

**S5-M-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D.  $2 \times (356 \times 12)$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 5 • MAIN • Q4/5

**True or false?**

$6 \times 25 + 4$  is 6 times as large as  $25 + 4$ .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q5

Grade 5 • Operations &amp; Algebraic Thinking • 5.OA.A.2 • Set 5 • MAIN • Q5/5

160

QUESTION ID

**S5-M-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q5

Grade 5 • Operations &amp; Algebraic Thinking • 5.OA.A.2 • Set 5 • MAIN • Q5/5



SET 5 • 5.OA.A.2

# BACKWARD 1

Multiplicative comparison • Grade 4 • Operations & Algebraic Thinking •  
4.OA.A.1

**True or false?**

35 is 5 times as many as 7.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

Grade 4 • Operations &amp; Algebraic Thinking • 4.OA.A.1 • Set 5 • BACKWARD 1 • Q1/5

162

QUESTION ID

**S5-B1-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

Grade 4 • Operations &amp; Algebraic Thinking • 4.OA.A.1 • Set 5 • BACKWARD 1 • Q1/5

Fill in the blank.

48 is \_\_\_\_ times as many as 6.

Blank: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 5 • BACKWARD 1 • Q2/5

163

QUESTION ID

**S5-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**8**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 5 • BACKWARD 1 • Q2/5

Which equation shows "36 is 4 times as many as 9"?

- A.  $9 = 4 \times 36$
- B.  $36 = 4 + 9$
- C.  $36 - 9 = 4$
- D.  $36 = 4 \times 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 5 • BACKWARD 1 • Q3/5

164

QUESTION ID

**S5-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D.  $36 = 4 \times 9$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 5 • BACKWARD 1 • Q3/5

**True or false?**

30 is 3 times as many as 27.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

Grade 4 • Operations &amp; Algebraic Thinking • 4.OA.A.1 • Set 5 • BACKWARD 1 • Q4/5

165

QUESTION ID

**S5-B1-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

Grade 4 • Operations &amp; Algebraic Thinking • 4.OA.A.1 • Set 5 • BACKWARD 1 • Q4/5

Tom has 6 cards. Ana has 3 times as many cards.

How many cards does Ana have?

Cards: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 5 • BACKWARD 1 • Q5/5

166

QUESTION ID

**S5-B1-Q5**

TYPE

Short answer

CORRECT ANSWER

**18**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • 5.OA.A.2

# BACKWARD 2

Groups of a sum • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

**True or false?**

$3 \times (4 + 1)$  means 3 groups of  $(4 + 1)$ .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.A.1 • Set 5 • BACKWARD 2 • Q1/5

168

QUESTION ID

**S5-B2-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.A.1 • Set 5 • BACKWARD 2 • Q1/5



Which expression means 6 groups of  $(10 + 2)$ ?

- A.  $6 \times (10 + 2)$
- B.  $(6 + 10) \times 2$
- C.  $6 \times 10 + 2$
- D.  $6 + (10 + 2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 5 • BACKWARD 2 • Q2/5

169

QUESTION ID

**S5-B2-Q2**

TYPE

Multiple choice

CORRECT ANSWER

**A.  $6 \times (10 + 2)$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 5 • BACKWARD 2 • Q2/5

How many groups of  $(5 + 4)$  are in  $7 \times (5 + 4)$ ?

Groups: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 5 • BACKWARD 2 • Q3/5

170

QUESTION ID

**S5-B2-Q3**

TYPE

Short answer

CORRECT ANSWER

**7**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 5 • BACKWARD 2 • Q3/5

**True or false?**

$$2 \times (8 + 1) = (8 + 1) + 2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 5 • BACKWARD 2 • Q4/5

171

QUESTION ID

**S5-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 5 • BACKWARD 2 • Q4/5

Write 4 groups of 9 as a multiplication expression.

Expression: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 5 • BACKWARD 2 • Q5/5

172

QUESTION ID

**S5-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**4 × 9**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 5 • BACKWARD 2 • Q5/5

SET 5 • 5.OA.A.2

# BACKWARD 3

Equal shares • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

**True or false?**

12 shared equally among 3 people is 4 each.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q1

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.A.2 • Set 5 • BACKWARD 3 • Q1/5

174

QUESTION ID

**S5-B3-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q1

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.A.2 • Set 5 • BACKWARD 3 • Q1/5

A 20-inch ribbon is cut into 5 equal pieces.

How long is each piece?

Length: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 5 • BACKWARD 3 • Q2/5

175

QUESTION ID

**S5-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**4 inches**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 5 • BACKWARD 3 • Q2/5

Which expression finds one of 4 equal shares of 28?

- A.  $28 - 4$
- B.  $28 \div 4$
- C.  $28 \times 4$
- D.  $4 \div 28$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 5 • BACKWARD 3 • Q3/5

176

QUESTION ID

**S5-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B.  $28 \div 4$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 5 • BACKWARD 3 • Q3/5



**True or false?**

Dividing a number by 2 splits it into 2 equal shares.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q4

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.A.2 • Set 5 • BACKWARD 3 • Q4/5

177

QUESTION ID

**S5-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q4

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.A.2 • Set 5 • BACKWARD 3 • Q4/5

30 crayons are shared equally among 6 children.

How many crayons does each child get?

Crayons: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 5 • BACKWARD 3 • Q5/5

178

QUESTION ID

**S5-B3-Q5**

TYPE

Short answer

CORRECT ANSWER

**5**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 5 • BACKWARD 3 • Q5/5

SET 5 • 5.OA.A.2

# FORWARD 1

Identify parts of an expression • Grade 6 • Expressions & Equations •  
6.EE.A.2.b

In  $5(x + 2)$ , what is 5?

- A.** A term of  $x + 2$
- B.** A quotient
- C.** A sum
- D.** A factor

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 5 • FORWARD 1 • Q1/5

180

QUESTION ID

**S5-F1-Q1**

TYPE

Multiple choice

CORRECT ANSWER

**D. A factor**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 5 • FORWARD 1 • Q1/5

How many terms are in  $4a + 7b - 1$ ?

Terms: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 5 • FORWARD 1 • Q2/5

181

QUESTION ID

**S5-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**3**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 5 • FORWARD 1 • Q2/5

What is the coefficient of  $m$  in  $9m + 4$ ?

Coefficient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 5 • FORWARD 1 • Q3/5

182

QUESTION ID

**S5-F1-Q3**

TYPE

Short answer

CORRECT ANSWER

**9**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 5 • FORWARD 1 • Q3/5

**True or false?**

In  $3(y + 6)$ , the quantity  $(y + 6)$  is a sum of two terms.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q4

Grade 6 • Expressions &amp; Equations • 6.EE.A.2.b • Set 5 • FORWARD 1 • Q4/5

183

QUESTION ID

**S5-F1-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q4

Grade 6 • Expressions &amp; Equations • 6.EE.A.2.b • Set 5 • FORWARD 1 • Q4/5

What is the expression  $12 \div (n + 1)$  called?

- A.** A coefficient
- B.** A quotient
- C.** A sum
- D.** A product

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 5 • FORWARD 1 • Q5/5

184

QUESTION ID

**S5-F1-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**B. A quotient**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 5 • FORWARD 1 • Q5/5



SET 5 • 5.OA.A.2

# FORWARD 2

Rewrite expressions to show how quantities are related • Grade 7 •  
Expressions & Equations • 7.EE.A.2

A price  $p$  is increased by 5%.

Which expression shows the new price?

- A.  $0.05p$
- B.  $1.05p$
- C.  $5p$
- D.  $p + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q1

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 5 • FORWARD 2 • Q1/5

186

QUESTION ID

**S5-F2-Q1**

TYPE

Multiple choice

CORRECT ANSWER

**B.  $1.05p$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q1

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 5 • FORWARD 2 • Q1/5

**True or false?**

$$x + 0.3x = 0.4x$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 5 • FORWARD 2 • Q2/5

187

QUESTION ID

**S5-F2-Q2**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 5 • FORWARD 2 • Q2/5

Write  $n - 0.4n$  as one term.

Expression: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q3

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 5 • FORWARD 2 • Q3/5

188

QUESTION ID

**S5-F2-Q3**

TYPE

Short answer

CORRECT ANSWER

**0.6n**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q3

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 5 • FORWARD 2 • Q3/5

Which expression is equivalent to  $2(3t)$ ?

- A.  $5t$
- B.  $2t + 3t$
- C.  $6t$
- D.  $3t + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 5 • FORWARD 2 • Q4/5

189

QUESTION ID

**S5-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C.  $6t$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 5 • FORWARD 2 • Q4/5

A sale price is  $0.75p$ , where  $p$  is the original price.

What percent of the original price is taken off?

Percent: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q5

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 5 • FORWARD 2 • Q5/5

190

QUESTION ID

**S5-F2-Q5**

TYPE

Short answer

CORRECT ANSWER

**25%**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q5

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 5 • FORWARD 2 • Q5/5

# Generate two numerical patterns and compare their terms

|                   |                                                        |                   |
|-------------------|--------------------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 5 main question                                  | <b>5.OA.B.3</b>   |
| <b>BACKWARD 1</b> | Patterns that follow a rule                            | <b>4.OA.C.5</b>   |
| <b>BACKWARD 2</b> | Multiplication facts                                   | <b>3.OA.C.7</b>   |
| <b>FORWARD 1</b>  | Equations relating dependent and independent variables | <b>6.EE.C.9</b>   |
| <b>FORWARD 2</b>  | Equations for proportional relationships               | <b>7.RP.A.2.c</b> |

SET 6 • 5.OA.B.3

# MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3



Pattern A starts at 0 and adds 3.

Pattern B starts at 0 and adds 6.

Write the first 5 terms of each pattern.

Pattern A: \_\_\_\_\_

Pattern B: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 6 • MAIN • Q1/5

193

QUESTION ID

**S6-M-Q1**

TYPE

Short answer

CORRECT ANSWER

**Pattern A: 0, 3, 6, 9, 12; Pattern B: 0, 6, 12, 18, 24**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 6 • MAIN • Q1/5

Pattern A: 0, 4, 8, 12, 16

Pattern B: 0, 12, 24, 36, 48

How does each term in Pattern B compare to the matching term in Pattern A?

- A.** 8 more                      **B.** 12 more  
**C.** 4 times as large        **D.** 3 times as large

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 6 • MAIN • Q2/5

194

QUESTION ID

**S6-M-Q2**

TYPE

Multiple choice

CORRECT ANSWER

**D. 3 times as large**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 6 • MAIN • Q2/5

Pattern A starts at 0 and adds 5. Pattern B starts at 0 and adds 10.

What is the 6th term of each pattern?

Pattern A: \_\_\_\_\_

Pattern B: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 6 • MAIN • Q3/5

195

QUESTION ID

**S6-M-Q3**

TYPE

Short answer

CORRECT ANSWER

**Pattern A: 25;  
Pattern B: 50**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 6 • MAIN • Q3/5

**True or false?**

Pattern A starts at 0 and adds 2. Pattern B starts at 0 and adds 8. Each term in Pattern B is 4 times the matching term in Pattern A.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

Grade 5 • Operations &amp; Algebraic Thinking • 5.OA.B.3 • Set 6 • MAIN • Q4/5

196

QUESTION ID

**S6-M-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

Grade 5 • Operations &amp; Algebraic Thinking • 5.OA.B.3 • Set 6 • MAIN • Q4/5

Pattern A starts at 0 and adds 7. Pattern B starts at 0 and adds 21.

When Pattern A is 28, what is the matching term in Pattern B?

Term: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 6 • MAIN • Q5/5

197

QUESTION ID

**S6-M-Q5**

TYPE

Short answer

CORRECT ANSWER

**84**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 6 • MAIN • Q5/5

SET 6 • 5.OA.B.3

# BACKWARD 1

Patterns that follow a rule • Grade 4 • Operations & Algebraic Thinking •  
4.OA.C.5

The rule is "add 6." The pattern starts at 0.

Write the first five terms.

Terms: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 6 • BACKWARD 1 • Q1/5

199

QUESTION ID

**S6-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**0, 6, 12, 18, 24**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 6 • BACKWARD 1 • Q1/5

**True or false?**

The rule "add 4," starting at 0, gives 0, 4, 8, 16.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 4 • Operations &amp; Algebraic Thinking • 4.OA.C.5 • Set 6 • BACKWARD 1 • Q2/5

200

QUESTION ID

**S6-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 4 • Operations &amp; Algebraic Thinking • 4.OA.C.5 • Set 6 • BACKWARD 1 • Q2/5



What is the rule for the pattern?

0, 9, 18, 27, 36

- A. Add 9
- B. Add 18
- C. Multiply by 9
- D. Add 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 6 • BACKWARD 1 • Q3/5

201

QUESTION ID

**S6-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. Add 9**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 6 • BACKWARD 1 • Q3/5

The rule is "add 11." The pattern starts at 0.

What is the 4th term?

Term: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 6 • BACKWARD 1 • Q4/5

202

QUESTION ID

**S6-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**33**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 6 • BACKWARD 1 • Q4/5

What comes next?

0, 15, 30, 45, \_\_\_\_

Next term: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 6 • BACKWARD 1 • Q5/5

203

QUESTION ID

**S6-B1-Q5**

TYPE

Short answer

CORRECT ANSWER

**60**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 6 • BACKWARD 1 • Q5/5

SET 6 • 5.OA.B.3

# BACKWARD 2

Multiplication facts • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Multiply.

$$4 \times 7$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 6 • BACKWARD 2 • Q1/5

205

QUESTION ID

**S6-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**28**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 6 • BACKWARD 2 • Q1/5

**True or false?**

$$6 \times 8 = 42$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 6 • BACKWARD 2 • Q2/5

206

QUESTION ID

**S6-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 6 • BACKWARD 2 • Q2/5

What number times 3 equals 24?

- A.** 8
- B.** 7
- C.** 21
- D.** 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 6 • BACKWARD 2 • Q3/5

207

QUESTION ID

**S6-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. 8**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 6 • BACKWARD 2 • Q3/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$5 \times 9$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.C.7 • Set 6 • BACKWARD 2 • Q4/5

208

QUESTION ID

**S6-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**45**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.C.7 • Set 6 • BACKWARD 2 • Q4/5

208



What number times 6 equals 42?

Number: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 6 • BACKWARD 2 • Q5/5

209

QUESTION ID

**S6-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**7**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 6 • BACKWARD 2 • Q5/5

SET 6 • 5.OA.B.3

# FORWARD 1

Equations relating dependent and independent variables • Grade 6 •  
Expressions & Equations • 6.EE.C.9

Each box holds 6 cans.

Write an equation for the number of cans  $c$  in  $b$  boxes.

Equation: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • FORWARD 1 • Q1/5

211

QUESTION ID

**S6-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**$c = 6b$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • FORWARD 1 • Q1/5

In  $c = 6b$ , which variable is the dependent variable?

- A.** b
- B.** c
- C.** 6
- D.** Neither

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • FORWARD 1 • Q2/5

212

QUESTION ID

**S6-F1-Q2**

TYPE

Multiple choice

CORRECT ANSWER

**B. c**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • FORWARD 1 • Q2/5

Write an equation for  $y$  in terms of  $x$ .

|          |   |    |    |    |
|----------|---|----|----|----|
| <b>x</b> | 1 | 2  | 3  | 4  |
| <b>y</b> | 5 | 10 | 15 | 20 |

Equation: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • FORWARD 1 • Q3/5

213

QUESTION ID

**S6-F1-Q3**

TYPE

Short answer

CORRECT ANSWER

**$y = 5x$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • FORWARD 1 • Q3/5

**True or false?**

In  $y = x + 4$ ,  $y$  is the independent variable.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

Grade 6 • Expressions &amp; Equations • 6.EE.C.9 • Set 6 • FORWARD 1 • Q4/5

214

QUESTION ID

**S6-F1-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

Grade 6 • Expressions &amp; Equations • 6.EE.C.9 • Set 6 • FORWARD 1 • Q4/5

A plant grows 2 cm each week.

Write an equation for its growth  $g$  after  $w$  weeks.

Equation: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • FORWARD 1 • Q5/5

215

QUESTION ID

**S6-F1-Q5**

TYPE

Short answer

CORRECT ANSWER

**$g = 2w$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • FORWARD 1 • Q5/5

SET 6 • 5.OA.B.3

# FORWARD 2

Equations for proportional relationships • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c



Pencils cost \$0.35 each.

Write an equation for the cost  $c$  of  $n$  pencils.

Equation: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 6 • FORWARD 2 • Q1/5

217

QUESTION ID

**S6-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**$c = 0.35n$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 6 • FORWARD 2 • Q1/5

The table shows a proportional relationship.

Write an equation for  $y$  in terms of  $x$ .

|     |    |    |    |
|-----|----|----|----|
| $x$ | 4  | 6  | 10 |
| $y$ | 10 | 15 | 25 |

Equation: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 6 • FORWARD 2 • Q2/5

218

QUESTION ID

**S6-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

$$y = 2.5x$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 6 • FORWARD 2 • Q2/5

Which equation represents a proportional relationship?

- A.  $y = 1.5$
- B.  $y = x + 1.5$
- C.  $y = 1.5x + 2$
- D.  $y = 1.5x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 6 • FORWARD 2 • Q3/5

219

QUESTION ID

**S6-F2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D.  $y = 1.5x$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 6 • FORWARD 2 • Q3/5

A train travels 150 miles in 2 hours at a constant speed.

Write an equation for the distance  $d$  after  $t$  hours.

Equation: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 6 • FORWARD 2 • Q4/5

220

QUESTION ID

**S6-F2-Q4**

TYPE

Short answer

CORRECT ANSWER

**$d = 75t$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 6 • FORWARD 2 • Q4/5

**True or false?**

$y = \frac{3}{4}x$  represents a proportional relationship.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q5

Grade 7 • Ratios &amp; Proportional Relationships • 7.RP.A.2.c • Set 6 • FORWARD 2 • Q5/5

221

QUESTION ID

**S6-F2-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q5

Grade 7 • Ratios &amp; Proportional Relationships • 7.RP.A.2.c • Set 6 • FORWARD 2 • Q5/5

# Form ordered pairs from two patterns and graph them

|                   |                                                              |                   |
|-------------------|--------------------------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 5 main question                                        | <b>5.OA.B.3</b>   |
| <b>BACKWARD 1</b> | Patterns that follow a rule                                  | <b>4.OA.C.5</b>   |
| <b>BACKWARD 2</b> | Ordered pairs and the coordinate plane                       | <b>5.G.A.1</b>    |
| <b>BACKWARD 3</b> | Reading the scale on an axis                                 | <b>3.MD.B.3</b>   |
| <b>FORWARD 1</b>  | Plot pairs of values from a ratio table                      | <b>6.RP.A.3.a</b> |
| <b>FORWARD 2</b>  | Use a graph to decide whether a relationship is proportional | <b>7.RP.A.2.a</b> |

SET 7 • 5.OA.B.3

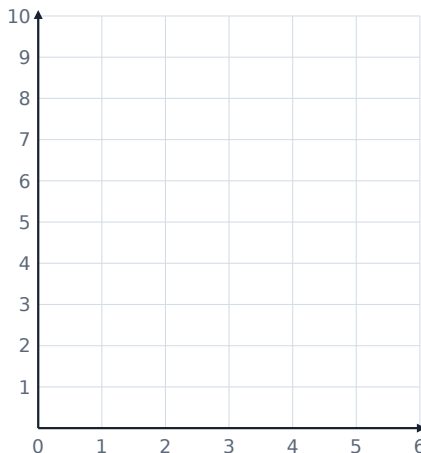
# MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A (x): start at 0 and add 1.

Pattern B (y): start at 0 and add 2.

Graph the ordered pairs for the first 5 terms.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 7 • MAIN • Q1/5

224

QUESTION ID

**S7-M-Q1**

TYPE

Drawing / plotting

CORRECT ANSWER

**Points plotted at (0, 0), (1, 2), (2, 4), (3, 6), (4, 8)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

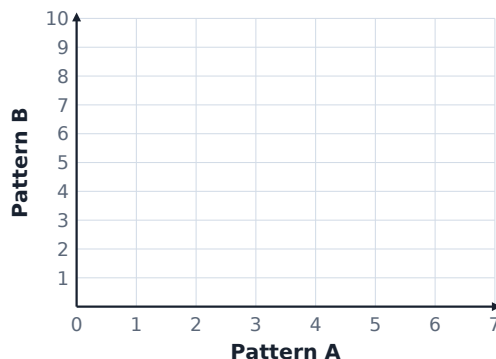
ID S7-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 7 • MAIN • Q1/5



Graph the ordered pairs from the two patterns.

|           |   |   |   |   |
|-----------|---|---|---|---|
| Pattern A | 0 | 2 | 4 | 6 |
| Pattern B | 0 | 3 | 6 | 9 |



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 7 • MAIN • Q2/5

225

QUESTION ID

**S7-M-Q2**

TYPE

Drawing / plotting

CORRECT ANSWER

**Points plotted at (0, 0), (2, 3), (4, 6), (6, 9)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 7 • MAIN • Q2/5

Pattern A starts at 0 and adds 3. Pattern B starts at 0 and adds 9.

Write the first four ordered pairs (A, B).

Ordered pairs: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 7 • MAIN • Q3/5

226

QUESTION ID

**S7-M-Q3**

TYPE

Short answer

CORRECT ANSWER

**(0, 0), (3, 9), (6, 18),  
(9, 27)**

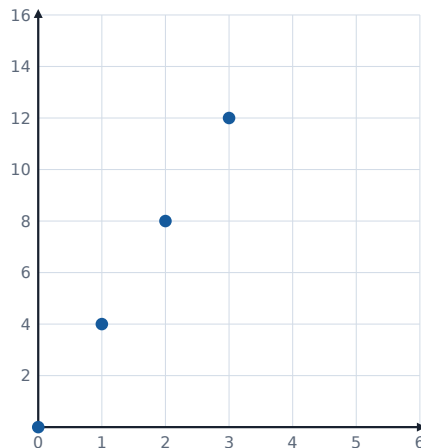
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 7 • MAIN • Q3/5

The points come from two patterns.  
Which statement is true?

- A.** The y-values increase by 1 each time.      **B.** Each y-value is 4 times its x-value.
- C.** Each x-value is 4 times its y-value.      **D.** Each y-value is 4 more than its x-value.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q4

Grade 5 • Operations &amp; Algebraic Thinking • 5.OA.B.3 • Set 7 • MAIN • Q4/5

227

QUESTION ID

**S7-M-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. Each y-value is 4 times its x-value.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

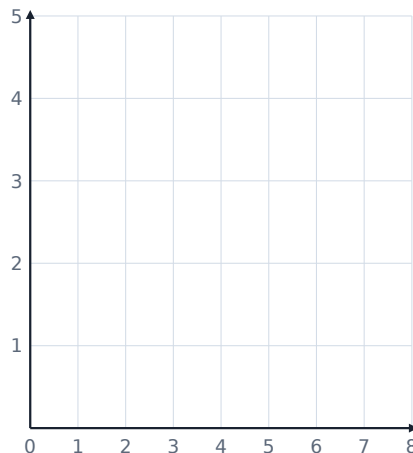
ID S7-M-Q4

Grade 5 • Operations &amp; Algebraic Thinking • 5.OA.B.3 • Set 7 • MAIN • Q4/5

Pattern A (x): start at 0 and add 2.

Pattern B (y): start at 0 and add 1.

Graph the ordered pairs for the first 4 terms.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 7 • MAIN • Q5/5

228

QUESTION ID

**S7-M-Q5**

TYPE

Drawing / plotting

CORRECT ANSWER

**Points plotted at (0, 0), (2, 1), (4, 2), (6, 3)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 7 • MAIN • Q5/5

SET 7 • 5.OA.B.3

# BACKWARD 1

Patterns that follow a rule • Grade 4 • Operations & Algebraic Thinking •  
4.OA.C.5

The rule is "add 3." The pattern starts at 0.

What is the 5th term?

Term: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 7 • BACKWARD 1 • Q1/5

230

QUESTION ID

**S7-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**12**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 7 • BACKWARD 1 • Q1/5

**True or false?**

The pattern 0, 5, 10, 15 follows the rule "add 5."

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

Grade 4 • Operations &amp; Algebraic Thinking • 4.OA.C.5 • Set 7 • BACKWARD 1 • Q2/5

231

QUESTION ID

**S7-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

Grade 4 • Operations &amp; Algebraic Thinking • 4.OA.C.5 • Set 7 • BACKWARD 1 • Q2/5

Which pattern follows the rule "add 2," starting at 0?

- A.** 0, 2, 4, 6
- B.** 2, 4, 6, 8
- C.** 0, 1, 2, 3
- D.** 0, 2, 4, 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 7 • BACKWARD 1 • Q3/5

232

QUESTION ID

**S7-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. 0, 2, 4, 6**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 7 • BACKWARD 1 • Q3/5



What is the rule for the pattern?

0, 8, 16, 24

Rule: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 7 • BACKWARD 1 • Q4/5

233

QUESTION ID

**S7-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**Add 8**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 7 • BACKWARD 1 • Q4/5

The rule is "add 10." The pattern starts at 0.

What is the 6th term?

Term: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 7 • BACKWARD 1 • Q5/5

234

QUESTION ID

**S7-B1-Q5**

TYPE

Short answer

CORRECT ANSWER

**50**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 7 • BACKWARD 1 • Q5/5

SET 7 • 5.OA.B.3

# BACKWARD 2

Ordered pairs and the coordinate plane • Grade 5 • Geometry • 5.G.A.1

**True or false?**

In (3, 6), the 3 tells how far to move along the x-axis.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

Grade 5 • Geometry • 5.G.A.1 • Set 7 • BACKWARD 2 • Q1/5

236

QUESTION ID

**S7-B2-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

Grade 5 • Geometry • 5.G.A.1 • Set 7 • BACKWARD 2 • Q1/5

In the ordered pair (4, 9), which number is the y-coordinate?

- A.** 9
- B.** 4
- C.** 13
- D.** 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 5 • Geometry • 5.G.A.1 • Set 7 • BACKWARD 2 • Q2/5

237

QUESTION ID

**S7-B2-Q2**

TYPE

Multiple choice

CORRECT ANSWER

**A. 9**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 5 • Geometry • 5.G.A.1 • Set 7 • BACKWARD 2 • Q2/5

Which ordered pair names the origin?

Ordered pair: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 5 • Geometry • 5.G.A.1 • Set 7 • BACKWARD 2 • Q3/5

238

QUESTION ID

**S7-B2-Q3**

TYPE

Short answer

CORRECT ANSWER

**(0, 0)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 5 • Geometry • 5.G.A.1 • Set 7 • BACKWARD 2 • Q3/5

**True or false?**

(2, 5) and (5, 2) name the same point.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 5 • Geometry • 5.G.A.1 • Set 7 • BACKWARD 2 • Q4/5

239

QUESTION ID

**S7-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

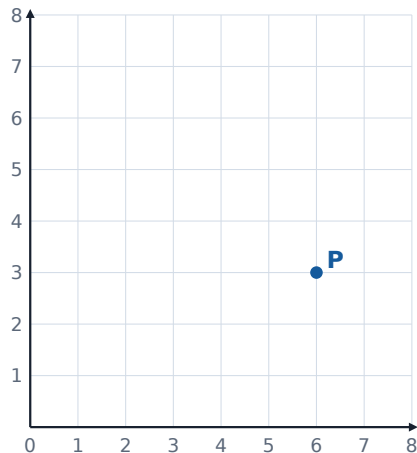
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 5 • Geometry • 5.G.A.1 • Set 7 • BACKWARD 2 • Q4/5

What are the coordinates of point P?

P = \_\_\_\_\_



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 5 • Geometry • 5.G.A.1 • Set 7 • BACKWARD 2 • Q5/5

240

QUESTION ID

**S7-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**(6, 3)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 5 • Geometry • 5.G.A.1 • Set 7 • BACKWARD 2 • Q5/5



SET 7 • 5.OA.B.3

# BACKWARD 3

Reading the scale on an axis • Grade 3 • Measurement & Data • 3.MD.B.3

An axis is numbered 0, 2, 4, 6, 8.

How much does each grid line stand for?

- A. 4
- B. 1
- C. 8
- D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 7 • BACKWARD 3 • Q1/5

242

QUESTION ID

**S7-B3-Q1**

TYPE

Multiple choice

CORRECT ANSWER

**D. 2**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 7 • BACKWARD 3 • Q1/5

**True or false?**

On an axis numbered 0, 4, 8, 12, the value 10 is halfway between 4 and 8.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q2

Grade 3 • Measurement &amp; Data • 3.MD.B.3 • Set 7 • BACKWARD 3 • Q2/5

243

QUESTION ID

**S7-B3-Q2**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q2

Grade 3 • Measurement &amp; Data • 3.MD.B.3 • Set 7 • BACKWARD 3 • Q2/5

An axis is numbered 0, 5, 10, 15.

Where does 20 go?

Answer: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 7 • BACKWARD 3 • Q3/5

244

QUESTION ID

**S7-B3-Q3**

TYPE

Short answer

CORRECT ANSWER

**One grid line above  
15**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 7 • BACKWARD 3 • Q3/5

On an axis numbered by 3s, which value is two grid lines above 0?

- A.** 6
- B.** 3
- C.** 9
- D.** 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q4

Grade 3 • Measurement &amp; Data • 3.MD.B.3 • Set 7 • BACKWARD 3 • Q4/5

245

QUESTION ID

**S7-B3-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. 6**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q4

Grade 3 • Measurement &amp; Data • 3.MD.B.3 • Set 7 • BACKWARD 3 • Q4/5

**True or false?**

On an axis numbered by 10s, the value 25 is on a grid line.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q5

Grade 3 • Measurement &amp; Data • 3.MD.B.3 • Set 7 • BACKWARD 3 • Q5/5

246

QUESTION ID

**S7-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q5

Grade 3 • Measurement &amp; Data • 3.MD.B.3 • Set 7 • BACKWARD 3 • Q5/5

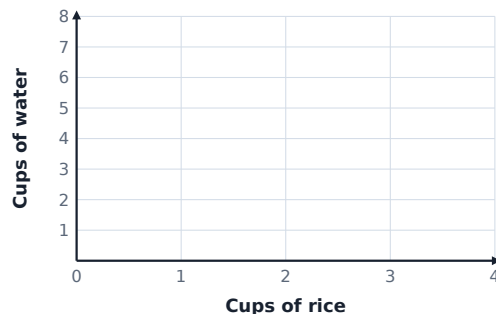
SET 7 • 5.OA.B.3

# FORWARD 1

Plot pairs of values from a ratio table • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a

Plot the pairs of values from the ratio table on the coordinate plane.

|               |   |   |   |
|---------------|---|---|---|
| Cups of rice  | 1 | 2 | 3 |
| Cups of water | 2 | 4 | 6 |



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q1

Grade 6 • Ratios &amp; Proportional Relationships • 6.RP.A.3.a • Set 7 • FORWARD 1 • Q1/5

248

QUESTION ID

**S7-F1-Q1**

TYPE

Drawing / plotting

CORRECT ANSWER

**Points plotted at (1, 2), (2, 4), (3, 6)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q1

Grade 6 • Ratios &amp; Proportional Relationships • 6.RP.A.3.a • Set 7 • FORWARD 1 • Q1/5



The table shows equivalent ratios.

Find the missing value.

|         |   |    |   |    |
|---------|---|----|---|----|
| Laps    | 2 | 4  | 6 | 8  |
| Minutes | 5 | 10 | ? | 20 |

Missing value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • FORWARD 1 • Q2/5

249

QUESTION ID

**S7-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**15**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • FORWARD 1 • Q2/5

Which point comes from the ratio table?

|       |   |    |    |
|-------|---|----|----|
| Boxes | 1 | 2  | 3  |
| Pens  | 6 | 12 | 18 |

- A. (2, 6)                                      B. (12, 2)  
C. (2, 12)                                    D. (3, 12)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • FORWARD 1 • Q3/5

250

QUESTION ID

**S7-F1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. (2, 12)**

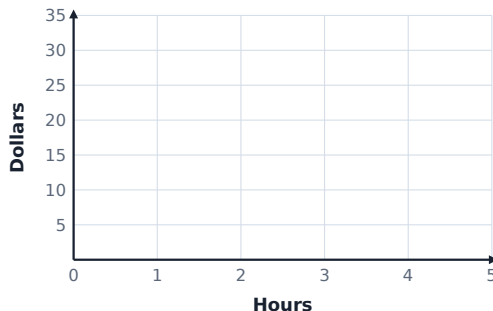
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • FORWARD 1 • Q3/5

Plot the pairs of values from the ratio table on the coordinate plane.

| Hours   | 1 | 2  | 3  | 4  |
|---------|---|----|----|----|
| Dollars | 8 | 16 | 24 | 32 |



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q4

Grade 6 • Ratios &amp; Proportional Relationships • 6.RP.A.3.a • Set 7 • FORWARD 1 • Q4/5

251

QUESTION ID

**S7-F1-Q4**

TYPE

Drawing / plotting

CORRECT ANSWER

**Points plotted at (1, 8), (2, 16), (3, 24), (4, 32)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q4

Grade 6 • Ratios &amp; Proportional Relationships • 6.RP.A.3.a • Set 7 • FORWARD 1 • Q4/5

**True or false?**

The points (1, 3), (2, 6), and (3, 9) come from a ratio table with the ratio 1 : 3.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q5

Grade 6 • Ratios &amp; Proportional Relationships • 6.RP.A.3.a • Set 7 • FORWARD 1 • Q5/5

252

QUESTION ID

**S7-F1-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q5

Grade 6 • Ratios &amp; Proportional Relationships • 6.RP.A.3.a • Set 7 • FORWARD 1 • Q5/5

SET 7 • 5.OA.B.3

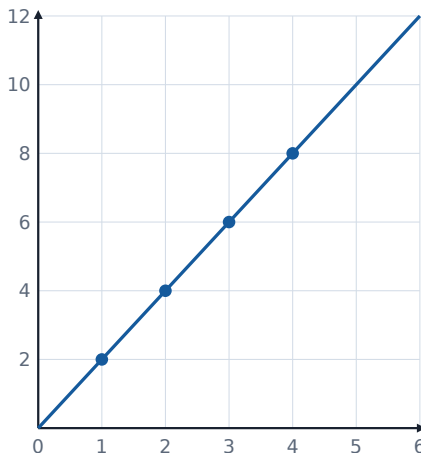
# FORWARD 2

Use a graph to decide whether a relationship is proportional • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

**True or false?**

The points lie on a straight line through the origin, so the relationship is proportional.

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q1

Grade 7 • Ratios &amp; Proportional Relationships • 7.RP.A.2.a • Set 7 • FORWARD 2 • Q1/5

254

QUESTION ID

**S7-F2-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

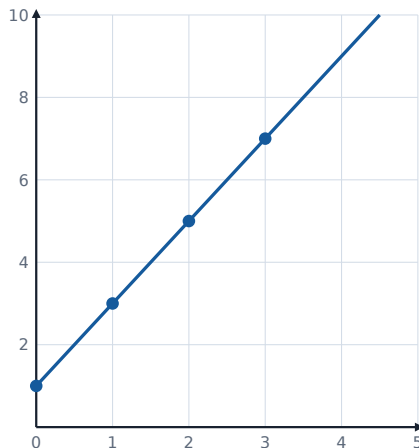
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q1

Grade 7 • Ratios &amp; Proportional Relationships • 7.RP.A.2.a • Set 7 • FORWARD 2 • Q1/5

Which statement about the graph is true?

- A. It is not proportional because the line does not pass through  $(0, 0)$ .
- B. It is not proportional because the points are not on a line.
- C. It is proportional because the points are on a line.
- D. It is proportional because  $y$  increases by 2.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 7 • FORWARD 2 • Q2/5

255

QUESTION ID

**S7-F2-Q2**

TYPE

Multiple choice

CORRECT ANSWER

**A. It is not proportional because the line does not pass through  $(0, 0)$ .**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

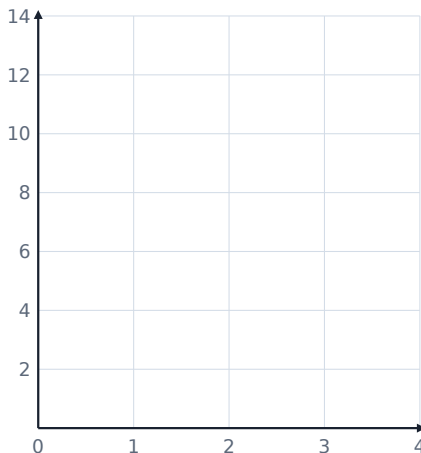
ID S7-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 7 • FORWARD 2 • Q2/5

Graph the pairs (1, 4), (2, 8),  
and (3, 12).

Is the relationship  
proportional?

Answer: \_\_\_\_\_



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 7 • FORWARD 2 • Q3/5

256

QUESTION ID

**S7-F2-Q3**

TYPE

Short answer

CORRECT ANSWER

**Yes**

**GRADING NOTE** The points lie on a straight line through (0, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 7 • FORWARD 2 • Q3/5



**True or false?**

The points (1, 2), (2, 5), and (3, 8) show a proportional relationship.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q4

Grade 7 • Ratios &amp; Proportional Relationships • 7.RP.A.2.a • Set 7 • FORWARD 2 • Q4/5

257

QUESTION ID

**S7-F2-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q4

Grade 7 • Ratios &amp; Proportional Relationships • 7.RP.A.2.a • Set 7 • FORWARD 2 • Q4/5

Which set of points shows a proportional relationship?

- A. (1, 1), (2, 4), (3, 9)
- B. (1, 2), (2, 3), (3, 4)
- C. (0, 3), (1, 6), (2, 9)
- D. (2, 5), (4, 10), (6, 15)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 7 • FORWARD 2 • Q5/5

258

QUESTION ID

**S7-F2-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**D. (2, 5), (4, 10), (6, 15)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 7 • FORWARD 2 • Q5/5

# A digit is worth 10 times as much as the same digit one place to its right

|                   |                                                     |                  |
|-------------------|-----------------------------------------------------|------------------|
| <b>MAIN</b>       | Grade 5 main question                               | <b>5.NBT.A.1</b> |
| <b>BACKWARD 1</b> | Each place is worth 10 times the place to its right | <b>4.NBT.A.1</b> |
| <b>BACKWARD 2</b> | Tenths and hundredths as decimals                   | <b>4.NF.C.6</b>  |
| <b>BACKWARD 3</b> | Tenths as hundredths                                | <b>4.NF.C.5</b>  |
| <b>FORWARD 1</b>  | Compute with multi-digit decimals                   | <b>6.NS.B.3</b>  |
| <b>FORWARD 2</b>  | Real-world problems with rational numbers           | <b>7.NS.A.3</b>  |

SET 8 • 5.NBT.A.1

# MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •  
5.NBT.A.1

In 5,550, how does the value of the 5 in the hundreds place compare to the value of the 5 in the tens place?

- A. It is 100 times as much.
- B. It is 10 times as much.
- C. It is the same.
- D. It is  $\frac{1}{10}$  as much.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 8 • MAIN • Q1/5

261

QUESTION ID

**S8-M-Q1**

TYPE

Multiple choice

CORRECT ANSWER

**B. It is 10 times as much.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 8 • MAIN • Q1/5

In 3.33, the first 3 is in the ones place and the second 3 is in the tenths place.

How many times as much is the first 3 worth as the second 3?

Times as much: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 8 • MAIN • Q2/5

262

QUESTION ID

**S8-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**10**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 8 • MAIN • Q2/5

**True or false?**

In 0.77, the 7 in the tenths place is worth 10 times as much as the 7 in the hundredths place.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.1 • Set 8 • MAIN • Q3/5

263

QUESTION ID

**S8-M-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.1 • Set 8 • MAIN • Q3/5

Find the value of the 4 in 642.18 and the value of the 4 in 6,421.8.

How many times as much is the second 4 worth?

4 in 642.18: \_\_\_\_\_

4 in 6,421.8: \_\_\_\_\_

Times as much: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 8 • MAIN • Q4/5

264

QUESTION ID

**S8-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**40; 400; 10 times as much**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 8 • MAIN • Q4/5



In which number is the 6 worth 10 times as much as the 6 in 3.6?

- A.** 0.36
- B.** 3.06
- C.** 46.1
- D.** 60.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 8 • MAIN • Q5/5

265

QUESTION ID

**S8-M-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**C. 46.1**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 8 • MAIN • Q5/5

SET 8 • 5.NBT.A.1

# BACKWARD 1

Each place is worth 10 times the place to its right • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1

**True or false?**

In 880, the 8 in the hundreds place is worth 10 times as much as the 8 in the tens place.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.A.1 • Set 8 • BACKWARD 1 • Q1/5

267

QUESTION ID

**S8-B1-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.A.1 • Set 8 • BACKWARD 1 • Q1/5

What is the value of the 7 in 2,740?

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 8 • BACKWARD 1 • Q2/5

268

QUESTION ID

**S8-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**700**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 8 • BACKWARD 1 • Q2/5

Which number is 10 times as much as 60?

- A.** 600
- B.** 6,000
- C.** 70
- D.** 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 8 • BACKWARD 1 • Q3/5

269

QUESTION ID

**S8-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. 600**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 8 • BACKWARD 1 • Q3/5

How many tens are in 1 hundred?

Tens: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 8 • BACKWARD 1 • Q4/5

270

QUESTION ID

**S8-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**10**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 8 • BACKWARD 1 • Q4/5

**True or false?**

70 is 10 times as much as 7.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.A.1 • Set 8 • BACKWARD 1 • Q5/5

271

QUESTION ID

**S8-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.A.1 • Set 8 • BACKWARD 1 • Q5/5

SET 8 • 5.NBT.A.1

# BACKWARD 2

Tenths and hundredths as decimals • Grade 4 • Number & Operations–Fractions • 4.NF.C.6



Write  $\frac{3}{10}$  as a decimal.

Decimal: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 8 • BACKWARD 2 • Q1/5

273

QUESTION ID

**S8-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**0.3**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 8 • BACKWARD 2 • Q1/5

Write  $\frac{47}{100}$  as a decimal.

Decimal: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 8 • BACKWARD 2 • Q2/5

274

QUESTION ID

**S8-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**0.47**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 8 • BACKWARD 2 • Q2/5

**True or false?**

$$0.6 = \frac{6}{100}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 8 • BACKWARD 2 • Q3/5

275

QUESTION ID

**S8-B2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 8 • BACKWARD 2 • Q3/5

Which decimal shows 5 hundredths?

- A.** 0.5
- B.** 0.05
- C.** 0.005
- D.** 5.0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 8 • BACKWARD 2 • Q4/5

276

QUESTION ID

**S8-B2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. 0.05**

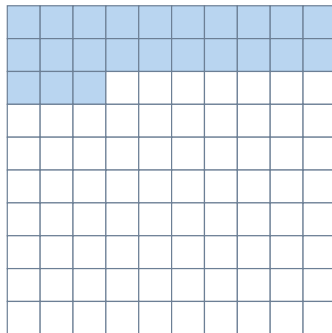
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 8 • BACKWARD 2 • Q4/5

What decimal does the shaded part of the grid show?

Decimal: \_\_\_\_\_



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

Grade 4 • Number &amp; Operations-Fractions • 4.NF.C.6 • Set 8 • BACKWARD 2 • Q5/5

277

QUESTION ID

**S8-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**0.23**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

Grade 4 • Number &amp; Operations-Fractions • 4.NF.C.6 • Set 8 • BACKWARD 2 • Q5/5

SET 8 • 5.NBT.A.1

# BACKWARD 3

Tenths as hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.5

**True or false?**

$$\frac{1}{10} = \frac{10}{100}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 8 • BACKWARD 3 • Q1/5

279

QUESTION ID

**S8-B3-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 8 • BACKWARD 3 • Q1/5

How many hundredths are in 1 tenth?

Hundredths: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 8 • BACKWARD 3 • Q2/5

280

QUESTION ID

**S8-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**10**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 8 • BACKWARD 3 • Q2/5



Which fraction is equal to  $\frac{4}{10}$ ?

A.  $\frac{4}{100}$

B.  $\frac{400}{100}$

C.  $\frac{14}{100}$

D.  $\frac{40}{100}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 8 • BACKWARD 3 • Q3/5

281

QUESTION ID

**S8-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. 40/100**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 8 • BACKWARD 3 • Q3/5

Find the missing number.

$$\frac{7}{10} = \frac{?}{100}$$

Missing number: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 8 • BACKWARD 3 • Q4/5

282

QUESTION ID

**S8-B3-Q4**

TYPE

Short answer

CORRECT ANSWER

**70**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 8 • BACKWARD 3 • Q4/5

**True or false?**

$$\frac{3}{10} + \frac{5}{100} = \frac{35}{100}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q5

Grade 4 • Number &amp; Operations-Fractions • 4.NF.C.5 • Set 8 • BACKWARD 3 • Q5/5

283

QUESTION ID

**S8-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q5

Grade 4 • Number &amp; Operations-Fractions • 4.NF.C.5 • Set 8 • BACKWARD 3 • Q5/5

SET 8 • 5.NBT.A.1

# FORWARD 1

Compute with multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Add.

$$27.6 + 4.38$$

Sum: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 1 • Q1/5

285

QUESTION ID

**S8-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**31.98**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 1 • Q1/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Subtract.

$$50.2 - 7.65$$

Difference: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 1 • Q2/5

286

QUESTION ID

**S8-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**42.55**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 1 • Q2/5

286

Multiply.

$$1.2 \times 0.4$$

- A.** 4.8
- B.** 0.48
- C.** 0.048
- D.** 1.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 1 • Q3/5

287

QUESTION ID

**S8-F1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. 0.48**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 1 • Q3/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Divide.

$$7.5 \div 2.5$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 1 • Q4/5

288

QUESTION ID

**S8-F1-Q4**

TYPE

Short answer

CORRECT ANSWER

**3**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 1 • Q4/5

288



**True or false?**

$$0.3 \times 0.3 = 0.9$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 1 • Q5/5

289

QUESTION ID

**S8-F1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 1 • Q5/5

SET 8 • 5.NBT.A.1

# FORWARD 2

Real-world problems with rational numbers • Grade 7 • The Number System •  
7.NS.A.3

A diver at  $-4.5$  m descends  $2.25$  m.

What is the diver's new position?

Position: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 8 • FORWARD 2 • Q1/5

291

QUESTION ID

**S8-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**$-6.75$  m**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 8 • FORWARD 2 • Q1/5

The temperature was  $3.5^{\circ}\text{F}$ . It fell  $6.2^{\circ}\text{F}$ .

What is the new temperature?

Temperature: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 8 • FORWARD 2 • Q2/5

292

QUESTION ID

**S8-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**$-2.7^{\circ}\text{F}$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 8 • FORWARD 2 • Q2/5

A stock's value changed by  $-\$0.75$  each day for 4 days.

What was the total change?

- A.  $-\$0.30$
- B.  $-\$4.75$
- C.  $\$3.00$
- D.  $-\$3.00$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 8 • FORWARD 2 • Q3/5

293

QUESTION ID

**S8-F2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D.  $-\$3.00$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 8 • FORWARD 2 • Q3/5

A bank balance of  $-\$12.40$  increases by  $\$20.00$ .

What is the new balance?

Balance: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 8 • FORWARD 2 • Q4/5

294

QUESTION ID

**S8-F2-Q4**

TYPE

Short answer

CORRECT ANSWER

**$\$7.60$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 8 • FORWARD 2 • Q4/5

**True or false?**

$$-1.5 \times 4 = 6$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 8 • FORWARD 2 • Q5/5

295

QUESTION ID

**S8-F2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 8 • FORWARD 2 • Q5/5

# A digit is worth $\frac{1}{10}$ of the same digit one place to its left

|                   |                                                        |                   |
|-------------------|--------------------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 5 main question                                  | <b>5.NBT.A.1</b>  |
| <b>BACKWARD 1</b> | One place to the right is worth $\frac{1}{10}$ as much | <b>4.NBT.A.1</b>  |
| <b>BACKWARD 2</b> | The unit fraction $\frac{1}{10}$                       | <b>3.NF.A.1</b>   |
| <b>BACKWARD 3</b> | Decimals for tenths and hundredths                     | <b>4.NF.C.6</b>   |
| <b>FORWARD 1</b>  | Divide multi-digit decimals                            | <b>6.NS.B.3</b>   |
| <b>FORWARD 2</b>  | Write fractions as decimals using long division        | <b>7.NS.A.2.d</b> |



SET 9 • 5.NBT.A.1

# MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •  
5.NBT.A.1

**True or false?**

In 44.4, the 4 in the tenths place is worth  $\frac{1}{10}$  of the 4 in the ones place.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.1 • Set 9 • MAIN • Q1/5

298

QUESTION ID

**S9-M-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.1 • Set 9 • MAIN • Q1/5

In 2.22, the 2 in the hundredths place is worth what part of the 2 in the tenths place?

- A. 10 times as much
- B.  $\frac{1}{10}$
- C. The same amount
- D.  $\frac{1}{100}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 9 • MAIN • Q2/5

299

QUESTION ID

**S9-M-Q2**

TYPE

Multiple choice

CORRECT ANSWER

**B. 1/10**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 9 • MAIN • Q2/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

What is  $\frac{1}{10}$  of 50?

Answer: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.1 • Set 9 • MAIN • Q3/5

300

QUESTION ID

**S9-M-Q3**

TYPE

Short answer

CORRECT ANSWER

**5**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.1 • Set 9 • MAIN • Q3/5

300

**ANSWER KEY**

Reduced copy of the question page, with its answer

What is  $\frac{1}{10}$  of 0.8?

Answer: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.1 • Set 9 • MAIN • Q4/5

301

QUESTION ID

**S9-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**0.08**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.1 • Set 9 • MAIN • Q4/5

301

In which number is the 9 worth  $\frac{1}{10}$  of the 9 in 19.4?

- A. 5.92
- B. 3.009
- C. 9.01
- D. 90.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 9 • MAIN • Q5/5

302

QUESTION ID

**S9-M-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**A. 5.92**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 9 • MAIN • Q5/5

SET 9 • 5.NBT.A.1

# BACKWARD 1

One place to the right is worth  $\frac{1}{10}$  as much • Grade 4 • Number & Operations  
in Base Ten • 4.NBT.A.1

**True or false?**

In 330, the 3 in the tens place is worth  $\frac{1}{10}$  of the 3 in the hundreds place.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.A.1 • Set 9 • BACKWARD 1 • Q1/5

304

QUESTION ID

**S9-B1-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.A.1 • Set 9 • BACKWARD 1 • Q1/5



**ANSWER KEY**

Reduced copy of the question page, with its answer

Divide.

$$400 \div 10$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.A.1 • Set 9 • BACKWARD 1 • Q2/5

305

QUESTION ID

**S9-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**40**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.A.1 • Set 9 • BACKWARD 1 • Q2/5

305

Which number is  $\frac{1}{10}$  of 3,000?

- A. 3
- B. 300
- C. 30,000
- D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 9 • BACKWARD 1 • Q3/5

306

QUESTION ID

**S9-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. 300**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 9 • BACKWARD 1 • Q3/5

How many hundreds are in 1 thousand?

Hundreds: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 9 • BACKWARD 1 • Q4/5

307

QUESTION ID

**S9-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**10**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 9 • BACKWARD 1 • Q4/5

**True or false?**

$$60 \div 10 = 600$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 9 • BACKWARD 1 • Q5/5

308

QUESTION ID

**S9-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 9 • BACKWARD 1 • Q5/5

SET 9 • 5.NBT.A.1

# BACKWARD 2

The unit fraction  $\frac{1}{10}$  • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

**True or false?**

$\frac{1}{10}$  means 1 of 10 equal parts.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q1

Grade 3 • Number &amp; Operations-Fractions • 3.NF.A.1 • Set 9 • BACKWARD 2 • Q1/5

310

QUESTION ID

**S9-B2-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q1

Grade 3 • Number &amp; Operations-Fractions • 3.NF.A.1 • Set 9 • BACKWARD 2 • Q1/5

A bar is cut into 10 equal parts.

What fraction of the bar is one part?

Fraction: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 9 • BACKWARD 2 • Q2/5

311

QUESTION ID

**S9-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**1/10**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 9 • BACKWARD 2 • Q2/5

Which fraction is greater?

A.  $\frac{1}{10}$

B.  $\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 9 • BACKWARD 2 • Q3/5

312

QUESTION ID

**S9-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. 1/2**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 9 • BACKWARD 2 • Q3/5



**True or false?**

A whole cut into 10 equal parts has parts that are each  $\frac{1}{10}$  of the whole.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 3 • Number &amp; Operations-Fractions • 3.NF.A.1 • Set 9 • BACKWARD 2 • Q4/5

313

QUESTION ID

**S9-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 3 • Number &amp; Operations-Fractions • 3.NF.A.1 • Set 9 • BACKWARD 2 • Q4/5

What fraction of the bar is shaded?



Fraction: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 9 • BACKWARD 2 • Q5/5

314

QUESTION ID

**S9-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**1/10**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 9 • BACKWARD 2 • Q5/5

SET 9 • 5.NBT.A.1

# BACKWARD 3

Decimals for tenths and hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write  $\frac{9}{100}$  as a decimal.

Decimal: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 9 • BACKWARD 3 • Q1/5

316

QUESTION ID

**S9-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**0.09**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 9 • BACKWARD 3 • Q1/5

Write 0.8 as a fraction.

Fraction: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 9 • BACKWARD 3 • Q2/5

317

QUESTION ID

**S9-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**8/10**

**GRADING NOTE** 4/5 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 9 • BACKWARD 3 • Q2/5

**True or false?**

0.4 and 0.04 have the same value.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q3

Grade 4 • Number &amp; Operations-Fractions • 4.NF.C.6 • Set 9 • BACKWARD 3 • Q3/5

318

QUESTION ID

**S9-B3-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q3

Grade 4 • Number &amp; Operations-Fractions • 4.NF.C.6 • Set 9 • BACKWARD 3 • Q3/5

Which decimal is equal to  $\frac{6}{10}$ ?

- A. 6.0
- B. 0.06
- C. 6.10
- D. 0.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 9 • BACKWARD 3 • Q4/5

319

QUESTION ID

**S9-B3-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. 0.6**

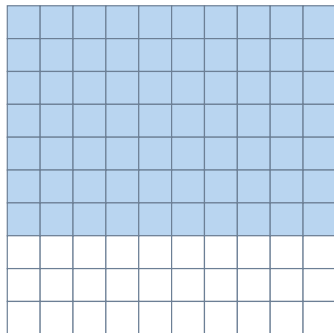
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 9 • BACKWARD 3 • Q4/5

What decimal does the shaded part of the grid show?

Decimal: \_\_\_\_\_



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 9 • BACKWARD 3 • Q5/5

320

QUESTION ID

**S9-B3-Q5**

TYPE

Short answer

CORRECT ANSWER

**0.70**

**GRADING NOTE** 0.7 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 9 • BACKWARD 3 • Q5/5



SET 9 • 5.NBT.A.1

# FORWARD 1

Divide multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

**ANSWER KEY**

Reduced copy of the question page, with its answer

Divide.

$$4.8 \div 0.4$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • FORWARD 1 • Q1/5

322

QUESTION ID

**S9-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**12**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • FORWARD 1 • Q1/5

322

**ANSWER KEY**

Reduced copy of the question page, with its answer

Divide.

$$9.36 \div 3$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • FORWARD 1 • Q2/5

323

QUESTION ID

**S9-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**3.12**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • FORWARD 1 • Q2/5

323

Divide.

$$0.56 \div 0.7$$

- A.** 80
- B.** 0.08
- C.** 8
- D.** 0.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • FORWARD 1 • Q3/5

324

QUESTION ID

**S9-F1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. 0.8**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • FORWARD 1 • Q3/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Divide.

$$15 \div 0.25$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • FORWARD 1 • Q4/5

325

QUESTION ID

**S9-F1-Q4**

TYPE

Short answer

CORRECT ANSWER

**60**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • FORWARD 1 • Q4/5

325

**True or false?**

$$6.3 \div 0.9 = 0.7$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • FORWARD 1 • Q5/5

326

QUESTION ID

**S9-F1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • FORWARD 1 • Q5/5

SET 9 • 5.NBT.A.1

# FORWARD 2

Write fractions as decimals using long division • Grade 7 • The Number System  
• 7.NS.A.2.d

Use long division to write  $\frac{7}{8}$  as a decimal.

Decimal: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 9 • FORWARD 2 • Q1/5

328

QUESTION ID

**S9-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**0.875**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 9 • FORWARD 2 • Q1/5



Use long division to write  $\frac{2}{9}$  as a decimal.

Decimal: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 9 • FORWARD 2 • Q2/5

329

QUESTION ID

**S9-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**0.222... (the 2 repeats)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 9 • FORWARD 2 • Q2/5

Which fraction has a terminating decimal?

A.  $\frac{5}{6}$

B.  $\frac{4}{7}$

C.  $\frac{1}{3}$

D.  $\frac{3}{20}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 9 • FORWARD 2 • Q3/5

330

QUESTION ID

**S9-F2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. 3/20**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 9 • FORWARD 2 • Q3/5

Use long division to write  $\frac{9}{20}$  as a decimal.

Decimal: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 9 • FORWARD 2 • Q4/5

331

QUESTION ID

**S9-F2-Q4**

TYPE

Short answer

CORRECT ANSWER

**0.45**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 9 • FORWARD 2 • Q4/5

**True or false?**

The decimal form of  $\frac{1}{11}$  repeats.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 9 • FORWARD 2 • Q5/5

332

QUESTION ID

**S9-F2-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 9 • FORWARD 2 • Q5/5

# Multiply and divide whole numbers by powers of 10; write powers of 10 with exponents

|                   |                                                                   |                   |
|-------------------|-------------------------------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 5 main question                                             | <b>5.NBT.A.2</b>  |
| <b>BACKWARD 1</b> | Multiplying and dividing by 10                                    | <b>4.NBT.A.1</b>  |
| <b>BACKWARD 2</b> | Multiplying by multiples of 10                                    | <b>3.NBT.A.3</b>  |
| <b>FORWARD 1</b>  | Write and evaluate whole-number exponents                         | <b>6.EE.A.1</b>   |
| <b>FORWARD 2</b>  | Multiply and divide negative numbers by powers of 10 — nearest... | <b>7.NS.A.2.c</b> |

SET 10 • 5.NBT.A.2

# MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •  
5.NBT.A.2

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$37 \times 1,000$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.2 • Set 10 • MAIN • Q1/5

335

QUESTION ID

**S10-M-Q1**

TYPE

Short answer

CORRECT ANSWER

**37,000**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.2 • Set 10 • MAIN • Q1/5

335

**ANSWER KEY**

Reduced copy of the question page, with its answer

Divide.

$$45,000 \div 100$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.2 • Set 10 • MAIN • Q2/5

336

QUESTION ID

**S10-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**450**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.2 • Set 10 • MAIN • Q2/5

336



Write  $10 \times 10 \times 10 \times 10$  as a power of 10.

Power of 10: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • MAIN • Q3/5

337

QUESTION ID

**S10-M-Q3**

TYPE

Short answer

CORRECT ANSWER

**$10^4$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • MAIN • Q3/5

$$6 \times 10^3 = ?$$

- A.** 60,000
- B.** 6,000
- C.** 18
- D.** 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q4

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.2 • Set 10 • MAIN • Q4/5

338

QUESTION ID

**S10-M-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. 6,000**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q4

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.2 • Set 10 • MAIN • Q4/5

**True or false?**

$$820 \div 10^2 = 82$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • MAIN • Q5/5

339

QUESTION ID

**S10-M-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • MAIN • Q5/5

SET 10 • 5.NBT.A.2

# BACKWARD 1

Multiplying and dividing by 10 • Grade 4 • Number & Operations in Base Ten •  
4.NBT.A.1

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$25 \times 10$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q1

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.A.1 • Set 10 • BACKWARD 1 • Q1/5

341

QUESTION ID

**S10-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**250**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q1

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.A.1 • Set 10 • BACKWARD 1 • Q1/5

**True or false?**

300 is 10 times as much as 30.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q2

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.A.1 • Set 10 • BACKWARD 1 • Q2/5

342

QUESTION ID

**S10-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q2

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.A.1 • Set 10 • BACKWARD 1 • Q2/5

Which number is 10 times as much as 450?

- A.** 4,500
- B.** 45,000
- C.** 460
- D.** 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 10 • BACKWARD 1 • Q3/5

343

QUESTION ID

**S10-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. 4,500**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 10 • BACKWARD 1 • Q3/5

How many tens are in 800?

Tens: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 10 • BACKWARD 1 • Q4/5

344

QUESTION ID

**S10-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**80**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 10 • BACKWARD 1 • Q4/5



**ANSWER KEY**

Reduced copy of the question page, with its answer

Divide.

$$7,000 \div 10$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.A.1 • Set 10 • BACKWARD 1 • Q5/5

345

QUESTION ID

**S10-B1-Q5**

TYPE

Short answer

CORRECT ANSWER

**700**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.A.1 • Set 10 • BACKWARD 1 • Q5/5

345

SET 10 • 5.NBT.A.2

# BACKWARD 2

Multiplying by multiples of 10 • Grade 3 • Number & Operations in Base Ten •  
3.NBT.A.3

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$4 \times 60$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q1

Grade 3 • Number &amp; Operations in Base Ten • 3.NBT.A.3 • Set 10 • BACKWARD 2 • Q1/5

347

QUESTION ID

**S10-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**240**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q1

Grade 3 • Number &amp; Operations in Base Ten • 3.NBT.A.3 • Set 10 • BACKWARD 2 • Q1/5

347

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$9 \times 80$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q2

Grade 3 • Number &amp; Operations in Base Ten • 3.NBT.A.3 • Set 10 • BACKWARD 2 • Q2/5

348

QUESTION ID

**S10-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**720**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q2

Grade 3 • Number &amp; Operations in Base Ten • 3.NBT.A.3 • Set 10 • BACKWARD 2 • Q2/5

348

**True or false?**

$$7 \times 30 = 2,100$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 10 • BACKWARD 2 • Q3/5

349

QUESTION ID

**S10-B2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 10 • BACKWARD 2 • Q3/5

$$5 \times 90 = ?$$

- A.** 4,500
- B.** 450
- C.** 45
- D.** 95

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 3 • Number &amp; Operations in Base Ten • 3.NBT.A.3 • Set 10 • BACKWARD 2 • Q4/5

350

QUESTION ID

**S10-B2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. 450**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 3 • Number &amp; Operations in Base Ten • 3.NBT.A.3 • Set 10 • BACKWARD 2 • Q4/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$8 \times 70$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 3 • Number &amp; Operations in Base Ten • 3.NBT.A.3 • Set 10 • BACKWARD 2 • Q5/5

351

QUESTION ID

**S10-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**560**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 3 • Number &amp; Operations in Base Ten • 3.NBT.A.3 • Set 10 • BACKWARD 2 • Q5/5

SET 10 • 5.NBT.A.2

# FORWARD 1

Write and evaluate whole-number exponents • Grade 6 • Expressions & Equations • 6.EE.A.1



Evaluate.

$3^4$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 10 • FORWARD 1 • Q1/5

353

QUESTION ID

**S10-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**81**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 10 • FORWARD 1 • Q1/5

Write  $5 \times 5 \times 5$  using an exponent.

Exponent form: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 10 • FORWARD 1 • Q2/5

354

QUESTION ID

**S10-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**$5^3$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 10 • FORWARD 1 • Q2/5

$$2^5 = ?$$

- A.** 25
- B.** 10
- C.** 64
- D.** 32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q3

Grade 6 • Expressions &amp; Equations • 6.EE.A.1 • Set 10 • FORWARD 1 • Q3/5

355

QUESTION ID

**S10-F1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. 32**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q3

Grade 6 • Expressions &amp; Equations • 6.EE.A.1 • Set 10 • FORWARD 1 • Q3/5

Evaluate.

$$10^3 + 2^2$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 10 • FORWARD 1 • Q4/5

356

QUESTION ID

**S10-F1-Q4**

TYPE

Short answer

CORRECT ANSWER

**1,004**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 10 • FORWARD 1 • Q4/5

**True or false?**

$$4^3 = 12$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 10 • FORWARD 1 • Q5/5

357

QUESTION ID

**S10-F1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 10 • FORWARD 1 • Q5/5

## FORWARD 2

Multiply and divide negative numbers by powers of 10 • Grade 7 • The Number System • 7.NS.A.2.c

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$-3.5 \times 100$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 10 • FORWARD 2 • Q1/5 • NEAREST RELATED

359

QUESTION ID

**S10-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**−350**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 10 • FORWARD 2 • Q1/5 • NEAREST RELATED

359

**ANSWER KEY**

Reduced copy of the question page, with its answer

Divide.

$$-420 \div 10$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 10 • FORWARD 2 • Q2/5 • NEAREST RELATED

360

QUESTION ID

**S10-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**-42**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 10 • FORWARD 2 • Q2/5 • NEAREST RELATED

360



$$-0.06 \times 1,000 = ?$$

- A.** -60
- B.** -0.006
- C.** -6
- D.** 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 10 • FORWARD 2 • Q3/5 • NEAREST RELATED

361

QUESTION ID

**S10-F2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. -60**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 10 • FORWARD 2 • Q3/5 • NEAREST RELATED

**ANSWER KEY**

Reduced copy of the question page, with its answer

Divide.

$$7.2 \div (-10)$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 10 • FORWARD 2 • Q4/5 • NEAREST RELATED

362

QUESTION ID

**S10-F2-Q4**

TYPE

Short answer

CORRECT ANSWER

**−0.72**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 10 • FORWARD 2 • Q4/5 • NEAREST RELATED

**True or false?**

$$(-10) \times (-10) \times (-10) = -1,000$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 10 • FORWARD 2 • Q5/5 • NEAREST RELATED

363

QUESTION ID

**S10-F2-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 10 • FORWARD 2 • Q5/5 • NEAREST RELATED

# Place the decimal point when multiplying or dividing a decimal by a power of 10

|                   |                                                                     |                   |
|-------------------|---------------------------------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 5 main question                                               | <b>5.NBT.A.2</b>  |
| <b>BACKWARD 1</b> | Decimal notation for fractions                                      | <b>4.NF.C.6</b>   |
| <b>BACKWARD 2</b> | Place value when multiplying by 10 or 100                           | <b>4.NBT.A.1</b>  |
| <b>FORWARD 1</b>  | Compute fluently with multi-digit decimals                          | <b>6.NS.B.3</b>   |
| <b>FORWARD 2</b>  | Powers of 10 with negative rational numbers — nearest related (n... | <b>7.NS.A.2.c</b> |

SET 11 • 5.NBT.A.2

# MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •  
5.NBT.A.2

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$3.82 \times 100$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q1

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.2 • Set 11 • MAIN • Q1/5

366

QUESTION ID

**S11-M-Q1**

TYPE

Short answer

CORRECT ANSWER

**382**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q1

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.2 • Set 11 • MAIN • Q1/5

366

**ANSWER KEY**

Reduced copy of the question page, with its answer

Divide.

$$56.4 \div 10$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.2 • Set 11 • MAIN • Q2/5

367

QUESTION ID

**S11-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**5.64**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.2 • Set 11 • MAIN • Q2/5

367

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$0.075 \times 10^2$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.2 • Set 11 • MAIN • Q3/5

368

QUESTION ID

**S11-M-Q3**

TYPE

Short answer

CORRECT ANSWER

**7.5**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.2 • Set 11 • MAIN • Q3/5

368



Divide.

$$2.9 \div 10^3$$

- A. 0.029
- B. 0.29
- C. 0.0029
- D. 2,900

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • MAIN • Q4/5

369

QUESTION ID

**S11-M-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. 0.0029**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • MAIN • Q4/5

**True or false?**

When you multiply a number by 1,000, each digit moves 3 places to the left.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q5

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.2 • Set 11 • MAIN • Q5/5

370

QUESTION ID

**S11-M-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q5

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.2 • Set 11 • MAIN • Q5/5

SET 11 • 5.NBT.A.2

# BACKWARD 1

Decimal notation for fractions • Grade 4 • Number & Operations–Fractions •  
4.NF.C.6

Write  $\frac{35}{100}$  as a decimal.

Decimal: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 11 • BACKWARD 1 • Q1/5

372

QUESTION ID

**S11-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**0.35**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 11 • BACKWARD 1 • Q1/5

**True or false?**

$$0.7 = \frac{7}{100}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 11 • BACKWARD 1 • Q2/5

373

QUESTION ID

**S11-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 11 • BACKWARD 1 • Q2/5

Which decimal is  $2\frac{4}{10}$ ?

- A. 24
- B. 2.4
- C. 0.24
- D. 2.04

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 11 • BACKWARD 1 • Q3/5

374

QUESTION ID

**S11-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. 2.4**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 11 • BACKWARD 1 • Q3/5

Write 0.09 as a fraction.

Fraction: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 11 • BACKWARD 1 • Q4/5

375

QUESTION ID

**S11-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**9/100**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 11 • BACKWARD 1 • Q4/5

**True or false?**

$$1.5 = 1\frac{5}{100}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 11 • BACKWARD 1 • Q5/5

376

QUESTION ID

**S11-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 11 • BACKWARD 1 • Q5/5



SET 11 • 5.NBT.A.2

# BACKWARD 2

Place value when multiplying by 10 or 100 • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1

**True or false?**

When a whole number is multiplied by 10, each digit moves one place to the left.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q1

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.A.1 • Set 11 • BACKWARD 2 • Q1/5

378

QUESTION ID

**S11-B2-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q1

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.A.1 • Set 11 • BACKWARD 2 • Q1/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$64 \times 100$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q2

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.A.1 • Set 11 • BACKWARD 2 • Q2/5

379

QUESTION ID

**S11-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**6,400**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q2

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.A.1 • Set 11 • BACKWARD 2 • Q2/5

$$3,500 \div 10 = ?$$

- A.** 350
- B.** 3,510
- C.** 35
- D.** 35,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.A.1 • Set 11 • BACKWARD 2 • Q3/5

380

QUESTION ID

**S11-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. 350**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.A.1 • Set 11 • BACKWARD 2 • Q3/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Divide.

$$9,000 \div 100$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.A.1 • Set 11 • BACKWARD 2 • Q4/5

381

QUESTION ID

**S11-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**90**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.A.1 • Set 11 • BACKWARD 2 • Q4/5

381

**True or false?**

$$500 \div 10 = 5$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 11 • BACKWARD 2 • Q5/5

382

QUESTION ID

**S11-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 11 • BACKWARD 2 • Q5/5

SET 11 • 5.NBT.A.2

# FORWARD 1

Compute fluently with multi-digit decimals • Grade 6 • The Number System •  
6.NS.B.3

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$2.4 \times 0.35$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 11 • FORWARD 1 • Q1/5

384

QUESTION ID

**S11-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**0.84**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 11 • FORWARD 1 • Q1/5



**ANSWER KEY**

Reduced copy of the question page, with its answer

Divide.

$$1.44 \div 0.12$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 11 • FORWARD 1 • Q2/5

385

QUESTION ID

**S11-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**12**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 11 • FORWARD 1 • Q2/5

385

Multiply.

$$12.5 \times 0.8$$

- A.** 100
- B.** 10
- C.** 0.1
- D.** 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 11 • FORWARD 1 • Q3/5

386

QUESTION ID

**S11-F1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. 10**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 11 • FORWARD 1 • Q3/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Add.

$$0.875 + 12.4$$

Sum: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 11 • FORWARD 1 • Q4/5

387

QUESTION ID

**S11-F1-Q4**

TYPE

Short answer

CORRECT ANSWER

**13.275**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 11 • FORWARD 1 • Q4/5

387

**True or false?**

$$4.5 \div 0.05 = 9$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 11 • FORWARD 1 • Q5/5

388

QUESTION ID

**S11-F1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 11 • FORWARD 1 • Q5/5

## FORWARD 2

Powers of 10 with negative rational numbers • Grade 7 • The Number System • 7.NS.A.2.c

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

**ANSWER KEY**

Reduced copy of the question page, with its answer

Find the value.

$$-0.45 \times 100$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 11 • FORWARD 2 • Q1/5 • NEAREST RELATED

390

QUESTION ID

**S11-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**-45**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 11 • FORWARD 2 • Q1/5 • NEAREST RELATED

**ANSWER KEY**

Reduced copy of the question page, with its answer

Find the value.

$$-63.5 \div 1,000$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 11 • FORWARD 2 • Q2/5 • NEAREST RELATED

391

QUESTION ID

**S11-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**−0.0635**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 11 • FORWARD 2 • Q2/5 • NEAREST RELATED

391

Find the value.

$$(-2.7) \times (-10)$$

- A.  $-0.27$
- B.  $-27$
- C.  $0.27$
- D.  $27$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 11 • FORWARD 2 • Q3/5 • NEAREST RELATED

392

QUESTION ID

**S11-F2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. 27**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 11 • FORWARD 2 • Q3/5 • NEAREST RELATED



Find the value.

$$0.8 \times (-100)$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 11 • FORWARD 2 • Q4/5 • NEAREST RELATED

393

QUESTION ID

**S11-F2-Q4**

TYPE

Short answer

CORRECT ANSWER

**−80**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 11 • FORWARD 2 • Q4/5 • NEAREST RELATED

**True or false?**

$$-5.2 \div 100 = -0.052$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 11 • FORWARD 2 • Q5/5 • NEAREST RELATED

394

QUESTION ID

**S11-F2-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 11 • FORWARD 2 • Q5/5 • NEAREST RELATED

# Read and write decimals to thousandths using numerals and words

|                   |                                                     |                    |
|-------------------|-----------------------------------------------------|--------------------|
| <b>MAIN</b>       | Grade 5 main question                               | <b>5.NBT.A.3.a</b> |
| <b>BACKWARD 1</b> | Decimals for tenths and hundredths                  | <b>4.NF.C.6</b>    |
| <b>BACKWARD 2</b> | Reading and writing whole numbers                   | <b>4.NBT.A.2</b>   |
| <b>BACKWARD 3</b> | Naming tenths and hundredths                        | <b>4.NF.C.5</b>    |
| <b>FORWARD 1</b>  | Find and position rational numbers on a number line | <b>6.NS.C.6.c</b>  |
| <b>FORWARD 2</b>  | Write fractions as decimals                         | <b>7.NS.A.2.d</b>  |

SET 12 • 5.NBT.A.3.a

# MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •  
5.NBT.A.3.a

Write the number in standard form.

four and twenty-six hundredths

Number: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 12 • MAIN • Q1/5

397

QUESTION ID

**S12-M-Q1**

TYPE

Short answer

CORRECT ANSWER

**4.26**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 12 • MAIN • Q1/5

Write 3.508 in words.

Words: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 12 • MAIN • Q2/5

398

QUESTION ID

**S12-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**three and five  
hundred eight  
thousandths**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 12 • MAIN • Q2/5

Which number is nine and forty-one thousandths?

- A. 9.410
- B. 9.41
- C. 9.0041
- D. 9.041

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 12 • MAIN • Q3/5

399

QUESTION ID

**S12-M-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. 9.041**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 12 • MAIN • Q3/5

Write the number in standard form.

seventeen and three tenths

Number: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 12 • MAIN • Q4/5

400

QUESTION ID

**S12-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**17.3**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 12 • MAIN • Q4/5



**True or false?**

0.072 is read "seventy-two hundredths."

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q5

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.3.a • Set 12 • MAIN • Q5/5

401

QUESTION ID

**S12-M-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q5

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.3.a • Set 12 • MAIN • Q5/5

SET 12 • 5.NBT.A.3.a

# BACKWARD 1

Decimals for tenths and hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write  $\frac{6}{100}$  as a decimal.

Decimal: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 12 • BACKWARD 1 • Q1/5

403

QUESTION ID

**S12-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**0.06**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 12 • BACKWARD 1 • Q1/5

**True or false?**

$$0.5 = \frac{5}{10}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 12 • BACKWARD 1 • Q2/5

404

QUESTION ID

**S12-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 12 • BACKWARD 1 • Q2/5

Which decimal is equal to  $\frac{12}{100}$ ?

- A. 1.2
- B. 12.0
- C. 0.12
- D. 0.012

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 12 • BACKWARD 1 • Q3/5

405

QUESTION ID

**S12-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. 0.12**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 12 • BACKWARD 1 • Q3/5

Write  $3\frac{7}{10}$  as a decimal.

Decimal: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 12 • BACKWARD 1 • Q4/5

406

QUESTION ID

**S12-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**3.7**

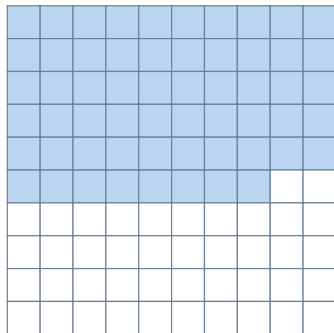
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 12 • BACKWARD 1 • Q4/5

What decimal does the shaded part of the grid show?

Decimal: \_\_\_\_\_



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 12 • BACKWARD 1 • Q5/5

407

QUESTION ID

**S12-B1-Q5**

TYPE

Short answer

CORRECT ANSWER

**0.58**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • 5.NBT.A.3.a

# BACKWARD 2

Reading and writing whole numbers • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2



Write the number in standard form.

two thousand, four hundred six

Number: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 12 • BACKWARD 2 • Q1/5

409

QUESTION ID

**S12-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**2,406**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 12 • BACKWARD 2 • Q1/5

**True or false?**

5,030 is read "five thousand three hundred."

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q2

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.A.2 • Set 12 • BACKWARD 2 • Q2/5

410

QUESTION ID

**S12-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q2

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.A.2 • Set 12 • BACKWARD 2 • Q2/5

Which number is eighty thousand, nine hundred?

- A.** 80,900
- B.** 8,900
- C.** 89,000
- D.** 80,090

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 12 • BACKWARD 2 • Q3/5

411

QUESTION ID

**S12-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. 80,900**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 12 • BACKWARD 2 • Q3/5

Write 607 in words.

Words: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 12 • BACKWARD 2 • Q4/5

412

QUESTION ID

**S12-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**six hundred seven**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 12 • BACKWARD 2 • Q4/5

Write the number in standard form.

twelve thousand, fifty

Number: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 12 • BACKWARD 2 • Q5/5

413

QUESTION ID

**S12-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**12,050**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 12 • BACKWARD 2 • Q5/5

SET 12 • 5.NBT.A.3.a

# BACKWARD 3

Naming tenths and hundredths • Grade 4 • Number & Operations–Fractions •  
4.NF.C.5

**True or false?**

Three tenths is equal to thirty hundredths.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q1

Grade 4 • Number &amp; Operations-Fractions • 4.NF.C.5 • Set 12 • BACKWARD 3 • Q1/5

415

QUESTION ID

**S12-B3-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q1

Grade 4 • Number &amp; Operations-Fractions • 4.NF.C.5 • Set 12 • BACKWARD 3 • Q1/5

How many hundredths are in 4 tenths?

Hundredths: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 12 • BACKWARD 3 • Q2/5

416

QUESTION ID

**S12-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**40**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 12 • BACKWARD 3 • Q2/5



Which fraction is "five hundredths"?

A.  $\frac{100}{5}$

B.  $\frac{5}{10}$

C.  $\frac{50}{100}$

D.  $\frac{5}{100}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 12 • BACKWARD 3 • Q3/5

417

QUESTION ID

**S12-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. 5/100**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 12 • BACKWARD 3 • Q3/5

Write "nine tenths" as a fraction.

Fraction: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 12 • BACKWARD 3 • Q4/5

418

QUESTION ID

**S12-B3-Q4**

TYPE

Short answer

CORRECT ANSWER

**9/10**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 12 • BACKWARD 3 • Q4/5

**True or false?**

$$\frac{60}{100} = \frac{6}{10}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 12 • BACKWARD 3 • Q5/5

419

QUESTION ID

**S12-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q5

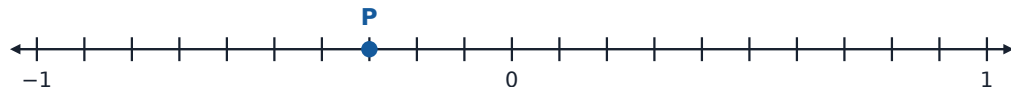
Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 12 • BACKWARD 3 • Q5/5

SET 12 • 5.NBT.A.3.a

# FORWARD 1

Find and position rational numbers on a number line • Grade 6 • The Number System • 6.NS.C.6.c

What number does point P represent?



P = \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 12 • FORWARD 1 • Q1/5

421

QUESTION ID

**S12-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**-0.3**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 12 • FORWARD 1 • Q1/5

Plot and label point Q at 0.45 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 12 • FORWARD 1 • Q2/5

422

QUESTION ID

**S12-F1-Q2**

TYPE

Drawing / plotting

CORRECT ANSWER

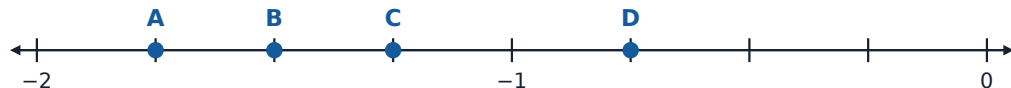
**Point Q plotted at 0.45, the fifth tick after 0.4 (halfway between 0.4 and 0.5), and labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 12 • FORWARD 1 • Q2/5

Which point is at  $-1.25$ ?



**A.** Point A

**B.** Point B

**C.** Point C

**D.** Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 12 • FORWARD 1 • Q3/5

423

QUESTION ID

**S12-F1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

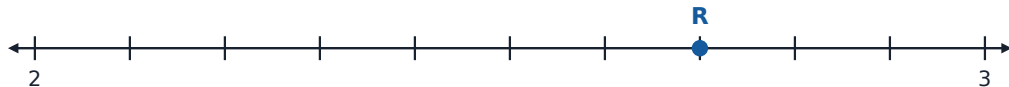
**C. Point C**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 12 • FORWARD 1 • Q3/5

What number does point R represent?



R = \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 12 • FORWARD 1 • Q4/5

424

QUESTION ID

**S12-F1-Q4**

TYPE

Short answer

CORRECT ANSWER

**2.7**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 12 • FORWARD 1 • Q4/5



**True or false?**

On a number line,  $-0.6$  is to the left of  $-0.5$ .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 12 • FORWARD 1 • Q5/5

425

QUESTION ID

**S12-F1-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 12 • FORWARD 1 • Q5/5

SET 12 • 5.NBT.A.3.a

# FORWARD 2

Write fractions as decimals • Grade 7 • The Number System • 7.NS.A.2.d

Use long division to write  $\frac{3}{8}$  as a decimal.

Decimal: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 12 • FORWARD 2 • Q1/5

427

QUESTION ID

**S12-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**0.375**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 12 • FORWARD 2 • Q1/5

Use long division to write  $\frac{5}{12}$  as a decimal.

Decimal: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 12 • FORWARD 2 • Q2/5

428

QUESTION ID

**S12-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**0.41666... (the 6 repeats)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 12 • FORWARD 2 • Q2/5

$$\frac{11}{20} = ?$$

- A.** 0.55
- B.** 5.5
- C.** 0.2
- D.** 0.11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 12 • FORWARD 2 • Q3/5

429

QUESTION ID

**S12-F2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. 0.55**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 12 • FORWARD 2 • Q3/5

Use long division to write  $\frac{4}{9}$  as a decimal.

Decimal: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 12 • FORWARD 2 • Q4/5

430

QUESTION ID

**S12-F2-Q4**

TYPE

Short answer

CORRECT ANSWER

**0.444... (the 4 repeats)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 12 • FORWARD 2 • Q4/5

**True or false?**

$$\frac{7}{25} = 0.32$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 12 • FORWARD 2 • Q5/5

431

QUESTION ID

**S12-F2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 12 • FORWARD 2 • Q5/5

# Write decimals to thousandths in expanded form

|                   |                                                                     |                    |
|-------------------|---------------------------------------------------------------------|--------------------|
| <b>MAIN</b>       | Grade 5 main question                                               | <b>5.NBT.A.3.a</b> |
| <b>BACKWARD 1</b> | Expanded form of whole numbers                                      | <b>4.NBT.A.2</b>   |
| <b>BACKWARD 2</b> | Decimals and fractions with denominators 10 and 100                 | <b>4.NF.C.6</b>    |
| <b>FORWARD 1</b>  | Write expanded form with exponents                                  | <b>6.EE.A.1</b>    |
| <b>FORWARD 2</b>  | Decimal expansions of fractions — nearest related (no direct Gra... | <b>7.NS.A.2.d</b>  |



SET 13 • 5.NBT.A.3.a

# MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •  
5.NBT.A.3.a

Write 352.68 in expanded form.

Expanded form: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 13 • MAIN • Q1/5

434

QUESTION ID

**S13-M-Q1**

TYPE

Short answer

CORRECT ANSWER

**$3 \times 100 + 5 \times 10 + 2 \times 1 + 6 \times 0.1 + 8 \times 0.01$**

**GRADING NOTE**  $300 + 50 + 2 + 0.6 + 0.08$  is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 13 • MAIN • Q1/5

Write the number in standard form.

$$4 \times 10 + 7 \times 1 + 3 \times \frac{1}{10} + 9 \times \frac{1}{1000}$$

Number: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 13 • MAIN • Q2/5

435

QUESTION ID

**S13-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**47.309**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 13 • MAIN • Q2/5

Which is the expanded form of 6.045?

- A.  $6 + 45$
- B.  $6 \times 10 + 4 \times \frac{1}{100} + 5 \times \frac{1}{1000}$
- C.  $6 \times 1 + 4 \times \frac{1}{100} + 5 \times \frac{1}{1000}$
- D.  $6 \times 1 + 4 \times \frac{1}{10} + 5 \times \frac{1}{100}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 13 • MAIN • Q3/5

436

QUESTION ID

**S13-M-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C.**

**$6 \times 1 + 4 \times \frac{1}{100} + 5 \times \frac{1}{1000}$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 13 • MAIN • Q3/5

Write the number in standard form.

$$8 \times 100 + 2 \times \frac{1}{10} + 6 \times \frac{1}{100}$$

Number: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 13 • MAIN • Q4/5

437

QUESTION ID

**S13-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**800.26**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 13 • MAIN • Q4/5

**True or false?**

$$0.908 = 9 \times \frac{1}{10} + 8 \times \frac{1}{1000}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 13 • MAIN • Q5/5

438

QUESTION ID

**S13-M-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 13 • MAIN • Q5/5

SET 13 • 5.NBT.A.3.a

# BACKWARD 1

Expanded form of whole numbers • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2

Write 4,371 in expanded form.

Expanded form: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 13 • BACKWARD 1 • Q1/5

440

QUESTION ID

**S13-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**4,000 + 300 + 70 + 1**

**GRADING NOTE**  $4 \times 1,000 + 3 \times 100 + 7 \times 10 + 1 \times 1$  is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 13 • BACKWARD 1 • Q1/5



**True or false?**

$$6,052 = 6,000 + 500 + 2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 13 • BACKWARD 1 • Q2/5

441

QUESTION ID

**S13-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 13 • BACKWARD 1 • Q2/5

Which number is  $7,000 + 400 + 9$ ?

- A.** 7,409
- B.** 7,490
- C.** 74,009
- D.** 7,049

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 13 • BACKWARD 1 • Q3/5

442

QUESTION ID

**S13-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. 7,409**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 13 • BACKWARD 1 • Q3/5

Write  $3 \times 1,000 + 8 \times 100 + 5 \times 1$  in standard form.

Number: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 13 • BACKWARD 1 • Q4/5

443

QUESTION ID

**S13-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**3,805**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 13 • BACKWARD 1 • Q4/5

**True or false?**

$$900 + 20 + 4 = 9,024$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 13 • BACKWARD 1 • Q5/5

444

QUESTION ID

**S13-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 13 • BACKWARD 1 • Q5/5

SET 13 • 5.NBT.A.3.a

# BACKWARD 2

Decimals and fractions with denominators 10 and 100 • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write 0.6 as a fraction.

Fraction: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 13 • BACKWARD 2 • Q1/5

446

QUESTION ID

**S13-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**6/10**

**GRADING NOTE** 3/5 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 13 • BACKWARD 2 • Q1/5

**True or false?**

$$\frac{4}{100} = 0.4$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 13 • BACKWARD 2 • Q2/5

447

QUESTION ID

**S13-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 13 • BACKWARD 2 • Q2/5

Which fraction is equal to 0.25?

A.  $\frac{1}{25}$

B.  $\frac{25}{100}$

C.  $\frac{25}{10}$

D.  $\frac{2}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 13 • BACKWARD 2 • Q3/5

448

QUESTION ID

**S13-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. 25/100**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 13 • BACKWARD 2 • Q3/5



Write  $2 + \frac{3}{10} + \frac{7}{100}$  as a decimal.

Decimal: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 13 • BACKWARD 2 • Q4/5

449

QUESTION ID

**S13-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**2.37**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 13 • BACKWARD 2 • Q4/5

**True or false?**

$$\frac{9}{10} = 0.09$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 13 • BACKWARD 2 • Q5/5

450

QUESTION ID

**S13-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 13 • BACKWARD 2 • Q5/5

SET 13 • 5.NBT.A.3.a

# FORWARD 1

Write expanded form with exponents • Grade 6 • Expressions & Equations •  
6.EE.A.1

Write  $5,000 + 300 + 20 + 7$  using powers of 10.

Expression: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • FORWARD 1 • Q1/5

452

QUESTION ID

**S13-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**$5 \times 10^3 + 3 \times 10^2 + 2 \times 10^1 + 7$**

**GRADING NOTE** Writing the last term as  $7 \times 10^0$  or  $7 \times 1$  is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • FORWARD 1 • Q1/5

Evaluate.

$$4 \times 10^3 + 6 \times 10^1$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • FORWARD 1 • Q2/5

453

QUESTION ID

**S13-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**4,060**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • FORWARD 1 • Q2/5

Which number is equal to  $10^4$ ?

- A. 40
- B. 100,000
- C. 1,000
- D. 10,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • FORWARD 1 • Q3/5

454

QUESTION ID

**S13-F1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. 10,000**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • FORWARD 1 • Q3/5

Write  $600 + 90 + 1$  using powers of 10.

Expression: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • FORWARD 1 • Q4/5

455

QUESTION ID

**S13-F1-Q4**

TYPE

Short answer

CORRECT ANSWER

**$6 \times 10^2 + 9 \times 10^1 + 1$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • FORWARD 1 • Q4/5

**True or false?**

$$2 \times 10^2 + 5 = 250$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • FORWARD 1 • Q5/5

456

QUESTION ID

**S13-F1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • FORWARD 1 • Q5/5



# FORWARD 2

Decimal expansions of fractions • Grade 7 • The Number System • 7.NS.A.2.d

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Write  $\frac{3}{4}$  as a decimal.

Decimal: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 13 • FORWARD 2 • Q1/5 • NEAREST RELATED

458

QUESTION ID

**S13-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**0.75**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 13 • FORWARD 2 • Q1/5 • NEAREST RELATED

Use long division to write  $\frac{1}{6}$  as a decimal.

Decimal: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 13 • FORWARD 2 • Q2/5 • NEAREST RELATED

459

QUESTION ID

**S13-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**0.1666... (the 6 repeats)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 13 • FORWARD 2 • Q2/5 • NEAREST RELATED

Which fraction is equal to 0.666... (the 6 repeats)?

A.  $\frac{3}{5}$

B.  $\frac{6}{10}$

C.  $\frac{2}{3}$

D.  $\frac{1}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 13 • FORWARD 2 • Q3/5 • NEAREST RELATED

460

QUESTION ID

**S13-F2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. 2/3**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 13 • FORWARD 2 • Q3/5 • NEAREST RELATED

Use long division to write  $\frac{5}{8}$  as a decimal.

Decimal: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 13 • FORWARD 2 • Q4/5 • NEAREST RELATED

461

QUESTION ID

**S13-F2-Q4**

TYPE

Short answer

CORRECT ANSWER

**0.625**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 13 • FORWARD 2 • Q4/5 • NEAREST RELATED

**True or false?**

$$\frac{2}{5} = 0.4$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 13 • FORWARD 2 • Q5/5 • NEAREST RELATED

462

QUESTION ID

**S13-F2-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 13 • FORWARD 2 • Q5/5 • NEAREST RELATED

# Compare decimals to thousandths

|                   |                                                         |                    |
|-------------------|---------------------------------------------------------|--------------------|
| <b>MAIN</b>       | Grade 5 main question                                   | <b>5.NBT.A.3.b</b> |
| <b>BACKWARD 1</b> | Comparing decimals to hundredths                        | <b>4.NF.C.7</b>    |
| <b>BACKWARD 2</b> | The symbols $>$ , $<$ , and $=$                         | <b>1.NBT.B.3</b>   |
| <b>FORWARD 1</b>  | Order rational numbers                                  | <b>6.NS.C.7.b</b>  |
| <b>FORWARD 2</b>  | Compare fractions and decimals using decimal expansions | <b>7.NS.A.2.d</b>  |

SET 14 • 5.NBT.A.3.b

# MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •  
5.NBT.A.3.b



Write  $>$ ,  $<$ , or  $=$  to compare.

0.47 \_\_\_\_ 0.405

Symbol: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 14 • MAIN • Q1/5

465

QUESTION ID

**S14-M-Q1**

TYPE

Short answer

CORRECT ANSWER

**$>$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 14 • MAIN • Q1/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Write  $>$ ,  $<$ , or  $=$  to compare.

3.06 \_\_\_\_ 3.060

Symbol: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.3.b • Set 14 • MAIN • Q2/5

466

QUESTION ID

**S14-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**=**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.3.b • Set 14 • MAIN • Q2/5

466

Which number is greatest?

- A.** 5.2
- B.** 5.19
- C.** 5.093
- D.** 5.199

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 14 • MAIN • Q3/5

467

QUESTION ID

**S14-M-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. 5.2**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 14 • MAIN • Q3/5

Order from least to greatest.

2.31, 2.301, 2.13, 2.3

Order: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 14 • MAIN • Q4/5

468

QUESTION ID

**S14-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**2.13, 2.3, 2.301,  
2.31**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 14 • MAIN • Q4/5

**True or false?**

$$0.8 < 0.79$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 14 • MAIN • Q5/5

469

QUESTION ID

**S14-M-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 14 • MAIN • Q5/5

SET 14 • 5.NBT.A.3.b

# BACKWARD 1

Comparing decimals to hundredths • Grade 4 • Number & Operations–Fractions  
• 4.NF.C.7

Write  $>$ ,  $<$ , or  $=$  to compare.

0.6 \_\_\_\_ 0.58

Symbol: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.7 • Set 14 • BACKWARD 1 • Q1/5

471

QUESTION ID

**S14-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**$>$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.7 • Set 14 • BACKWARD 1 • Q1/5

**True or false?**

$$0.3 = 0.03$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.7 • Set 14 • BACKWARD 1 • Q2/5

472

QUESTION ID

**S14-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.7 • Set 14 • BACKWARD 1 • Q2/5



Which number is less?

**A.** 0.09

**B.** 0.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.7 • Set 14 • BACKWARD 1 • Q3/5

473

QUESTION ID

**S14-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. 0.09**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.7 • Set 14 • BACKWARD 1 • Q3/5

Write  $>$ ,  $<$ , or  $=$  to compare.

0.75 \_\_\_\_ 0.8

Symbol: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.7 • Set 14 • BACKWARD 1 • Q4/5

474

QUESTION ID

**S14-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

$<$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.7 • Set 14 • BACKWARD 1 • Q4/5

**True or false?**

$0.42 > 0.5$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.7 • Set 14 • BACKWARD 1 • Q5/5

475

QUESTION ID

**S14-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.7 • Set 14 • BACKWARD 1 • Q5/5

SET 14 • 5.NBT.A.3.b

# BACKWARD 2

The symbols  $>$ ,  $<$ , and  $=$  • Grade 1 • Number & Operations in Base Ten •  
1.NBT.B.3

Which symbol means "is less than"?

**A.**  $<$

**B.**  $>$

**C.**  $=$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q1

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 14 • BACKWARD 2 • Q1/5

477

QUESTION ID

**S14-B2-Q1**

TYPE

Multiple choice

CORRECT ANSWER

**A.**  $<$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q1

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 14 • BACKWARD 2 • Q1/5

**True or false?**

$$9 > 6$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q2

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 14 • BACKWARD 2 • Q2/5

478

QUESTION ID

**S14-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q2

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 14 • BACKWARD 2 • Q2/5

Write  $>$ ,  $<$ , or  $=$  to compare.

52 \_\_\_\_ 25

Symbol: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q3

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 14 • BACKWARD 2 • Q3/5

479

QUESTION ID

**S14-B2-Q3**

TYPE

Short answer

CORRECT ANSWER

**$>$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q3

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 14 • BACKWARD 2 • Q3/5

**True or false?**

The symbol  $>$  means "is less than."

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q4

Grade 1 • Number &amp; Operations in Base Ten • 1.NBT.B.3 • Set 14 • BACKWARD 2 • Q4/5

480

QUESTION ID

**S14-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q4

Grade 1 • Number &amp; Operations in Base Ten • 1.NBT.B.3 • Set 14 • BACKWARD 2 • Q4/5



Which statement is true?

- A.**  $14 < 41$
- B.**  $14 = 41$
- C.**  $14 > 41$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 14 • BACKWARD 2 • Q5/5

481

QUESTION ID

**S14-B2-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**A.  $14 < 41$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 14 • BACKWARD 2 • Q5/5

SET 14 • 5.NBT.A.3.b

# FORWARD 1

Order rational numbers • Grade 6 • The Number System • 6.NS.C.7.b

Write  $>$ ,  $<$ , or  $=$  to compare.

$-0.4$  \_\_\_\_  $-0.45$

Symbol: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q1

Grade 6 • The Number System • 6.NS.C.7.b • Set 14 • FORWARD 1 • Q1/5

483

QUESTION ID

**S14-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**$>$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q1

Grade 6 • The Number System • 6.NS.C.7.b • Set 14 • FORWARD 1 • Q1/5

Which number is least?

- A.  $-2$
- B.  $-2.5$
- C.  $0.5$
- D.  $-2.05$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q2

Grade 6 • The Number System • 6.NS.C.7.b • Set 14 • FORWARD 1 • Q2/5

484

QUESTION ID

**S14-F1-Q2**

TYPE

Multiple choice

CORRECT ANSWER

**B.  $-2.5$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q2

Grade 6 • The Number System • 6.NS.C.7.b • Set 14 • FORWARD 1 • Q2/5

Order from least to greatest.

$-1.2$ ,  $0.3$ ,  $-1.25$ ,  $0$

Order: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q3

Grade 6 • The Number System • 6.NS.C.7.b • Set 14 • FORWARD 1 • Q3/5

485

QUESTION ID

**S14-F1-Q3**

TYPE

Short answer

CORRECT ANSWER

**$-1.25$ ,  $-1.2$ ,  $0$ ,  $0.3$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q3

Grade 6 • The Number System • 6.NS.C.7.b • Set 14 • FORWARD 1 • Q3/5

**True or false?**

$$-3.7 > -3.6$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q4

Grade 6 • The Number System • 6.NS.C.7.b • Set 14 • FORWARD 1 • Q4/5

486

QUESTION ID

**S14-F1-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q4

Grade 6 • The Number System • 6.NS.C.7.b • Set 14 • FORWARD 1 • Q4/5

A temperature of  $-2.5^{\circ}\text{C}$  is colder than a temperature of  $-1.8^{\circ}\text{C}$ .

Write an inequality to show this.

Inequality: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q5

Grade 6 • The Number System • 6.NS.C.7.b • Set 14 • FORWARD 1 • Q5/5

487

QUESTION ID

**S14-F1-Q5**

TYPE

Short answer

CORRECT ANSWER

**$-2.5 < -1.8$**

**GRADING NOTE**  $-1.8 > -2.5$  is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q5

Grade 6 • The Number System • 6.NS.C.7.b • Set 14 • FORWARD 1 • Q5/5

SET 14 • 5.NBT.A.3.b

# FORWARD 2

Compare fractions and decimals using decimal expansions • Grade 7 • The Number System • 7.NS.A.2.d



Write  $\frac{5}{8}$  as a decimal.

Is  $\frac{5}{8}$  greater than or less than 0.6?

Decimal: \_\_\_\_\_

Answer: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 14 • FORWARD 2 • Q1/5

489

QUESTION ID

**S14-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**0.625; greater than  
0.6**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 14 • FORWARD 2 • Q1/5

Which number is greatest?

- A.  $\frac{4}{5}$
- B. 0.7
- C.  $\frac{3}{4}$
- D. 0.78

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 14 • FORWARD 2 • Q2/5

490

QUESTION ID

**S14-F2-Q2**

TYPE

Multiple choice

CORRECT ANSWER

**A. 4/5**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 14 • FORWARD 2 • Q2/5

Order from least to greatest.

$\frac{2}{3}$ , 0.6,  $\frac{5}{8}$

Order: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 14 • FORWARD 2 • Q3/5

491

QUESTION ID

**S14-F2-Q3**

TYPE

Short answer

CORRECT ANSWER

**0.6, 5/8, 2/3**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 14 • FORWARD 2 • Q3/5

**True or false?**

$$\frac{1}{3} > 0.33$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 14 • FORWARD 2 • Q4/5

492

QUESTION ID

**S14-F2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 14 • FORWARD 2 • Q4/5

Write  $>$ ,  $<$ , or  $=$  to compare.

$$\frac{7}{20} \text{ \_\_\_\_ } 0.4$$

Symbol: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 14 • FORWARD 2 • Q5/5

493

QUESTION ID

**S14-F2-Q5**

TYPE

Short answer

CORRECT ANSWER

$<$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 14 • FORWARD 2 • Q5/5

# Round decimals to any place

## MAIN

Grade 5 main question

**5.NBT.A.4**

## BACKWARD 1

Rounding whole numbers

**4.NBT.A.3**

## BACKWARD 2

Decimals on a number line

**4.NF.C.6**

## FORWARD 1

Estimate to check computations with decimals

**6.NS.B.3**

## FORWARD 2

Use estimation to judge whether an answer is reasonable

**7.EE.B.3**

SET 15 • 5.NBT.A.4

# MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •  
5.NBT.A.4

Round 6.483 to the nearest tenth.

Rounded: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 15 • MAIN • Q1/5

496

QUESTION ID

**S15-M-Q1**

TYPE

Short answer

CORRECT ANSWER

**6.5**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 15 • MAIN • Q1/5



Round 12.749 to the nearest hundredth.

Rounded: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 15 • MAIN • Q2/5

497

QUESTION ID

**S15-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**12.75**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 15 • MAIN • Q2/5

Round 0.865 to the nearest whole number.

- A.** 0.9
- B.** 1
- C.** 0
- D.** 0.87

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 15 • MAIN • Q3/5

498

QUESTION ID

**S15-M-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. 1**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 15 • MAIN • Q3/5

Round 39.96 to the nearest tenth.

Rounded: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 15 • MAIN • Q4/5

499

QUESTION ID

**S15-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**40.0**

**GRADING NOTE** 40 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 15 • MAIN • Q4/5

**True or false?**

4.351 rounded to the nearest hundredth is 4.36.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.4 • Set 15 • MAIN • Q5/5

500

QUESTION ID

**S15-M-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.4 • Set 15 • MAIN • Q5/5

SET 15 • 5.NBT.A.4

# BACKWARD 1

Rounding whole numbers • Grade 4 • Number & Operations in Base Ten •  
4.NBT.A.3

Round 4,682 to the nearest hundred.

Rounded: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 15 • BACKWARD 1 • Q1/5

502

QUESTION ID

**S15-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**4,700**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 15 • BACKWARD 1 • Q1/5

**True or false?**

350 rounded to the nearest hundred is 300.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q2

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.A.3 • Set 15 • BACKWARD 1 • Q2/5

503

QUESTION ID

**S15-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q2

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.A.3 • Set 15 • BACKWARD 1 • Q2/5

Round 7,249 to the nearest thousand.

- A.** 7,200
- B.** 7,000
- C.** 7,250
- D.** 8,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 15 • BACKWARD 1 • Q3/5

504

QUESTION ID

**S15-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. 7,000**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 15 • BACKWARD 1 • Q3/5



Round 85 to the nearest ten.

Rounded: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 15 • BACKWARD 1 • Q4/5

505

QUESTION ID

**S15-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**90**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 15 • BACKWARD 1 • Q4/5

**True or false?**

1,349 rounded to the nearest hundred is 1,400.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.A.3 • Set 15 • BACKWARD 1 • Q5/5

506

QUESTION ID

**S15-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.A.3 • Set 15 • BACKWARD 1 • Q5/5

SET 15 • 5.NBT.A.4

# BACKWARD 2

Decimals on a number line • Grade 4 • Number & Operations–Fractions •  
4.NF.C.6

What decimal is at point A?



A = \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 15 • BACKWARD 2 • Q1/5

508

QUESTION ID

**S15-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**0.6**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 15 • BACKWARD 2 • Q1/5

**True or false?**

0.7 is closer to 1 than to 0.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 4 • Number &amp; Operations-Fractions • 4.NF.C.6 • Set 15 • BACKWARD 2 • Q2/5

509

QUESTION ID

**S15-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 4 • Number &amp; Operations-Fractions • 4.NF.C.6 • Set 15 • BACKWARD 2 • Q2/5

Is 0.43 closer to 0.4 or to 0.5?

**A.** 0.4

**B.** 0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 15 • BACKWARD 2 • Q3/5

510

QUESTION ID

**S15-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

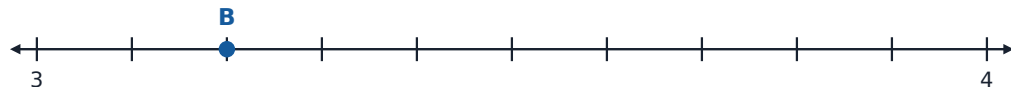
**A. 0.4**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 15 • BACKWARD 2 • Q3/5

What decimal is at point B?



B = \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 15 • BACKWARD 2 • Q4/5

511

QUESTION ID

**S15-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**3.2**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 15 • BACKWARD 2 • Q4/5

**True or false?**

2.5 is halfway between 2 and 3.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 4 • Number &amp; Operations-Fractions • 4.NF.C.6 • Set 15 • BACKWARD 2 • Q5/5

512

QUESTION ID

**S15-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 4 • Number &amp; Operations-Fractions • 4.NF.C.6 • Set 15 • BACKWARD 2 • Q5/5



SET 15 • 5.NBT.A.4

# FORWARD 1

Estimate to check computations with decimals • Grade 6 • The Number System  
• 6.NS.B.3

Round each number to the nearest whole number. Then add to estimate.

$$18.7 + 6.25$$

Estimate: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 1 • Q1/5

514

QUESTION ID

**S15-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**25**

GRADING NOTE  $19 + 6 = 25$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 1 • Q1/5

Which is the best estimate of  $4.9 \times 6.1$ ?

- A.** 36
- B.** 30
- C.** 24
- D.** 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 1 • Q2/5

515

QUESTION ID

**S15-F1-Q2**

TYPE

Multiple choice

CORRECT ANSWER

**B. 30**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 1 • Q2/5

Estimate  $31.8 \div 7.9$  by rounding each number to the nearest whole number.

Estimate: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 1 • Q3/5

516

QUESTION ID

**S15-F1-Q3**

TYPE

Short answer

CORRECT ANSWER

**4**

GRADING NOTE  $32 \div 8 = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 1 • Q3/5

**True or false?**

29.6 – 10.2 is about 40.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 1 • Q4/5

517

QUESTION ID

**S15-F1-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 1 • Q4/5

Kim says  $3.9 \times 2.1 = 81.9$ .

Use an estimate to decide whether her answer is reasonable. Write yes or no.

Answer: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 1 • Q5/5

518

QUESTION ID

**S15-F1-Q5**

TYPE

Short answer

CORRECT ANSWER

**No**

**GRADING NOTE**  $3.9 \times 2.1$  is about  $4 \times 2 = 8$ , so 81.9 is far too large (the exact product is 8.19).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 1 • Q5/5

SET 15 • 5.NBT.A.4

# FORWARD 2

Use estimation to judge whether an answer is reasonable • Grade 7 •  
Expressions & Equations • 7.EE.B.3

A \$24.75 shirt is 20% off.

Estimate the sale price.

Estimate: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 15 • FORWARD 2 • Q1/5

520

QUESTION ID

**S15-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**About \$20**

**GRADING NOTE** The exact sale price is \$19.80; any estimate from \$19 to \$21 is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 15 • FORWARD 2 • Q1/5



A worker earns \$19.85 per hour for 9.75 hours.

Which is the best estimate of her pay?

- A. \$20
- B. \$2,000
- C. \$200
- D. \$100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 15 • FORWARD 2 • Q2/5

521

QUESTION ID

**S15-F2-Q2**

TYPE

Multiple choice

CORRECT ANSWER

**C. \$200**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 15 • FORWARD 2 • Q2/5

**True or false?**

A 6.2% tax on \$49.50 is about \$3.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q3

Grade 7 • Expressions &amp; Equations • 7.EE.B.3 • Set 15 • FORWARD 2 • Q3/5

522

QUESTION ID

**S15-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q3

Grade 7 • Expressions &amp; Equations • 7.EE.B.3 • Set 15 • FORWARD 2 • Q3/5

Nina says  $\frac{3}{4} \times 39.8$  is about 40.

Is her estimate reasonable? Give a better estimate.

Reasonable? \_\_\_\_\_

Better estimate: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 15 • FORWARD 2 • Q4/5

523

QUESTION ID

**S15-F2-Q4**

TYPE

Short answer

CORRECT ANSWER

**No; about 30**

**GRADING NOTE**  $\frac{3}{4}$  of 40 is 30. The exact value is 29.85.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 15 • FORWARD 2 • Q4/5

A board is  $8\frac{1}{4}$  feet long. It is cut into pieces that are  $1\frac{7}{8}$  feet long.

Estimate how many whole pieces can be cut.

Estimate: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 15 • FORWARD 2 • Q5/5

524

QUESTION ID

**S15-F2-Q5**

TYPE

Short answer

CORRECT ANSWER

**4 pieces**

**GRADING NOTE** About  $8 \div 2 = 4$ ; the exact quotient is 4.4, so 4 whole pieces.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 15 • FORWARD 2 • Q5/5

# Multiply multi-digit whole numbers using the standard algorithm

|                   |                                                |                   |
|-------------------|------------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 5 main question                          | <b>5.NBT.B.5</b>  |
| <b>BACKWARD 1</b> | Multiplying by a one-digit or two-digit number | <b>4.NBT.B.5</b>  |
| <b>BACKWARD 2</b> | Multiplication facts                           | <b>3.OA.C.7</b>   |
| <b>BACKWARD 3</b> | Adding partial products                        | <b>4.NBT.B.4</b>  |
| <b>FORWARD 1</b>  | Multiply multi-digit decimals                  | <b>6.NS.B.3</b>   |
| <b>FORWARD 2</b>  | Multiply integers                              | <b>7.NS.A.2.c</b> |

SET 16 • 5.NBT.B.5

# MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •  
5.NBT.B.5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$246 \times 37$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.5 • Set 16 • MAIN • Q1/5

527

QUESTION ID

**S16-M-Q1**

TYPE

Short answer

CORRECT ANSWER

**9,102**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.5 • Set 16 • MAIN • Q1/5

527

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$1,385 \times 24$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.5 • Set 16 • MAIN • Q2/5

528

QUESTION ID

**S16-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**33,240**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.5 • Set 16 • MAIN • Q2/5

528



Multiply.

$$508 \times 63$$

- A. 3,556
- B. 32,104
- C. 32,004
- D. 31,904

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 16 • MAIN • Q3/5

529

QUESTION ID

**S16-M-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. 32,004**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 16 • MAIN • Q3/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$79 \times 86$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q4

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.5 • Set 16 • MAIN • Q4/5

530

QUESTION ID

**S16-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**6,794**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q4

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.5 • Set 16 • MAIN • Q4/5

530

A theater has 48 rows with 125 seats in each row.

How many seats are in the theater?

Seats: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 16 • MAIN • Q5/5

531

QUESTION ID

**S16-M-Q5**

TYPE

Short answer

CORRECT ANSWER

**6,000**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 16 • MAIN • Q5/5

SET 16 • 5.NBT.B.5

# BACKWARD 1

Multiplying by a one-digit or two-digit number • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$346 \times 7$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q1

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.5 • Set 16 • BACKWARD 1 • Q1/5

533

QUESTION ID

**S16-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**2,422**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q1

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.5 • Set 16 • BACKWARD 1 • Q1/5

533

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$23 \times 45$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q2

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.5 • Set 16 • BACKWARD 1 • Q2/5

534

QUESTION ID

**S16-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**1,035**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q2

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.5 • Set 16 • BACKWARD 1 • Q2/5

**True or false?**

$$1,203 \times 4 = 4,082$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 16 • BACKWARD 1 • Q3/5

535

QUESTION ID

**S16-B1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 16 • BACKWARD 1 • Q3/5

$$62 \times 18 = ?$$

- A.** 1,126
- B.** 496
- C.** 1,116
- D.** 1,006

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.5 • Set 16 • BACKWARD 1 • Q4/5

536

QUESTION ID

**S16-B1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. 1,116**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.5 • Set 16 • BACKWARD 1 • Q4/5



**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$58 \times 30$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.5 • Set 16 • BACKWARD 1 • Q5/5

537

QUESTION ID

**S16-B1-Q5**

TYPE

Short answer

CORRECT ANSWER

**1,740**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.5 • Set 16 • BACKWARD 1 • Q5/5

537

SET 16 • 5.NBT.B.5

# BACKWARD 2

Multiplication facts • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Multiply.

$$7 \times 6$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 16 • BACKWARD 2 • Q1/5

539

QUESTION ID

**S16-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**42**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 16 • BACKWARD 2 • Q1/5

Multiply.

$$8 \times 9$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 16 • BACKWARD 2 • Q2/5

540

QUESTION ID

**S16-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**72**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 16 • BACKWARD 2 • Q2/5

**True or false?**

$$6 \times 9 = 56$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 16 • BACKWARD 2 • Q3/5

541

QUESTION ID

**S16-B2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 16 • BACKWARD 2 • Q3/5

$$4 \times 8 = ?$$

- A.** 24
- B.** 36
- C.** 32
- D.** 28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.C.7 • Set 16 • BACKWARD 2 • Q4/5

542

QUESTION ID

**S16-B2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. 32**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.C.7 • Set 16 • BACKWARD 2 • Q4/5

Multiply.

$$9 \times 9$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 16 • BACKWARD 2 • Q5/5

543

QUESTION ID

**S16-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**81**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • 5.NBT.B.5

# BACKWARD 3

Adding partial products • Grade 4 • Number & Operations in Base Ten •  
4.NBT.B.4



Add.

$$7,380 + 1,722$$

Sum: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 3 • Q1/5

545

QUESTION ID

**S16-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**9,102**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 3 • Q1/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Add.

$$27,700 + 5,540$$

Sum: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q2

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 3 • Q2/5

546

QUESTION ID

**S16-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**33,240**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q2

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 3 • Q2/5

546

**True or false?**

$$30,480 + 1,524 = 31,904$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 3 • Q3/5

547

QUESTION ID

**S16-B3-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 3 • Q3/5

$$6,400 + 394 = ?$$

- A.** 6,784
- B.** 6,794
- C.** 7,794
- D.** 6,694

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q4

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 3 • Q4/5

548

QUESTION ID

**S16-B3-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. 6,794**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q4

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 3 • Q4/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Add.

$$4,800 + 1,200$$

Sum: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q5

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 3 • Q5/5

549

QUESTION ID

**S16-B3-Q5**

TYPE

Short answer

CORRECT ANSWER

**6,000**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q5

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 3 • Q5/5

549

SET 16 • 5.NBT.B.5

# FORWARD 1

Multiply multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Multiply.

$$2.46 \times 3.7$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • FORWARD 1 • Q1/5

551

QUESTION ID

**S16-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**9.102**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • FORWARD 1 • Q1/5

Multiply.

$$13.85 \times 2.4$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • FORWARD 1 • Q2/5

552

QUESTION ID

**S16-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**33.24**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • FORWARD 1 • Q2/5



Multiply.

$$5.08 \times 6.3$$

- A.** 31.904
- B.** 320.04
- C.** 3.2004
- D.** 32.004

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • FORWARD 1 • Q3/5

553

QUESTION ID

**S16-F1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. 32.004**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • FORWARD 1 • Q3/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$0.79 \times 8.6$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • FORWARD 1 • Q4/5

554

QUESTION ID

**S16-F1-Q4**

TYPE

Short answer

CORRECT ANSWER

**6.794**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • FORWARD 1 • Q4/5

554

**True or false?**

$$1.25 \times 4.8 = 60$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • FORWARD 1 • Q5/5

555

QUESTION ID

**S16-F1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • FORWARD 1 • Q5/5

SET 16 • 5.NBT.B.5

# FORWARD 2

Multiply integers • Grade 7 • The Number System • 7.NS.A.2.c

Find the product.

$$(-24) \times 15$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 16 • FORWARD 2 • Q1/5

557

QUESTION ID

**S16-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**−360**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 16 • FORWARD 2 • Q1/5

Find the product.

$$(-12) \times (-35)$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 16 • FORWARD 2 • Q2/5

558

QUESTION ID

**S16-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**420**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 16 • FORWARD 2 • Q2/5

Find the product.

$$18 \times (-21)$$

- A. 378
- B. -378
- C. -39
- D. -388

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 16 • FORWARD 2 • Q3/5

559

QUESTION ID

**S16-F2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. -378**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 16 • FORWARD 2 • Q3/5

Find the product.

$$(-3) \times (-4) \times (-25)$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 16 • FORWARD 2 • Q4/5

560

QUESTION ID

**S16-F2-Q4**

TYPE

Short answer

CORRECT ANSWER

**−300**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 16 • FORWARD 2 • Q4/5



**True or false?**

$$(-6)(-7) = -42$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 16 • FORWARD 2 • Q5/5

561

QUESTION ID

**S16-F2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 16 • FORWARD 2 • Q5/5

# Divide with up to four-digit dividends and two-digit divisors

|                   |                                                         |                   |
|-------------------|---------------------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 5 main question                                   | <b>5.NBT.B.6</b>  |
| <b>BACKWARD 1</b> | Dividing by a one-digit number                          | <b>4.NBT.B.6</b>  |
| <b>BACKWARD 2</b> | Multiplying to check a quotient                         | <b>4.NBT.B.5</b>  |
| <b>BACKWARD 3</b> | Division facts                                          | <b>3.OA.C.7</b>   |
| <b>FORWARD 1</b>  | Divide multi-digit numbers using the standard algorithm | <b>6.NS.B.2</b>   |
| <b>FORWARD 2</b>  | Divide integers                                         | <b>7.NS.A.2.b</b> |

SET 17 • 5.NBT.B.6

# MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •  
5.NBT.B.6

**ANSWER KEY**

Reduced copy of the question page, with its answer

Divide.

$$2,016 \div 24$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.6 • Set 17 • MAIN • Q1/5

564

QUESTION ID

**S17-M-Q1**

TYPE

Short answer

CORRECT ANSWER

**84**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.6 • Set 17 • MAIN • Q1/5

564

**ANSWER KEY**

Reduced copy of the question page, with its answer

Divide.

$$7,830 \div 45$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.6 • Set 17 • MAIN • Q2/5

565

QUESTION ID

**S17-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**174**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.6 • Set 17 • MAIN • Q2/5

565

Divide.

$$952 \div 17$$

- A.** 54
- B.** 46
- C.** 56
- D.** 57

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 17 • MAIN • Q3/5

566

QUESTION ID

**S17-M-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. 56**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 17 • MAIN • Q3/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Divide.

$$3,724 \div 38$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.6 • Set 17 • MAIN • Q4/5

567

QUESTION ID

**S17-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**98**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.6 • Set 17 • MAIN • Q4/5

567

Divide. Write the quotient and the remainder.

$$1,000 \div 32$$

Quotient and remainder: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 17 • MAIN • Q5/5

568

QUESTION ID

**S17-M-Q5**

TYPE

Short answer

CORRECT ANSWER

**31 R 8**

**GRADING NOTE** 31  $\frac{8}{32}$  = 31  $\frac{1}{4}$  is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 17 • MAIN • Q5/5



SET 17 • 5.NBT.B.6

# BACKWARD 1

Dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten •  
4.NBT.B.6

Divide.

$$852 \div 6$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 17 • BACKWARD 1 • Q1/5

570

QUESTION ID

**S17-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**142**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 17 • BACKWARD 1 • Q1/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Divide.

$$1,935 \div 5$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.6 • Set 17 • BACKWARD 1 • Q2/5

571

QUESTION ID

**S17-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**387**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.6 • Set 17 • BACKWARD 1 • Q2/5

571

**True or false?**

$$728 \div 8 = 901$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 17 • BACKWARD 1 • Q3/5

572

QUESTION ID

**S17-B1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 17 • BACKWARD 1 • Q3/5

$$504 \div 7 = ?$$

- A.** 82
- B.** 71
- C.** 62
- D.** 72

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.6 • Set 17 • BACKWARD 1 • Q4/5

573

QUESTION ID

**S17-B1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. 72**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.6 • Set 17 • BACKWARD 1 • Q4/5

Divide. Write the quotient and the remainder.

$$95 \div 4$$

Quotient and remainder: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 17 • BACKWARD 1 • Q5/5

574

QUESTION ID

**S17-B1-Q5**

TYPE

Short answer

CORRECT ANSWER

**23 R 3**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 17 • BACKWARD 1 • Q5/5

SET 17 • 5.NBT.B.6

# BACKWARD 2

Multiplying to check a quotient • Grade 4 • Number & Operations in Base Ten •  
4.NBT.B.5

**True or false?**

To check  $2,016 \div 24 = 84$ , you can multiply  $24 \times 84$ .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q1

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.5 • Set 17 • BACKWARD 2 • Q1/5

576

QUESTION ID

**S17-B2-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q1

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.5 • Set 17 • BACKWARD 2 • Q1/5



**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$45 \times 174$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q2

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.5 • Set 17 • BACKWARD 2 • Q2/5

577

QUESTION ID

**S17-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**7,830**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q2

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.5 • Set 17 • BACKWARD 2 • Q2/5

577

Which product checks  $952 \div 17 = 56$ ?

- A.  $17 + 56$
- B.  $17 \times 56$
- C.  $56 - 17$
- D.  $952 \times 17$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 17 • BACKWARD 2 • Q3/5

578

QUESTION ID

**S17-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B.  $17 \times 56$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 17 • BACKWARD 2 • Q3/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$38 \times 98$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.5 • Set 17 • BACKWARD 2 • Q4/5

579

QUESTION ID

**S17-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**3,724**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.5 • Set 17 • BACKWARD 2 • Q4/5

579

**True or false?**

$$32 \times 31 + 8 = 1,000$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 17 • BACKWARD 2 • Q5/5

580

QUESTION ID

**S17-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 17 • BACKWARD 2 • Q5/5

SET 17 • 5.NBT.B.6

# BACKWARD 3

Division facts • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Divide.

$$72 \div 8$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 17 • BACKWARD 3 • Q1/5

582

QUESTION ID

**S17-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**9**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 17 • BACKWARD 3 • Q1/5

Divide.

$$54 \div 9$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 17 • BACKWARD 3 • Q2/5

583

QUESTION ID

**S17-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**6**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 17 • BACKWARD 3 • Q2/5

**True or false?**

$$63 \div 7 = 8$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 17 • BACKWARD 3 • Q3/5

584

QUESTION ID

**S17-B3-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 17 • BACKWARD 3 • Q3/5



$$40 \div 5 = ?$$

- A.** 6
- B.** 7
- C.** 8
- D.** 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q4

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.C.7 • Set 17 • BACKWARD 3 • Q4/5

585

QUESTION ID

**S17-B3-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. 8**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q4

Grade 3 • Operations &amp; Algebraic Thinking • 3.OA.C.7 • Set 17 • BACKWARD 3 • Q4/5

Divide.

$$35 \div 7$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 17 • BACKWARD 3 • Q5/5

586

QUESTION ID

**S17-B3-Q5**

TYPE

Short answer

CORRECT ANSWER

**5**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 17 • BACKWARD 3 • Q5/5

SET 17 • 5.NBT.B.6

# FORWARD 1

Divide multi-digit numbers using the standard algorithm • Grade 6 • The Number System • 6.NS.B.2

Use the standard algorithm to divide.

$$6,432 \div 48$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q1

Grade 6 • The Number System • 6.NS.B.2 • Set 17 • FORWARD 1 • Q1/5

588

QUESTION ID

**S17-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**134**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q1

Grade 6 • The Number System • 6.NS.B.2 • Set 17 • FORWARD 1 • Q1/5

Use the standard algorithm to divide.

$$25,875 \div 69$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q2

Grade 6 • The Number System • 6.NS.B.2 • Set 17 • FORWARD 1 • Q2/5

589

QUESTION ID

**S17-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**375**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q2

Grade 6 • The Number System • 6.NS.B.2 • Set 17 • FORWARD 1 • Q2/5

Use the standard algorithm to divide.

$$14,040 \div 52$$

- A.** 207
- B.** 2,700
- C.** 270
- D.** 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q3

Grade 6 • The Number System • 6.NS.B.2 • Set 17 • FORWARD 1 • Q3/5

590

QUESTION ID

**S17-F1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. 270**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q3

Grade 6 • The Number System • 6.NS.B.2 • Set 17 • FORWARD 1 • Q3/5

Use the standard algorithm to divide.

$$9,594 \div 41$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

Grade 6 • The Number System • 6.NS.B.2 • Set 17 • FORWARD 1 • Q4/5

591

QUESTION ID

**S17-F1-Q4**

TYPE

Short answer

CORRECT ANSWER

**234**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

Grade 6 • The Number System • 6.NS.B.2 • Set 17 • FORWARD 1 • Q4/5

**True or false?**

$$11,700 \div 45 = 2,600$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q5

Grade 6 • The Number System • 6.NS.B.2 • Set 17 • FORWARD 1 • Q5/5

592

QUESTION ID

**S17-F1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q5

Grade 6 • The Number System • 6.NS.B.2 • Set 17 • FORWARD 1 • Q5/5



SET 17 • 5.NBT.B.6

# FORWARD 2

Divide integers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-2,016 \div 24$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 17 • FORWARD 2 • Q1/5

594

QUESTION ID

**S17-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**-84**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 17 • FORWARD 2 • Q1/5

Find the quotient.

$$(-952) \div (-17)$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 17 • FORWARD 2 • Q2/5

595

QUESTION ID

**S17-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**56**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 17 • FORWARD 2 • Q2/5

Find the quotient.

$$780 \div (-15)$$

- A. -52
- B. -53
- C. -765
- D. 52

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 17 • FORWARD 2 • Q3/5

596

QUESTION ID

**S17-F2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. -52**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 17 • FORWARD 2 • Q3/5

Find the quotient.

$$-(144 \div 12)$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 17 • FORWARD 2 • Q4/5

597

QUESTION ID

**S17-F2-Q4**

TYPE

Short answer

CORRECT ANSWER

**-12**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 17 • FORWARD 2 • Q4/5

**True or false?**

$$(-90) \div 18 = 90 \div (-18)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 17 • FORWARD 2 • Q5/5

598

QUESTION ID

**S17-F2-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 17 • FORWARD 2 • Q5/5

# Explain division using area models and equations

|                   |                                                               |                   |
|-------------------|---------------------------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 5 main question                                         | <b>5.NBT.B.6</b>  |
| <b>BACKWARD 1</b> | Area models for dividing by a one-digit number                | <b>4.NBT.B.6</b>  |
| <b>BACKWARD 2</b> | Area as length times width                                    | <b>3.MD.C.7.b</b> |
| <b>BACKWARD 3</b> | Splitting a rectangle to find its area                        | <b>3.MD.C.7.c</b> |
| <b>FORWARD 1</b>  | Use the distributive property with the greatest common factor | <b>6.NS.B.4</b>   |
| <b>FORWARD 2</b>  | Factor and expand linear expressions                          | <b>7.EE.A.1</b>   |

SET 18 • 5.NBT.B.6

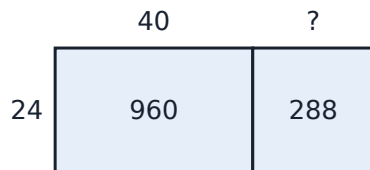
# MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •  
5.NBT.B.6



The area model shows  
 $1,248 \div 24$ .

What is the quotient?



Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 18 • MAIN • Q1/5

601

QUESTION ID

**S18-M-Q1**

TYPE

Short answer

CORRECT ANSWER

**52**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

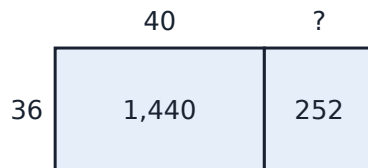
ID S18-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 18 • MAIN • Q1/5

The area model shows

$$1,692 \div 36.$$

What is the quotient?



Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 18 • MAIN • Q2/5

602

QUESTION ID

**S18-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**47**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 18 • MAIN • Q2/5

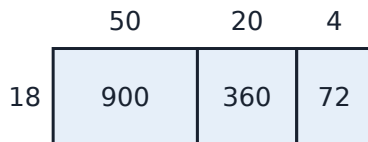
Which equation matches the area model?

A.  $1,332 \div 74 = 20$

B.  $1,332 \div 18 = 74$

C.  $18 \times 50 = 1,332$

D.  $900 \div 18 = 74$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 18 • MAIN • Q3/5

603

QUESTION ID

**S18-M-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B.  $1,332 \div 18 = 74$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 18 • MAIN • Q3/5

Use an area model or equations to divide.

$$2,184 \div 28$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 18 • MAIN • Q4/5

604

QUESTION ID

**S18-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**78**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 18 • MAIN • Q4/5

**True or false?**

$$1,248 \div 24 = (960 \div 24) + (288 \div 24)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 18 • MAIN • Q5/5

605

QUESTION ID

**S18-M-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 18 • MAIN • Q5/5

SET 18 • 5.NBT.B.6

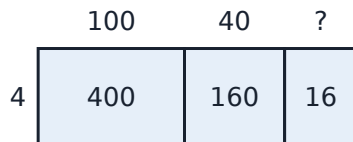
# BACKWARD 1

Area models for dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

The area model shows

$$576 \div 4.$$

What is the quotient?



Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 18 • BACKWARD 1 • Q1/5

607

QUESTION ID

**S18-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**144**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 18 • BACKWARD 1 • Q1/5

**True or false?**

$$468 \div 3 = (300 \div 3) + (150 \div 3) + (18 \div 3)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.6 • Set 18 • BACKWARD 1 • Q2/5

608

QUESTION ID

**S18-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.6 • Set 18 • BACKWARD 1 • Q2/5



Which shows  $245 \div 5$  split into easier parts?

- A.  $(200 \div 5) + (45 \div 5)$
- B.  $(245 \div 2) + (245 \div 3)$
- C.  $200 \div (5 + 45)$
- D.  $(200 \div 5) \times (45 \div 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 18 • BACKWARD 1 • Q3/5

609

QUESTION ID

**S18-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A.**

**$(200 \div 5) + (45 \div 5)$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 18 • BACKWARD 1 • Q3/5

Divide.

$$735 \div 7$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 18 • BACKWARD 1 • Q4/5

610

QUESTION ID

**S18-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**105**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 18 • BACKWARD 1 • Q4/5

A rectangle has an area of 312 square units and a width of 6 units.

What is its length?

Length: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 18 • BACKWARD 1 • Q5/5

611

QUESTION ID

**S18-B1-Q5**

TYPE

Short answer

CORRECT ANSWER

**52 units**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • 5.NBT.B.6

# BACKWARD 2

Area as length times width • Grade 3 • Measurement & Data • 3.MD.C.7.b

A rectangle is 9 units long and 7 units wide.

What is its area?

Area: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 18 • BACKWARD 2 • Q1/5

613

QUESTION ID

**S18-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**63 square units**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 18 • BACKWARD 2 • Q1/5

**True or false?**

A rectangle with an area of 40 square units and a side of 5 units has another side of 8 units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

Grade 3 • Measurement &amp; Data • 3.MD.C.7.b • Set 18 • BACKWARD 2 • Q2/5

614

QUESTION ID

**S18-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

Grade 3 • Measurement &amp; Data • 3.MD.C.7.b • Set 18 • BACKWARD 2 • Q2/5

Which equation gives the area of a 6-by-8 rectangle?

- A.  $8 - 6 = 2$
- B.  $6 + 8 = 14$
- C.  $6 \times 8 = 48$
- D.  $2 \times (6 + 8) = 28$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 18 • BACKWARD 2 • Q3/5

615

QUESTION ID

**S18-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C.  $6 \times 8 = 48$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 18 • BACKWARD 2 • Q3/5

A rectangle has an area of 36 square units. One side is 4 units long.

How long is the other side?

Length: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 18 • BACKWARD 2 • Q4/5

616

QUESTION ID

**S18-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**9 units**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 18 • BACKWARD 2 • Q4/5



Find the area of the rectangle.

Area: \_\_\_\_\_



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 18 • BACKWARD 2 • Q5/5

617

QUESTION ID

**S18-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**30 square cm**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 18 • BACKWARD 2 • Q5/5

SET 18 • 5.NBT.B.6

# BACKWARD 3

Splitting a rectangle to find its area • Grade 3 • Measurement & Data •  
3.MD.C.7.c

**True or false?**

The area of a rectangle that is 4 by  $(10 + 3)$  equals  $4 \times 10 + 4 \times 3$ .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q1

Grade 3 • Measurement &amp; Data • 3.MD.C.7.c • Set 18 • BACKWARD 3 • Q1/5

619

QUESTION ID

**S18-B3-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q1

Grade 3 • Measurement &amp; Data • 3.MD.C.7.c • Set 18 • BACKWARD 3 • Q1/5

A 5-by-12 rectangle is split into a 5-by-10 rectangle and a 5-by-2 rectangle.

What is the total area?

Area: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.c • Set 18 • BACKWARD 3 • Q2/5

620

QUESTION ID

**S18-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**60 square units**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.c • Set 18 • BACKWARD 3 • Q2/5

A 7-by-13 rectangle is split into a 7-by-10 part and a 7-by-3 part.

Which expression gives its area?

**A.**  $7 \times 10 + 7 \times 3$

**B.**  $7 \times 10 + 3$

**C.**  $7 + 10 + 3$

**D.**  $7 \times 10 \times 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.c • Set 18 • BACKWARD 3 • Q3/5

621

QUESTION ID

**S18-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A.  $7 \times 10 + 7 \times 3$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.c • Set 18 • BACKWARD 3 • Q3/5

$$6 \times 14 = 6 \times 10 + 6 \times \underline{\quad}$$

Blank: \_\_\_\_\_

ID S18-B3-Q4

622

**S18-B3-Q4**

### Short answer

4

ID S18-B3-Q4

622

**True or false?**

$$8 \times 12 = 8 \times 10 + 2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.c • Set 18 • BACKWARD 3 • Q5/5

623

QUESTION ID

**S18-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.c • Set 18 • BACKWARD 3 • Q5/5

SET 18 • 5.NBT.B.6

# FORWARD 1

Use the distributive property with the greatest common factor • Grade 6 • The Number System • 6.NS.B.4



Use the greatest common factor to write  $36 + 48$  as a product.

Expression: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 18 • FORWARD 1 • Q1/5

625

QUESTION ID

**S18-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**12(3 + 4)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 18 • FORWARD 1 • Q1/5

Find the greatest common factor of 32 and 56.

GCF: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 18 • FORWARD 1 • Q2/5

626

QUESTION ID

**S18-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**8**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 18 • FORWARD 1 • Q2/5

Which expression is equal to  $45 + 60$  and uses the greatest common factor?

- A.  $15(3 + 6)$
- B.  $15(3 + 4)$
- C.  $3(15 + 20)$
- D.  $5(9 + 12)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 18 • FORWARD 1 • Q3/5

627

QUESTION ID

**S18-F1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B.  $15(3 + 4)$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 18 • FORWARD 1 • Q3/5

Complete the equation.

$$24 + 40 = 8(\underline{\quad} + \underline{\quad})$$

First blank: \_\_\_\_\_

Second blank: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 18 • FORWARD 1 • Q4/5

628

QUESTION ID

**S18-F1-Q4**

TYPE

Short answer

CORRECT ANSWER

**3 and 5**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 18 • FORWARD 1 • Q4/5

**True or false?**

$$18 + 27 = 9(2 + 3)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 18 • FORWARD 1 • Q5/5

629

QUESTION ID

**S18-F1-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 18 • FORWARD 1 • Q5/5

SET 18 • 5.NBT.B.6

# FORWARD 2

Factor and expand linear expressions • Grade 7 • Expressions & Equations •  
7.EE.A.1

Factor using the greatest common factor.

$$12x + 30$$

Expression: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 18 • FORWARD 2 • Q1/5

631

QUESTION ID

**S18-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**$6(2x + 5)$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 18 • FORWARD 2 • Q1/5

Expand.

$$7(4y - 3)$$

Expression: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 18 • FORWARD 2 • Q2/5

632

QUESTION ID

**S18-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**$28y - 21$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 18 • FORWARD 2 • Q2/5



Which expression is equivalent to  $18a - 24$ ?

- A.  $3(6a - 4)$
- B.  $6(3a - 24)$
- C.  $18(a - 6)$
- D.  $6(3a - 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 18 • FORWARD 2 • Q3/5

633

QUESTION ID

**S18-F2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D.  $6(3a - 4)$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 18 • FORWARD 2 • Q3/5

Factor out  $-4$ .

$$-8n - 20$$

Expression: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 18 • FORWARD 2 • Q4/5

634

QUESTION ID

**S18-F2-Q4**

TYPE

Short answer

CORRECT ANSWER

**$-4(2n + 5)$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 18 • FORWARD 2 • Q4/5

**True or false?**

$$4(2.5x + 3) = 10x + 3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 18 • FORWARD 2 • Q5/5

635

QUESTION ID

**S18-F2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 18 • FORWARD 2 • Q5/5

# Add and subtract decimals to hundredths

## MAIN

Grade 5 main question

**5.NBT.B.7**

## BACKWARD 1

Adding and subtracting multi-digit whole numbers

**4.NBT.B.4**

## BACKWARD 2

Adding tenths and hundredths

**4.NF.C.5**

## BACKWARD 3

Writing decimals with the same number of places

**4.NF.C.6**

## FORWARD 1

Add and subtract multi-digit decimals

**6.NS.B.3**

## FORWARD 2

Add and subtract rational numbers

**7.NS.A.1.d**

SET 19 • 5.NBT.B.7

# MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •  
5.NBT.B.7

**ANSWER KEY**

Reduced copy of the question page, with its answer

Add.

$$23.45 + 8.7$$

Sum: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q1

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.7 • Set 19 • MAIN • Q1/5

638

QUESTION ID

**S19-M-Q1**

TYPE

Short answer

CORRECT ANSWER

**32.15**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q1

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.7 • Set 19 • MAIN • Q1/5

638

**ANSWER KEY**

Reduced copy of the question page, with its answer

Subtract.

$$15.3 - 6.48$$

Difference: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q2

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.7 • Set 19 • MAIN • Q2/5

639

QUESTION ID

**S19-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**8.82**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q2

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.7 • Set 19 • MAIN • Q2/5

639

Add.

$$4.06 + 12.9$$

- A. 16.15
- B. 16.069
- C. 5.35
- D. 16.96

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • MAIN • Q3/5

640

QUESTION ID

**S19-M-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. 16.96**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • MAIN • Q3/5



**ANSWER KEY**

Reduced copy of the question page, with its answer

Subtract.

$$20 - 7.35$$

Difference: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.7 • Set 19 • MAIN • Q4/5

641

QUESTION ID

**S19-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**12.65**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.7 • Set 19 • MAIN • Q4/5

641

Maya had \$18.50. She spent \$6.75.

How much money does she have left?

Money left: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • MAIN • Q5/5

642

QUESTION ID

**S19-M-Q5**

TYPE

Short answer

CORRECT ANSWER

**\$11.75**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • MAIN • Q5/5

SET 19 • 5.NBT.B.7

# BACKWARD 1

Adding and subtracting multi-digit whole numbers • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4

Add.

$$2,345 + 870$$

Sum: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 19 • BACKWARD 1 • Q1/5

644

QUESTION ID

**S19-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**3,215**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 19 • BACKWARD 1 • Q1/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Subtract.

$$1,530 - 648$$

Difference: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q2

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.4 • Set 19 • BACKWARD 1 • Q2/5

645

QUESTION ID

**S19-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**882**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q2

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.4 • Set 19 • BACKWARD 1 • Q2/5

**True or false?**

$$406 + 1,290 = 1,596$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 19 • BACKWARD 1 • Q3/5

646

QUESTION ID

**S19-B1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 19 • BACKWARD 1 • Q3/5

$$2,000 - 735 = ?$$

- A.** 2,735
- B.** 1,265
- C.** 1,275
- D.** 1,365

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q4

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.4 • Set 19 • BACKWARD 1 • Q4/5

647

QUESTION ID

**S19-B1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. 1,265**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q4

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.4 • Set 19 • BACKWARD 1 • Q4/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Subtract.

$$1,850 - 675$$

Difference: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q5

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.4 • Set 19 • BACKWARD 1 • Q5/5

648

QUESTION ID

**S19-B1-Q5**

TYPE

Short answer

CORRECT ANSWER

**1,175**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q5

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.4 • Set 19 • BACKWARD 1 • Q5/5



SET 19 • 5.NBT.B.7

# BACKWARD 2

Adding tenths and hundredths • Grade 4 • Number & Operations–Fractions •  
4.NF.C.5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Add.

$$\frac{3}{10} + \frac{45}{100}$$

Sum: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

Grade 4 • Number &amp; Operations-Fractions • 4.NF.C.5 • Set 19 • BACKWARD 2 • Q1/5

650

QUESTION ID

**S19-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**75/100****GRADING NOTE** 3/4 or 0.75 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

Grade 4 • Number &amp; Operations-Fractions • 4.NF.C.5 • Set 19 • BACKWARD 2 • Q1/5

**True or false?**

$$\frac{6}{10} + \frac{2}{100} = \frac{8}{100}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 19 • BACKWARD 2 • Q2/5

651

QUESTION ID

**S19-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 19 • BACKWARD 2 • Q2/5

$$\frac{7}{10} + \frac{15}{100} = ?$$

A.  $\frac{85}{100}$

B.  $\frac{22}{100}$

C.  $\frac{22}{110}$

D.  $\frac{85}{10}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 19 • BACKWARD 2 • Q3/5

652

QUESTION ID

**S19-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. 85/100**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 19 • BACKWARD 2 • Q3/5

Write  $\frac{4}{10}$  as hundredths.

Fraction: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 19 • BACKWARD 2 • Q4/5

653

QUESTION ID

**S19-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**40/100**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 19 • BACKWARD 2 • Q4/5

**True or false?**

$$\frac{5}{10} + \frac{5}{100} = \frac{10}{100}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 19 • BACKWARD 2 • Q5/5

654

QUESTION ID

**S19-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • 5.NBT.B.7

# BACKWARD 3

Writing decimals with the same number of places • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

**True or false?**

$$8.7 = 8.70$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 19 • BACKWARD 3 • Q1/5

656

QUESTION ID

**S19-B3-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 19 • BACKWARD 3 • Q1/5



Write 15.3 with two decimal places.

Decimal: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 19 • BACKWARD 3 • Q2/5

657

QUESTION ID

**S19-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**15.30**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 19 • BACKWARD 3 • Q2/5

Which decimal is equal to 20?

- A.** 20.00
- B.** 0.20
- C.** 2.00
- D.** 200.0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 19 • BACKWARD 3 • Q3/5

658

QUESTION ID

**S19-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. 20.00**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 19 • BACKWARD 3 • Q3/5

**True or false?**

$$6.4 = 6.04$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 19 • BACKWARD 3 • Q4/5

659

QUESTION ID

**S19-B3-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 19 • BACKWARD 3 • Q4/5

Write 12.9 with a digit in the hundredths place, without changing its value.

Decimal: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 19 • BACKWARD 3 • Q5/5

660

QUESTION ID

**S19-B3-Q5**

TYPE

Short answer

CORRECT ANSWER

**12.90**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 19 • BACKWARD 3 • Q5/5

SET 19 • 5.NBT.B.7

# FORWARD 1

Add and subtract multi-digit decimals • Grade 6 • The Number System •  
6.NS.B.3

**ANSWER KEY**

Reduced copy of the question page, with its answer

Add.

$$128.47 + 9.605$$

Sum: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 1 • Q1/5

662

QUESTION ID

**S19-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**138.075**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 1 • Q1/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Subtract.

$$54.2 - 18.376$$

Difference: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 1 • Q2/5

663

QUESTION ID

**S19-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**35.824**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 1 • Q2/5

663

Add.

$$0.952 + 3.48 + 11.6$$

- A.** 16.132
- B.** 4.53
- C.** 15.032
- D.** 16.032

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 1 • Q3/5

664

QUESTION ID

**S19-F1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. 16.032**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 1 • Q3/5



**ANSWER KEY**

Reduced copy of the question page, with its answer

Subtract.

$$100 - 0.485$$

Difference: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 1 • Q4/5

665

QUESTION ID

**S19-F1-Q4**

TYPE

Short answer

CORRECT ANSWER

**99.515**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 1 • Q4/5

665

**True or false?**

$$7.3 - 2.145 = 5.245$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 1 • Q5/5

666

QUESTION ID

**S19-F1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 1 • Q5/5

SET 19 • 5.NBT.B.7

# FORWARD 2

Add and subtract rational numbers • Grade 7 • The Number System •  
7.NS.A.1.d

Find the value.

$$-12.5 + 4.75$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 19 • FORWARD 2 • Q1/5

668

QUESTION ID

**S19-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**-7.75**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 19 • FORWARD 2 • Q1/5

Find the value.

$$3.2 - 8.45$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 19 • FORWARD 2 • Q2/5

669

QUESTION ID

**S19-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**−5.25**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 19 • FORWARD 2 • Q2/5

Find the value.

$$-6.3 + (-2.85)$$

- A. 9.15
- B. -9.15
- C. -3.45
- D. 3.45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 19 • FORWARD 2 • Q3/5

670

QUESTION ID

**S19-F2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. -9.15**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 19 • FORWARD 2 • Q3/5

Find the value.

$$-4.6 - (-10.25)$$

Value: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 19 • FORWARD 2 • Q4/5

671

QUESTION ID

**S19-F2-Q4**

TYPE

Short answer

CORRECT ANSWER

**5.65**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 19 • FORWARD 2 • Q4/5

**True or false?**

$$-1.5 + 1.5 = 0$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 19 • FORWARD 2 • Q5/5

672

QUESTION ID

**S19-F2-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 19 • FORWARD 2 • Q5/5



# Multiply decimals to hundredths

|                   |                                                   |                   |
|-------------------|---------------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 5 main question                             | <b>5.NBT.B.7</b>  |
| <b>BACKWARD 1</b> | Multiplying whole numbers                         | <b>4.NBT.B.5</b>  |
| <b>BACKWARD 2</b> | Decimals as tenths and hundredths                 | <b>4.NF.C.6</b>   |
| <b>BACKWARD 3</b> | Dividing by 10 and 100 to place the decimal point | <b>5.NBT.A.2</b>  |
| <b>FORWARD 1</b>  | Multiply multi-digit decimals                     | <b>6.NS.B.3</b>   |
| <b>FORWARD 2</b>  | Multiply rational numbers                         | <b>7.NS.A.2.c</b> |

SET 20 • 5.NBT.B.7

# MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •  
5.NBT.B.7

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$3.4 \times 6$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.7 • Set 20 • MAIN • Q1/5

675

QUESTION ID

**S20-M-Q1**

TYPE

Short answer

CORRECT ANSWER

**20.4**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.7 • Set 20 • MAIN • Q1/5

675

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$0.7 \times 0.5$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.7 • Set 20 • MAIN • Q2/5

676

QUESTION ID

**S20-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**0.35**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.7 • Set 20 • MAIN • Q2/5

676

Multiply.

$$2.5 \times 1.2$$

- A.** 3
- B.** 30
- C.** 3.7
- D.** 0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • MAIN • Q3/5

677

QUESTION ID

**S20-M-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. 3**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • MAIN • Q3/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$0.06 \times 8$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.7 • Set 20 • MAIN • Q4/5

678

QUESTION ID

**S20-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**0.48**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.7 • Set 20 • MAIN • Q4/5

678

A pen costs \$1.25.

How much do 7 pens cost?

Cost: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • MAIN • Q5/5

679

QUESTION ID

**S20-M-Q5**

TYPE

Short answer

CORRECT ANSWER

**\$8.75**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • MAIN • Q5/5

SET 20 • 5.NBT.B.7

# BACKWARD 1

Multiplying whole numbers • Grade 4 • Number & Operations in Base Ten •  
4.NBT.B.5



Multiply.

$$34 \times 6$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 20 • BACKWARD 1 • Q1/5

681

QUESTION ID

**S20-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**204**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 20 • BACKWARD 1 • Q1/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$25 \times 12$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q2

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.5 • Set 20 • BACKWARD 1 • Q2/5

682

QUESTION ID

**S20-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**300**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q2

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.5 • Set 20 • BACKWARD 1 • Q2/5

**True or false?**

$$125 \times 7 = 785$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 20 • BACKWARD 1 • Q3/5

683

QUESTION ID

**S20-B1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 20 • BACKWARD 1 • Q3/5

$$18 \times 4 = ?$$

- A.** 72
- B.** 82
- C.** 48
- D.** 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.5 • Set 20 • BACKWARD 1 • Q4/5

684

QUESTION ID

**S20-B1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. 72**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.5 • Set 20 • BACKWARD 1 • Q4/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$106 \times 8$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.5 • Set 20 • BACKWARD 1 • Q5/5

685

QUESTION ID

**S20-B1-Q5**

TYPE

Short answer

CORRECT ANSWER

**848**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.5 • Set 20 • BACKWARD 1 • Q5/5

685

SET 20 • 5.NBT.B.7

# BACKWARD 2

Decimals as tenths and hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write 35 hundredths as a decimal.

Decimal: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 20 • BACKWARD 2 • Q1/5

687

QUESTION ID

**S20-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**0.35**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 20 • BACKWARD 2 • Q1/5

**True or false?**

0.4 is 4 tenths.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 20 • BACKWARD 2 • Q2/5

688

QUESTION ID

**S20-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 20 • BACKWARD 2 • Q2/5



Which decimal is 204 tenths?

- A.** 204
- B.** 20.4
- C.** 2.04
- D.** 0.204

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 20 • BACKWARD 2 • Q3/5

689

QUESTION ID

**S20-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. 20.4**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 20 • BACKWARD 2 • Q3/5

How many hundredths are in 0.48?

Hundredths: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 20 • BACKWARD 2 • Q4/5

690

QUESTION ID

**S20-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**48**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 20 • BACKWARD 2 • Q4/5

**True or false?**

0.06 is 6 tenths.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 20 • BACKWARD 2 • Q5/5

691

QUESTION ID

**S20-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • 5.NBT.B.7

# BACKWARD 3

Dividing by 10 and 100 to place the decimal point • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Divide.

$$204 \div 10$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 3 • Q1/5

693

QUESTION ID

**S20-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**20.4**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 3 • Q1/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Divide.

$$35 \div 100$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q2

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 3 • Q2/5

694

QUESTION ID

**S20-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**0.35**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q2

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 3 • Q2/5

**True or false?**

$$300 \div 100 = 30$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 3 • Q3/5

695

QUESTION ID

**S20-B3-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 3 • Q3/5

$$875 \div 100 = ?$$

- A.** 875
- B.** 8.75
- C.** 0.875
- D.** 87.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q4

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 3 • Q4/5

696

QUESTION ID

**S20-B3-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. 8.75**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q4

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 3 • Q4/5



**ANSWER KEY**

Reduced copy of the question page, with its answer

Divide.

$$48 \div 100$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q5

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 3 • Q5/5

697

QUESTION ID

**S20-B3-Q5**

TYPE

Short answer

CORRECT ANSWER

**0.48**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q5

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 3 • Q5/5

697

SET 20 • 5.NBT.B.7

# FORWARD 1

Multiply multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$3.45 \times 0.6$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 1 • Q1/5

699

QUESTION ID

**S20-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**2.07**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 1 • Q1/5

699

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$12.08 \times 2.5$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 1 • Q2/5

700

QUESTION ID

**S20-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**30.2**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 1 • Q2/5

700

Multiply.

$$0.125 \times 0.4$$

- A.** 0.5
- B.** 0.05
- C.** 5
- D.** 0.005

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 1 • Q3/5

701

QUESTION ID

**S20-F1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. 0.05**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 1 • Q3/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$7.2 \times 1.15$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 1 • Q4/5

702

QUESTION ID

**S20-F1-Q4**

TYPE

Short answer

CORRECT ANSWER

**8.28**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 1 • Q4/5

702

**True or false?**

$$0.03 \times 0.03 = 0.09$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 1 • Q5/5

703

QUESTION ID

**S20-F1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 1 • Q5/5

SET 20 • 5.NBT.B.7

# FORWARD 2

Multiply rational numbers • Grade 7 • The Number System • 7.NS.A.2.c



Find the product.

$$-3.4 \times 6$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 20 • FORWARD 2 • Q1/5

705

QUESTION ID

**S20-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**-20.4**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 20 • FORWARD 2 • Q1/5

Find the product.

$$(-0.7) \times (-0.5)$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 20 • FORWARD 2 • Q2/5

706

QUESTION ID

**S20-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**0.35**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 20 • FORWARD 2 • Q2/5

Find the product.

$$2.5 \times (-1.2)$$

- A.  $-0.3$
- B.  $-30$
- C.  $3$
- D.  $-3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 20 • FORWARD 2 • Q3/5

707

QUESTION ID

**S20-F2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D.  $-3$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 20 • FORWARD 2 • Q3/5

Find the product.

$$(-0.06) \times 8$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 20 • FORWARD 2 • Q4/5

708

QUESTION ID

**S20-F2-Q4**

TYPE

Short answer

CORRECT ANSWER

**-0.48**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 20 • FORWARD 2 • Q4/5

**True or false?**

$$(-1.25) \times (-4) = -5$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 20 • FORWARD 2 • Q5/5

709

QUESTION ID

**S20-F2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 20 • FORWARD 2 • Q5/5

# Divide decimals to hundredths

|                   |                                                              |                   |
|-------------------|--------------------------------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 5 main question                                        | <b>5.NBT.B.7</b>  |
| <b>BACKWARD 1</b> | Dividing whole numbers                                       | <b>4.NBT.B.6</b>  |
| <b>BACKWARD 2</b> | Division as an unknown-factor problem                        | <b>3.OA.B.6</b>   |
| <b>BACKWARD 3</b> | Multiplying both numbers by a power of 10 keeps the quotient | <b>5.NBT.A.2</b>  |
| <b>FORWARD 1</b>  | Divide multi-digit decimals                                  | <b>6.NS.B.3</b>   |
| <b>FORWARD 2</b>  | Divide rational numbers                                      | <b>7.NS.A.2.b</b> |

SET 21 • 5.NBT.B.7

# MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •  
5.NBT.B.7

**ANSWER KEY**

Reduced copy of the question page, with its answer

Divide.

$$8.4 \div 4$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q1

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.7 • Set 21 • MAIN • Q1/5

712

QUESTION ID

**S21-M-Q1**

TYPE

Short answer

CORRECT ANSWER

**2.1**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q1

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.7 • Set 21 • MAIN • Q1/5

712



**ANSWER KEY**

Reduced copy of the question page, with its answer

Divide.

$$3.6 \div 0.6$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q2

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.7 • Set 21 • MAIN • Q2/5

713

QUESTION ID

**S21-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**6**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q2

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.7 • Set 21 • MAIN • Q2/5

713

Divide.

$$0.45 \div 0.05$$

- A.** 0.09
- B.** 0.9
- C.** 90
- D.** 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 21 • MAIN • Q3/5

714

QUESTION ID

**S21-M-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. 9**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 21 • MAIN • Q3/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Divide.

$$7.5 \div 0.3$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q4

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.7 • Set 21 • MAIN • Q4/5

715

QUESTION ID

**S21-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**25**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q4

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.7 • Set 21 • MAIN • Q4/5

715

A rope is 4.8 meters long. It is cut into pieces that are each 0.6 meter long.

How many pieces are there?

Pieces: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 21 • MAIN • Q5/5

716

QUESTION ID

**S21-M-Q5**

TYPE

Short answer

CORRECT ANSWER

**8**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 21 • MAIN • Q5/5

SET 21 • 5.NBT.B.7

# BACKWARD 1

Dividing whole numbers • Grade 4 • Number & Operations in Base Ten •  
4.NBT.B.6

Divide.

$$84 \div 4$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 21 • BACKWARD 1 • Q1/5

718

QUESTION ID

**S21-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**21**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 21 • BACKWARD 1 • Q1/5

Divide.

$$360 \div 6$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 21 • BACKWARD 1 • Q2/5

719

QUESTION ID

**S21-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**60**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 21 • BACKWARD 1 • Q2/5

**True or false?**

$$450 \div 5 = 9$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 21 • BACKWARD 1 • Q3/5

720

QUESTION ID

**S21-B1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 21 • BACKWARD 1 • Q3/5



$$750 \div 3 = ?$$

- A.** 25
- B.** 250
- C.** 2,500
- D.** 260

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.6 • Set 21 • BACKWARD 1 • Q4/5

721

QUESTION ID

**S21-B1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. 250**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

Grade 4 • Number &amp; Operations in Base Ten • 4.NBT.B.6 • Set 21 • BACKWARD 1 • Q4/5

Divide.

$$48 \div 6$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 21 • BACKWARD 1 • Q5/5

722

QUESTION ID

**S21-B1-Q5**

TYPE

Short answer

CORRECT ANSWER

**8**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • 5.NBT.B.7

# BACKWARD 2

Division as an unknown-factor problem • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6

Find  $36 \div 6$  by thinking  $6 \times \square = 36$ .

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 21 • BACKWARD 2 • Q1/5

724

QUESTION ID

**S21-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**6**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 21 • BACKWARD 2 • Q1/5

**True or false?**

$7 \times 9 = 63$ , so  $63 \div 7 = 9$ .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 21 • BACKWARD 2 • Q2/5

725

QUESTION ID

**S21-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 21 • BACKWARD 2 • Q2/5

Which fact helps you find  $56 \div 8$ ?

- A.  $56 \times 8 = 448$
- B.  $8 \times 7 = 56$
- C.  $8 + 48 = 56$
- D.  $8 - 7 = 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 21 • BACKWARD 2 • Q3/5

726

QUESTION ID

**S21-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B.  $8 \times 7 = 56$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 21 • BACKWARD 2 • Q3/5

What number goes in the box?

$$\square \times 5 = 45$$

Number: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 21 • BACKWARD 2 • Q4/5

727

QUESTION ID

**S21-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**9**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 21 • BACKWARD 2 • Q4/5

**True or false?**

$24 \div 3 = 6$  because  $3 \times 6 = 24$ .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 21 • BACKWARD 2 • Q5/5

728

QUESTION ID

**S21-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 21 • BACKWARD 2 • Q5/5



SET 21 • 5.NBT.B.7

# BACKWARD 3

Multiplying both numbers by a power of 10 keeps the quotient • Grade 5 •  
Number & Operations in Base Ten • 5.NBT.A.2

**True or false?**

$3.6 \div 0.6$  has the same quotient as  $36 \div 6$ .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q1

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.2 • Set 21 • BACKWARD 3 • Q1/5

730

QUESTION ID

**S21-B3-Q1**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q1

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.2 • Set 21 • BACKWARD 3 • Q1/5

Which has the same quotient as  $0.45 \div 0.05$ ?

- A.  $4.5 \div 5$
- B.  $0.45 \div 5$
- C.  $45 \div 5$
- D.  $45 \div 0.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 21 • BACKWARD 3 • Q2/5

731

QUESTION ID

**S21-B3-Q2**

TYPE

Multiple choice

CORRECT ANSWER

**C.  $45 \div 5$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 21 • BACKWARD 3 • Q2/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$0.6 \times 10$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q3

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.2 • Set 21 • BACKWARD 3 • Q3/5

732

QUESTION ID

**S21-B3-Q3**

TYPE

Short answer

CORRECT ANSWER

**6**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q3

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.2 • Set 21 • BACKWARD 3 • Q3/5

732

**ANSWER KEY**

Reduced copy of the question page, with its answer

Multiply.

$$0.05 \times 100$$

Product: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q4

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.2 • Set 21 • BACKWARD 3 • Q4/5

733

QUESTION ID

**S21-B3-Q4**

TYPE

Short answer

CORRECT ANSWER

**5**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q4

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.2 • Set 21 • BACKWARD 3 • Q4/5

733

**True or false?**

$7.5 \div 0.3$  has the same quotient as  $75 \div 30$ .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q5

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.2 • Set 21 • BACKWARD 3 • Q5/5

734

QUESTION ID

**S21-B3-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q5

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.A.2 • Set 21 • BACKWARD 3 • Q5/5

SET 21 • 5.NBT.B.7

# FORWARD 1

Divide multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Divide.

$$13.65 \div 0.35$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 1 • Q1/5

736

QUESTION ID

**S21-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**39**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 1 • Q1/5



Divide.

$$9.072 \div 2.4$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 1 • Q2/5

737

QUESTION ID

**S21-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**3.78**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 1 • Q2/5

Divide.

$$0.081 \div 0.09$$

- A.** 90
- B.** 0.09
- C.** 0.9
- D.** 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 1 • Q3/5

738

QUESTION ID

**S21-F1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. 0.9**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 1 • Q3/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Divide.

$$52.5 \div 1.25$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 1 • Q4/5

739

QUESTION ID

**S21-F1-Q4**

TYPE

Short answer

CORRECT ANSWER

**42**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 1 • Q4/5

739

**True or false?**

$$1.8 \div 0.006 = 3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 1 • Q5/5

740

QUESTION ID

**S21-F1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 1 • Q5/5

SET 21 • 5.NBT.B.7

# FORWARD 2

Divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-8.4 \div 4$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • FORWARD 2 • Q1/5

742

QUESTION ID

**S21-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**-2.1**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • FORWARD 2 • Q1/5

Find the quotient.

$$(-3.6) \div (-0.6)$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • FORWARD 2 • Q2/5

743

QUESTION ID

**S21-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**6**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • FORWARD 2 • Q2/5

Find the quotient.

$$0.45 \div (-0.05)$$

- A.** -9
- B.** -0.9
- C.** 0.9
- D.** 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • FORWARD 2 • Q3/5

744

QUESTION ID

**S21-F2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. -9**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • FORWARD 2 • Q3/5



Find the quotient.

$$-7.5 \div 0.3$$

Quotient: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • FORWARD 2 • Q4/5

745

QUESTION ID

**S21-F2-Q4**

TYPE

Short answer

CORRECT ANSWER

**-25**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • FORWARD 2 • Q4/5

**True or false?**

$$-(4.8 \div 0.6) = (-4.8) \div 0.6$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • FORWARD 2 • Q5/5

746

QUESTION ID

**S21-F2-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • FORWARD 2 • Q5/5

# Solve real-world problems with decimal operations

|                   |                                               |                  |
|-------------------|-----------------------------------------------|------------------|
| <b>MAIN</b>       | Grade 5 main question                         | <b>5.NBT.B.7</b> |
| <b>BACKWARD 1</b> | Word problems with money and measurement      | <b>4.MD.A.2</b>  |
| <b>BACKWARD 2</b> | Hundredths on a grid                          | <b>4.NF.C.6</b>  |
| <b>FORWARD 1</b>  | Real-world problems with multi-digit decimals | <b>6.NS.B.3</b>  |
| <b>FORWARD 2</b>  | Real-world problems with rational numbers     | <b>7.NS.A.3</b>  |

SET 22 • 5.NBT.B.7

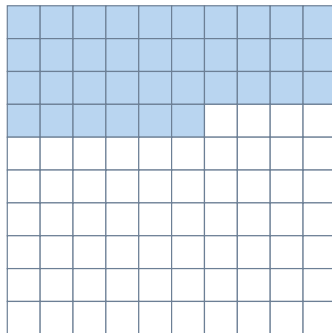
# MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •  
5.NBT.B.7

The grid shows 0.36 shaded.

How much more must be shaded to show 0.5?

Amount: \_\_\_\_\_



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 22 • MAIN • Q1/5

749

QUESTION ID

**S22-M-Q1**

TYPE

Short answer

CORRECT ANSWER

**0.14**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 22 • MAIN • Q1/5

Juice costs \$2.35 a bottle. Ben buys 4 bottles and pays with a \$20 bill.

How much change does he get?

Change: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 22 • MAIN • Q2/5

750

QUESTION ID

**S22-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**\$10.60**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 22 • MAIN • Q2/5

A 2.5-pound bag of nuts is shared equally among 5 friends.

How many pounds does each friend get?

- A.** 0.05 pound                      **B.** 0.5 pound  
**C.** 2 pounds                        **D.** 12.5 pounds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 22 • MAIN • Q3/5

751

QUESTION ID

**S22-M-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. 0.5 pound**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 22 • MAIN • Q3/5

A runner ran 3.75 km on Monday and 4.6 km on Tuesday.

How much farther did she run on Tuesday?

Distance: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 22 • MAIN • Q4/5

752

QUESTION ID

**S22-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**0.85 km**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 22 • MAIN • Q4/5



**ANSWER KEY**

Reduced copy of the question page, with its answer

Tiles cost \$0.45 each.

How much do 12 tiles cost?

Cost: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q5

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.7 • Set 22 • MAIN • Q5/5

753

QUESTION ID

**S22-M-Q5**

TYPE

Short answer

CORRECT ANSWER

**\$5.40**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q5

Grade 5 • Number &amp; Operations in Base Ten • 5.NBT.B.7 • Set 22 • MAIN • Q5/5

753

SET 22 • 5.NBT.B.7

# BACKWARD 1

Word problems with money and measurement • Grade 4 • Measurement & Data • 4.MD.A.2

Pens cost 75 cents each.

How much do 4 pens cost, in dollars?

Cost: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 22 • BACKWARD 1 • Q1/5

755

QUESTION ID

**S22-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**\$3.00**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 22 • BACKWARD 1 • Q1/5

A board is 120 cm long. It is cut into 4 equal pieces.

How long is each piece?

Length: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 22 • BACKWARD 1 • Q2/5

756

QUESTION ID

**S22-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**30 cm**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 22 • BACKWARD 1 • Q2/5

**True or false?**

Two quarters and three dimes are worth 65 cents.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q3

Grade 4 • Measurement &amp; Data • 4.MD.A.2 • Set 22 • BACKWARD 1 • Q3/5

757

QUESTION ID

**S22-B1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q3

Grade 4 • Measurement &amp; Data • 4.MD.A.2 • Set 22 • BACKWARD 1 • Q3/5

Ann has \$5. She spends \$2.25.

How much money is left?

- A. \$7.25
- B. \$3.25
- C. \$2.75
- D. \$2.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 22 • BACKWARD 1 • Q4/5

758

QUESTION ID

**S22-B1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. \$2.75**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 22 • BACKWARD 1 • Q4/5

A jug holds 2 liters.

How many milliliters is that?

Milliliters: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q5

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 22 • BACKWARD 1 • Q5/5

759

QUESTION ID

**S22-B1-Q5**

TYPE

Short answer

CORRECT ANSWER

**2,000 mL**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q5

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 22 • BACKWARD 1 • Q5/5

SET 22 • 5.NBT.B.7

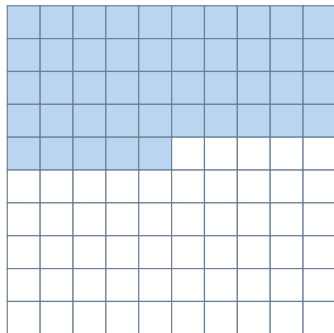
# BACKWARD 2

Hundredths on a grid • Grade 4 • Number & Operations–Fractions • 4.NF.C.6



What decimal does the shaded part of the grid show?

Decimal: \_\_\_\_\_



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 22 • BACKWARD 2 • Q1/5

761

QUESTION ID

**S22-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**0.45**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

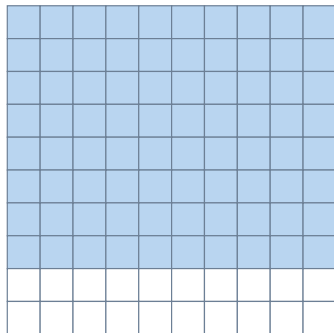
ID S22-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 22 • BACKWARD 2 • Q1/5

**True or false?**

The shaded part of the grid shows 0.8.

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q2

Grade 4 • Number &amp; Operations-Fractions • 4.NF.C.6 • Set 22 • BACKWARD 2 • Q2/5

762

QUESTION ID

**S22-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**A. True**

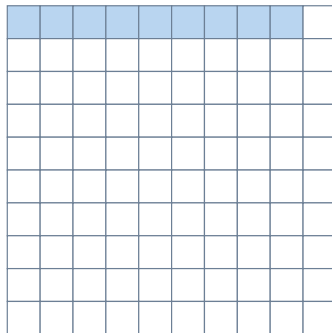
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q2

Grade 4 • Number &amp; Operations-Fractions • 4.NF.C.6 • Set 22 • BACKWARD 2 • Q2/5

Which decimal does the shaded part of the grid show?

- A. 0.9
- B. 9
- C. 0.009
- D. 0.09



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q3

Grade 4 • Number &amp; Operations-Fractions • 4.NF.C.6 • Set 22 • BACKWARD 2 • Q3/5

763

QUESTION ID

**S22-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. 0.09**

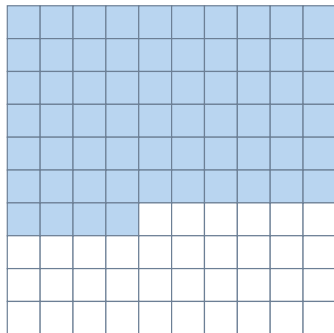
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q3

Grade 4 • Number &amp; Operations-Fractions • 4.NF.C.6 • Set 22 • BACKWARD 2 • Q3/5

How many more squares must be shaded to fill the whole grid?

Squares: \_\_\_\_\_



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 22 • BACKWARD 2 • Q4/5

764

QUESTION ID

**S22-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**36**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 22 • BACKWARD 2 • Q4/5

**True or false?**

0.5 of a hundredths grid is 5 squares.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q5

Grade 4 • Number &amp; Operations-Fractions • 4.NF.C.6 • Set 22 • BACKWARD 2 • Q5/5

765

QUESTION ID

**S22-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q5

Grade 4 • Number &amp; Operations-Fractions • 4.NF.C.6 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • 5.NBT.B.7

# FORWARD 1

Real-world problems with multi-digit decimals • Grade 6 • The Number System  
• 6.NS.B.3

Gas costs \$3.85 per gallon.

How much do 12 gallons cost?

Cost: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 1 • Q1/5

767

QUESTION ID

**S22-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**\$46.20**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 1 • Q1/5

A board is 6.75 m long. It is cut into pieces that are 0.45 m long.

How many pieces are there?

Pieces: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 1 • Q2/5

768

QUESTION ID

**S22-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**15**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 1 • Q2/5



Rice costs \$1.20 per pound.

How much do 2.5 pounds cost?

- A.** \$3.00
- B.** \$30.00
- C.** \$2.70
- D.** \$3.70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 1 • Q3/5

769

QUESTION ID

**S22-F1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. \$3.00**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 1 • Q3/5

Lee buys items that cost \$12.49, \$3.05, and \$7.60.

What is the total cost?

Total: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 1 • Q4/5

770

QUESTION ID

**S22-F1-Q4**

TYPE

Short answer

CORRECT ANSWER

**\$23.14**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 1 • Q4/5

**True or false?**

A 2.4-pound bag split into 0.3-pound portions makes 8 portions.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 1 • Q5/5

771

QUESTION ID

**S22-F1-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 1 • Q5/5

SET 22 • 5.NBT.B.7

# FORWARD 2

Real-world problems with rational numbers • Grade 7 • The Number System •  
7.NS.A.3

A submarine at  $-45.5$  m rises  $12.75$  m and then dives  $20$  m.

What is its final position?

Position: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 22 • FORWARD 2 • Q1/5

773

QUESTION ID

**S22-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**$-52.75$  m**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 22 • FORWARD 2 • Q1/5

A store's profit was  $-\$125.40$  in May and  $\$310.15$  in June.

What was the total profit for the two months?

Total: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 22 • FORWARD 2 • Q2/5

774

QUESTION ID

**S22-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**$\$184.75$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 22 • FORWARD 2 • Q2/5

The temperature was  $4^{\circ}\text{F}$ . It dropped  $1.5^{\circ}\text{F}$  each hour for 6 hours.

What was the final temperature?

- A.**  $-5^{\circ}\text{F}$                       **B.**  $13^{\circ}\text{F}$   
**C.**  $5^{\circ}\text{F}$                       **D.**  $-9^{\circ}\text{F}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 22 • FORWARD 2 • Q3/5

775

QUESTION ID

**S22-F2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A.  $-5^{\circ}\text{F}$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 22 • FORWARD 2 • Q3/5

An account changes by  $-\$2.40$  each week for 5 weeks.

What is the total change?

Total change: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 22 • FORWARD 2 • Q4/5

776

QUESTION ID

**S22-F2-Q4**

TYPE

Short answer

CORRECT ANSWER

**$-\$12.00$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 22 • FORWARD 2 • Q4/5



**True or false?**

$$-3.6 \div 0.4 = -0.9$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 22 • FORWARD 2 • Q5/5

777

QUESTION ID

**S22-F2-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 22 • FORWARD 2 • Q5/5

# Add fractions with unlike denominators

|                   |                                         |                   |
|-------------------|-----------------------------------------|-------------------|
| <b>MAIN</b>       | Grade 5 main question                   | <b>5.NF.A.1</b>   |
| <b>BACKWARD 1</b> | Equivalent fractions                    | <b>4.NF.A.1</b>   |
| <b>BACKWARD 2</b> | Adding fractions with like denominators | <b>4.NF.B.3.a</b> |
| <b>BACKWARD 3</b> | Common multiples                        | <b>4.OA.B.4</b>   |
| <b>FORWARD 1</b>  | Find the least common multiple          | <b>6.NS.B.4</b>   |
| <b>FORWARD 2</b>  | Add rational numbers                    | <b>7.NS.A.1.d</b> |

SET 23 • 5.NF.A.1

# MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Add.

$$\frac{2}{3} + \frac{1}{4}$$

Sum: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 23 • MAIN • Q1/5

780

QUESTION ID

**S23-M-Q1**

TYPE

Short answer

CORRECT ANSWER

**11/12**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 23 • MAIN • Q1/5

Add.

$$\frac{3}{5} + \frac{1}{10}$$

Sum: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 23 • MAIN • Q2/5

781

QUESTION ID

**S23-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**7/10**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 23 • MAIN • Q2/5

Add.

$$\frac{5}{6} + \frac{3}{8}$$

A.  $1\frac{5}{24}$

B.  $1\frac{1}{24}$

C.  $\frac{8}{24}$

D.  $\frac{8}{14}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 23 • MAIN • Q3/5

782

QUESTION ID

**S23-M-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. 1 5/24**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 23 • MAIN • Q3/5

Add.

$$\frac{1}{2} + \frac{2}{7}$$

Sum: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 23 • MAIN • Q4/5

783

QUESTION ID

**S23-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**11/14**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 23 • MAIN • Q4/5

**ANSWER KEY**

Reduced copy of the question page, with its answer

Add.

$$\frac{3}{4} + \frac{5}{12} + \frac{1}{6}$$

Sum: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q5

Grade 5 • Number &amp; Operations-Fractions • 5.NF.A.1 • Set 23 • MAIN • Q5/5

784

QUESTION ID

**S23-M-Q5**

TYPE

Short answer

CORRECT ANSWER

**1 1/3****GRADING NOTE** 16/12 or 4/3 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q5

Grade 5 • Number &amp; Operations-Fractions • 5.NF.A.1 • Set 23 • MAIN • Q5/5

784



SET 23 • 5.NF.A.1

# BACKWARD 1

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{2}{3} = \frac{?}{12}$$

Missing number: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 23 • BACKWARD 1 • Q1/5

786

QUESTION ID

**S23-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**8**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 23 • BACKWARD 1 • Q1/5

**True or false?**

$$\frac{1}{4} = \frac{4}{12}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 23 • BACKWARD 1 • Q2/5

787

QUESTION ID

**S23-B1-Q2**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 23 • BACKWARD 1 • Q2/5

Which fraction is equivalent to  $\frac{3}{5}$ ?

A.  $\frac{6}{5}$

B.  $\frac{5}{3}$

C.  $\frac{3}{10}$

D.  $\frac{6}{10}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 23 • BACKWARD 1 • Q3/5

788

QUESTION ID

**S23-B1-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**D. 6/10**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 23 • BACKWARD 1 • Q3/5

Find the missing number.

$$\frac{5}{6} = \frac{20}{?}$$

Missing number: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 23 • BACKWARD 1 • Q4/5

789

QUESTION ID

**S23-B1-Q4**

TYPE

Short answer

CORRECT ANSWER

**24**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 23 • BACKWARD 1 • Q4/5

**True or false?**

$$\frac{2}{7} = \frac{4}{21}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 23 • BACKWARD 1 • Q5/5

790

QUESTION ID

**S23-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • 5.NF.A.1

# BACKWARD 2

Adding fractions with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.a

Add.

$$\frac{3}{8} + \frac{2}{8}$$

Sum: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 23 • BACKWARD 2 • Q1/5

792

QUESTION ID

**S23-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**5/8**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 23 • BACKWARD 2 • Q1/5



**True or false?**

$$\frac{4}{12} + \frac{5}{12} = \frac{9}{24}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 23 • BACKWARD 2 • Q2/5

793

QUESTION ID

**S23-B2-Q2**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 23 • BACKWARD 2 • Q2/5

$$\frac{7}{10} + \frac{6}{10} = ?$$

A.  $1\frac{6}{10}$

B.  $1\frac{3}{10}$

C.  $\frac{13}{20}$

D.  $\frac{1}{10}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 23 • BACKWARD 2 • Q3/5

794

QUESTION ID

**S23-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**B. 1 3/10**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 23 • BACKWARD 2 • Q3/5

Add.

$$\frac{1}{6} + \frac{4}{6}$$

Sum: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 23 • BACKWARD 2 • Q4/5

795

QUESTION ID

**S23-B2-Q4**

TYPE

Short answer

CORRECT ANSWER

**5/6**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 23 • BACKWARD 2 • Q4/5

**True or false?**

$$\frac{2}{5} + \frac{3}{5} = 1$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 23 • BACKWARD 2 • Q5/5

796

QUESTION ID

**S23-B2-Q5**

TYPE

True or false

CORRECT ANSWER

**A. True**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 23 • BACKWARD 2 • Q5/5

SET 23 • 5.NF.A.1

# BACKWARD 3

Common multiples • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List the first five multiples of 4.

Multiples: \_\_\_\_\_

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 3 • Q1/5

798

QUESTION ID

**S23-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**4, 8, 12, 16, 20**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 3 • Q1/5

**True or false?**

10 is a multiple of both 3 and 4.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q2

Grade 4 • Operations &amp; Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 3 • Q2/5

799

QUESTION ID

**S23-B3-Q2**

TYPE

True or false

CORRECT ANSWER

**B. False**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q2

Grade 4 • Operations &amp; Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 3 • Q2/5

Which number is a multiple of both 6 and 8?

- A.** 16
- B.** 30
- C.** 24
- D.** 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 3 • Q3/5

800

QUESTION ID

**S23-B3-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. 24**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 3 • Q3/5